Recall Performance, Beliefs and Behavior

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Abstract

The US Health and Retirement Study measures recall performance (RP), the ability to recall words from a list, a standard memory measure in psychology. Higher RP people are more optimistic about stock returns, their home's price, and their survival. We hypothesize that RP captures lower interference in a general-purpose recall technology. Accordingly, beliefs of higher RP people in each domain are more sensitive to domain-specific experiences and cues, but also to irrelevant yet similar non-domain specific experiences. Higher sensitivity explains the optimism-gap and why higher RP people are not necessarily more accurate. It also predicts investment, housing, and retirement choices.

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1 Introduction

For over a century, psychologists have studied peoples' ability to recall words from a list (Ebbinghaus, 1885). The US Health and Retirement Study (HRS) has measured this ability, which we refer to as recall performance (RP) (Unsworth and Engle, 2007), for tens of thousands of Americans over two decades. RP declines with age, but differences across people are large, stable, and distinct from numerical ability (IQ) (Krueger and Salthouse, 2011). This trait is economically relevant: HRS respondents with higher RP are more optimistic about the stock market, the price of their home, and their own life expectancy. Their beliefs also update more swiftly when conditions change. Why should remembering words from a list be linked to how a person thinks about the stock market or her own lifespan?

Based on models of mnemonic beliefs (Bordalo et al., 2025, 2026a), we hypothesize that this link may arise because recalling a word list and forecasting the future rely on the same memory technology. When a person is cued to recall words studied minutes earlier, forgetting is due to *interference* from other words or thoughts that come to mind (Kahana, 2012). Higher RP people face less interference: they recall more selectively what is similar to their memory task. Forecasting also relies on this selection. When cued to forecast an outcome (e.g. stock returns), a person does not just consult a statistical model. She also recalls experiences similar to this task, and use them to imagine the future (Bordalo et al., 2025; Schacter et al., 2007). A high RP person recalls such experiences better, so her beliefs are more sensitive to them. Because these recollections are driven by similarity, they include experiences from the target domain, which we call domain-specific (DS), but also superficially similar non-domain-specific (NDS) experiences that may be uninformative.

We test this hypothesis using a standard model of mnemonic beliefs, in which higher RP captures sharper similarity-based retrieval, and hence less interference. The model yields five predictions that should hold in any belief domain X.

1. Beliefs of higher RP people about X are more sensitive to their average past DS experiences. Thus, having lived through higher average stock returns should raise stock market optimism more for higher RP people. The same should hold for home price and survival

beliefs based on average lived housing returns and health conditions, respectively.

2. Beliefs of higher RP people about X are more sensitive to current DS cues. A current stock return that resembles the good part of a person's experienced stock returns should raise stock market optimism more for higher RP people. The same should occur to home price and survival beliefs with respect to current housing returns and health, respectively.

3. The beliefs of higher RP people are not necessarily more accurate than those of lower RP ones. Depending on average experiences and time varying cues, higher RP people may have a larger systematic bias or volatility. For instance, a high RP person may be too optimistic about the stock market if she experienced high average returns, but become too pessimistic during a market crash (and similarly for home prices and survival).

4. The beliefs of higher RP people about X are more sensitive to NDS experiences similar to X, even if uninformative. For instance, better economic conditions during childhood should raise stock market or home price optimism more for higher RP people.

5. Higher RP people have higher confidence and a lower likelihood to answer "I do not know", because more material comes to their mind to imagine the future.

Overall, then, the general purpose nature of memory produces a rich set of cross-equation restrictions: the beliefs of higher RP people should be more sensitive to both DS and NDS experiences, as well as to cues, across different domains.

We test the model in 11 waves of the HRS, covering more than 37,000 respondents. For each respondent and wave we construct: i) the database (lifetime S&P 500 returns, regional home price growth, and health history), ii) the current cue in each domain (current stock return, current home price appreciation, current health state), and iii) a proxy for the similarity of DS experiences to the cue. In a separate survey we elicit the perceived similarity between stock and home price changes to NDS experiences such as childhood health and socioeconomic status.

The evidence supports our predictions, with very similar patterns across domains. Consistent with Predictions 1 and 2, RP amplifies the sensitivity of beliefs to DS experiences and cues. Moving from the bottom to the top quartile of RP raises the effect of average DS ex-

periences by a factor of roughly 2.7 for stocks (from 0.5 to 1.3 pp per interquartile shift in experiences), by 53% for home prices (from 2.9 to 4.4 pp), and by 37% for survival (from 3.3 to 4.5 pp). Because the average DS experiences of HRS subjects have been good, this pattern explains the greater average optimism of high RP people compared to low RP ones. The same shift in RP more than triples the effect of the current cue in all domains: an interquartile shift in the similarity-weighted cue moves stock market beliefs by 0.3 pp in the bottom RP quartile and 1.1 pp in the top, home price beliefs by 2.6 and 8.4 pp, and survival beliefs by 0.06 and 0.2 pp. The effects of RP are comparable in size to variation in standard demographics.

Consistent with Prediction 3, greater sensitivity to DS experiences and cues does not mean greater accuracy: while the beliefs of high RP people are on average more accurate for stocks, they are not so for home prices, and they are less accurate (too optimistic) for survival. In all three domains, beliefs of high RP people also exhibit excessive volatility in response to changing DS conditions. Moreover, and consistent with Prediction 4, RP amplifies irrelevant experiences: the beliefs of high RP respondents are more sensitive to uninformative but similar NDS experiences. In the bottom RP quartile, childhood health and childhood socioeconomic status have no detectable effect on beliefs. In the top quartile, an interquartile improvement in childhood socioeconomic status (childhood health) raises the subjective probability that stocks go up by 1 pp (0.3 pp) and that home prices go up by 0.8 pp (0.2 pp), while an interquartile rise in lifetime stock returns raises survival beliefs by 1.9 pp. Finally, in line with Prediction 5, high RP people are 13 percentage points less likely than low RP people to say "I do not know" about stock returns, 7 points about their home price, and 6 points about their survival.

These results support our hypothesis that RP captures a general purpose recall technology. High RP people are not simply more informed or more sensitive to information; in that case, they would rely *less* on their own DS and especially NDS experiences, and rely more on objective metrics, exhibiting more accurate beliefs compared to low RP people. The data reject this view. Related to our finding, Hartzmark et al. (2021) show that more accurate recall of the return on an asset (due to ownership) fosters belief extrapolation.

The impact of RP on beliefs shapes choice. The greater sensitivity of high RP people to DS and NDS experiences predicts higher stock investments, equity share in retirement ac-

counts, home purchases, plans to work past 70, and pension contributions. This finding holds controlling for risk tolerance and for direct effects of RP on choices. Thus, RP helps explain choice heterogeneity and instability as a product of memory.

Growing work shows that memory affects beliefs within specific domains (Bordalo et al., 2020, 2026a; Conlon, 2024; Enke et al., 2024; Gennaioli et al., 2025; Jiang et al., 2025; Sial et al., 2023; Taubinsky et al., 2026). This work founds the DS experience effects shown in Malmendier and Nagel (2011, 2016)'s seminal work, but also stresses that their effect: i) can be catalyzed by time-varying cues, and ii) can be dampened due to interference from NDS experiences, which are then used for forecasting. Our innovation is to show that a memory parameter measured in a task with no economic content scales the interaction between cues, DS and NDS experiences across three different domains. The structure of memory can offer a general theory linking experience-based biases and excess volatility of beliefs to economic behavior, with cross equation restrictions across different domains.

Our second contribution is to identify RP as a new and easily measurable source of belief and choice heterogeneity. Neoclassical and behavioral economics have made progress in explaining heterogeneity using variation in tastes such as risk aversion, present bias, etc. We show that heterogeneity in RP, a purely cognitive trait, offers a complementary source by shaping the sensitivity of behavior to lived experiences and time varying cues, including uninformative ones. In the conclusions we discuss the broader implications of this finding.

Section 2 describes the data and links RP to beliefs, Section 3 presents the model, Section 4 tests it, and Section 5 studies choices. Section 6 concludes.

2 Motivating Facts

2.1 The HRS survey

The Sample. The University of Michigan Health and Retirement Study (HRS) is a representative biennial panel of U.S. adults over age 50. It adds a new cohort roughly every six years and follows respondents until death, with some attrition. We consider 11 waves, starting in 2002 when our belief variables start being collected, resulting in 37,438 unique respondents and

204,885 person-wave observations. Appendix Table B.1 and Table B.4 describe the survey and our sample. Among the respondents, 54% are retired, 63% are married, 59% are female. The average age, education, and income are 67.6 years (11.4 s.d.), 12.6 years (3.3 s.d.), and $64,496 ($129,635 s.d.), respectively.[1] We use age, gender, marital status, retirement status, income, and education as regression controls.

Outcomes: Beliefs and Choices. The HRS elicits beliefs about social and personal events. We focus on the stock market, the price of one's home, and life expectancy. These beliefs play a key role in economic models.[2] The stock market question elicits the believed percent chance that mutual fund shares invested in blue chip stocks will be worth more a year from now. The home price question asks homeowners to assess the percent chance that their own home will be worth more a year from now. The life expectancy question asks respondents to assess the percent chance that they will live to a target age, equal to 75 for respondents under 65, to 80 for those aged 65–69, to 85 for those aged 70–74, and so on.[3]

The HRS also measures choices related to these beliefs. In the stock market domain, we consider whether the respondent has made a stock investment in the past 2 years (yes/no) and the percentage of the respondent's IRA (if positive) invested in stocks or mutual funds. In the home price domain, we consider whether the respondent has bought a new home in the past 2 years. For life expectancy, we measure whether she expects to work past the age of 70 (if currently working) or is retired, and the % of the respondent's income invested in their pension (if they have a defined contribution plan).

2.2 Recall Performance and IQ

The HRS measures a range of cognitive faculties. We focus on memory but also consider mathematical ability. Memory is measured using three tasks. The first two are based on word lists, a standard psychology proxy for recall performance (Kahana, 2012). Respondents study a

[1] We combine salaried income, self-employment income, bonuses, social security benefits, workers compensation, and pensions. The combination of different incomes creates a high variance.

[2] The HRS also elicits expectations about US inflation and economic performance, but only in a few waves (2002-2008). We later show that our key results extend to these measures. Other beliefs are hard to trace to measured experiences (e.g. beliefs about inheritance), so we neglect them.

[3] In 2002 and 2004 respondents aged 69 or below were asked about age 80, with no separate group below 65. The wording of each belief question, and the full target-age schedule by wave, is in Appendix Table B.1.

list of 10 words and are asked to recall them: i) right after the study period (Immediate Memory task), and ii) after a five minutes questionnaire on mental health (Delayed Memory task). Recall performance is the number of successful recollections. In the third task, available in waves 2010-2020, respondents must name as many animals as possible in 60 seconds. Performance is the number of recalled animals. We construct an index of *Recall Performance* (RP) by standardizing memory scores across waves and by averaging, in each wave, a respondent's available components.[4] It captures recall by abstracting from differences in education and world knowledge (the words are common), and does not entail symbol manipulation or reasoning.

Mathematical ability is measured in the HRS using three tasks. In "counting backwards" respondents must perform subtractions starting from a given number. In "number series" they guess a rule by filling the last item of a series. In "numeracy" they convert percentages into absolute frequencies. Higher performance indicates higher math ability. Counting backward is available in all years, number series is elicited in the 2010, 2012, 2016 and 2020 waves, numeracy is only asked to first-time respondents. Math ability affects the use of statistical information, a key driver of beliefs in standard models. Controlling for it allows us to isolate memory.[5] Our index IQ is constructed by standardizing math performance in each task across waves and by averaging, in each wave, a respondent's available components.

Figure 1 plots the distribution of person-level average RP (residualized by IQ and demographics, Panel A) and IQ (residualized by RP an demographics, Panel B). Memory tasks do not require math ability but some math tasks (e.g. counting backwards) benefit from memory. Consistent with the positive correlation of cognitive faculties (Spearman, 1904; Unsworth et al., 2024), the correlation between RP and IQ is 0.4 (see Table C.1 and Figure C.5). There is however large independent variation: regressing RP (IQ) on demographics yields an R^2 of only 0.24 (0.21). Adding IQ (RP) to this regression raises the R^2 to 0.3 (0.28). Pooling

[4] Appendix Table B.2 lists the components and aggregation of the recall performance and IQ indices, along with every other index used in the paper.

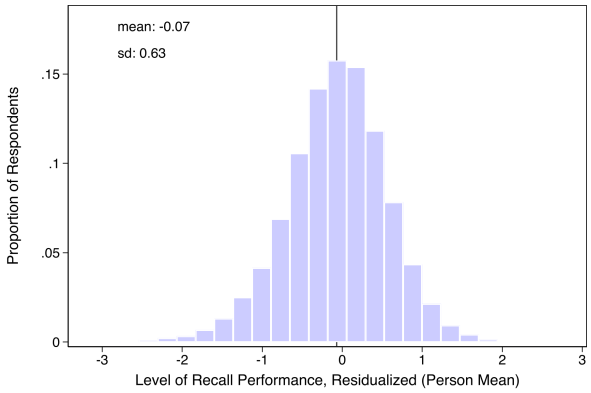
[5] D'Acunto et al. (2023) show that IQ predicts inflation expectations. The HRS also elicits: 1) General Knowledge, consisting in factual recall (e.g., identifying the current President or today's date), 2) word meaning, requiring definition of words (e.g., "repair," "fabric," "plagiarize"), and 3) analogies such as filling "Mother is to Daughter as Father is to [blank]". The first two measures align with the crystallized intelligence psychometric, verbal analogies are less clear. We do not incorporate these measures in our IQ index because verbal analogies are available in only three survey waves (2012, 2014, and 2018), general knowledge is administered to only half of respondents in each wave, and word meanings is administered only to first-time respondents.

a person's scores across waves, residualizing reduces the standard deviation of RP by only 0.17 and that of IQ by only 0.15 (Appendix Figure C.4).

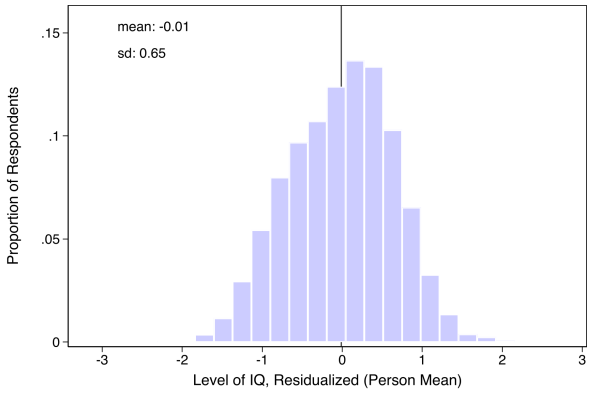
In Figure 1, what is the relative importance of stable interpersonal differences versus cognitive aging? Figure 2 reports, for the 3,169 respondents present in all 2002–2020 waves, the average of the index over time (black line), the average among respondents in the top 25% of the index in 2002 (blue line) and that among those in the bottom 25% in 2002 (red line). RP is in Panel A, IQ in Panel B (restricting to the counting-backwards task, which is available throughout the sample). Appendix Figure C.2 shows the full IQ index alongside this sub-index.

Figure 1: Distribution of Recall Performance and IQ.

Panel A: Recall Performance (Residualized)



Panel B: IQ (Residualized)



Notes: Each panel plots the distribution across respondents of a person-level average, where for each respondent the relevant index is averaged over all survey waves in which they appear (one observation per respondent). Panel A shows person-level average Recall Performance (RP) after residualizing on IQ, age, income, education, gender, marital status, and retirement status; Panel B shows person-level average IQ after residualizing on RP and the same demographics. Both indices are constructed as the standardized mean of their components and are expressed in standard deviation units. The raw (un-residualized) person-level distributions are reported in Appendix Figure C.3, and the corresponding pooled (person-wave) distributions in Appendix Figure C.4 (residualized) and Figure C.5 (raw).

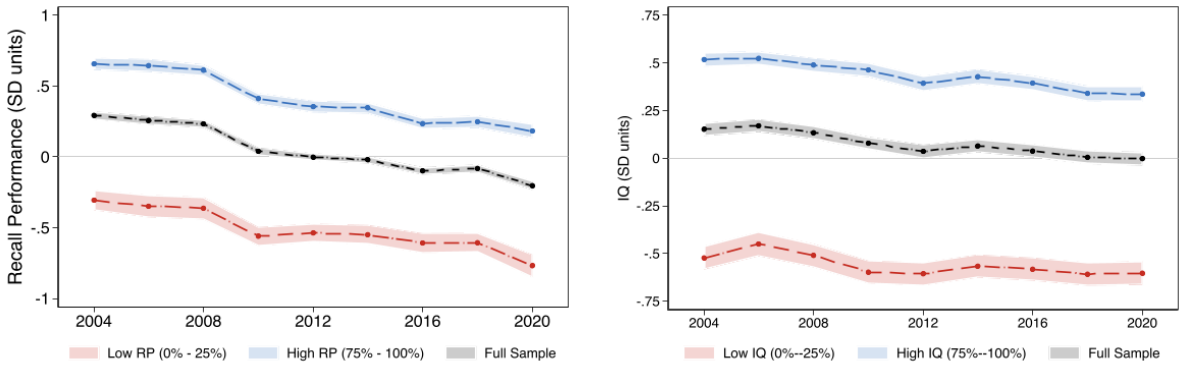
Interpersonal differences in RP are stable, large, and dominate the average decay over time. The interquartile range in RP is 0.96 standard deviations in 2004, and 0.95 in 2020 (in comparison, the average RP declines by 0.57 standard deviations between 2002 and 2020).[6] Stable differences emerge at the level of individual measures: Appendix Figure C.1 plots the

[6]We cannot compare quartiles in 2002 variation in this year is partly driven by random shocks that revert in 2004. The "within" standard deviation due to age is 0.55, the "between" standard deviation due to average differences is 0.70, so the latter accounts for 62% of the total variance in RP, see Table C.2. If we residualize respondents' average RP by average IQ, the standard deviation drops by 0.13, again showing the significant orthogonal variation in the two measures.

Spearman rank correlation between each pair of memory sub-components at lags of zero to five survey waves, for a horizon of up to ten years. Correlations remain high across all lags, showing stable performance rankings. Similar patterns appear, but are less pronounced, for IQ.

Figure 2: Recall Performance and IQ Over Time, by Quartile

Panel A: Recall Performance *Panel B: IQ*



Notes: The figure plots the average level of Recall Performance (Panel A) and IQ (Panel B) in each survey wave from 2004 to 2020. Panel A uses the full memory index. For IQ (Panel B) we plot only the counting-backwards sub-index, the single IQ component administered in every wave, because the number series and numeracy components enter only from 2010; the full IQ index is shown alongside the always-available sub-indices in Appendix Figure C.2. Blue lines show respondents in the top 25% of the respective index in 2002; red lines show the bottom 25%; black lines show the full sample. Shaded bands are 95% confidence intervals. The sample is restricted to the 3,169 respondents present in all ten survey waves (2002–2020). 2002 is omitted as it is the base year for group assignment. Memory and IQ are expressed in standard deviation units.

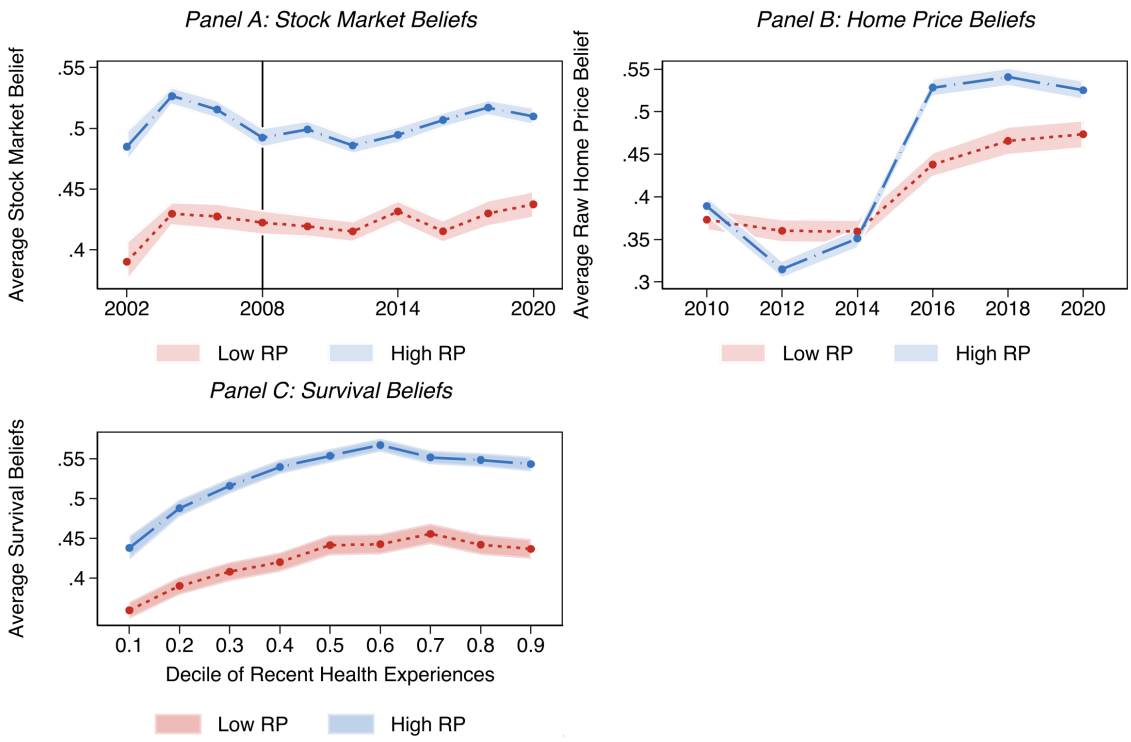
In sum, RP is a stable personal trait, that can be studied controlling for IQ. We next show that it correlates with beliefs about aggregate and personal events.

2.3 Recall Performance and Beliefs

Panel A of Figure 3 reports the raw average estimate of the percent chance that stocks will be worth more in 1 year, for the top (blue) and bottom (red) RP quartiles and across survey waves. There is a large difference between high and low RP people, in both the level and time variation of beliefs. High RP people on average estimate a stock market increase to be 8 percentage points more likely than low RP people. Differences in time variation are also large: high RP people exhibit a sharper reaction to the financial crisis, with beliefs falling by 3.4 percentage points from the 2004 peak, and recovering by 2.5 percentage points from the 2008

trough; over the same periods, low RP beliefs fall by 0.7 p.p. and recover by 0.8 p.p.

Figure 3: Average Beliefs Over Time by Recall Performance.



Notes: Each panel plots the average raw (un-residualized) belief for the top (blue) and bottom (red) quartile of Recall Performance; the residualized counterpart is Appendix Figure C.7. Panel A: stock market beliefs ("chance stocks will be worth more in one year") by survey wave. Panel B: home-price beliefs ("chance home will be worth more in one year") by survey wave from 2010 onward, pooling both question framings. Panel C: survival beliefs ("chance live 10+ years") by decile of recent health experiences, i.e. the change in the current-health index between consecutive surveys ($\Delta\text{HealthPCA}_{t,t-1}$) described in Section 4. Shaded bands are 95% confidence intervals. The raw belief distributions are shown in Appendix Figure C.9.

Panel B of Figure 3 reports, for the two RP groups, the estimated percent chance that the respondent's own home will be worth more in one year. The question is asked to homeowners— who fully own or are paying off their mortgage—corresponding to 44% of the sample, and is available only from 2010 onwards, after the financial crisis.[7] The beliefs of high RP people again exhibit larger time variation. Right after the crisis, high RP people are relatively more pessimistic about their home prices (their beliefs drop by 6.1 p.p. more than

[7]The home-price question uses two framings: some homeowners report the probability that their home will be worth *more* in a year, others the probability it will be worth *less*. To use all homeowners we pool the two —reverse-coding negative-frame responses onto the probability-of-increase scale and adding a constant equal to the difference in mean belief between the two framings. This adjustment affects levels, not the sensitivity of beliefs to experiences and cues. Appendix Figure C.8 shows the series separately by question framing.

those of low RP individuals during 2010–2012). As home prices increase, however, their beliefs recover (rising by 21.3 percentage points from 2012 to 2016, versus 7.8 for low RP people) and they become more optimistic than low RP people.

Finally, Panel C of Figure 3 reports beliefs about own life expectancy as a function of "health fundamentals", measured as the current change in health status (see Section 4). Higher RP people are more optimistic about their life expectancy than low RP people at every level of health fundamentals. They are also more sensitive to changing conditions: high RP optimism rises more steeply as health improves. The cumulative increase in beliefs across the health range is to 10.5 p.p. for the high RP group compared to 7.7 p.p. for the low RP group.

We thus see a strong and surprising regularity: performance in abstract word recall tasks is associated with sharp differences in both the level and the sensitivity of beliefs about macroeconomic and personal events. Why is this the case? One possibility is omitted variables: high RP people may have different tastes or information than low RP ones (note however that word recall tasks do not rely on information or on preferences). A second possibility is that RP captures a parameter of a person's "recall technology", which affects her recall in laboratory tasks but also her recall of experiences in the field, and thus her beliefs. To assess this hypothesis, we introduce a standard model of mnemonic beliefs to obtain distinctive predictions about the link between RP and beliefs. In the regression analysis, we also control for variation in tastes and information by controlling for a rich set of demographics and IQ (the patterns of Figure 3 are unchanged when we do so, see Figure C.7).

3 The Model

We present a model of mnemonic expectations (Bordalo et al., 2025, 2026a, Bonaglia and Gennaioli, 2025). We first describe the recall technology and link RP to one of its parameters. We then derive the predictions of the recall technology for the link between RP and expectations.

3.1 The Recall Technology

Memory hinges on: i) a database of experiences, ii) a cue, and iii) a function mapping the cue to recall from the database. Extensive evidence from word lists (Baddeley and Hitch, 1974; Kahana, 2012) shows that recall follows frequency, similarity, and interference regularities.

The database. A category of experiences is a trace e in a mnemonic space E. The database is a frequency distribution over E, denoted $\mathbf{f} = \{f_e\} \in \Delta(E)$. A vector $e \equiv (s, v, c) \in E$ reports the objective realization or signal $s \subset \mathbb{R}$ of the experience, its emotional valence $v \subset \mathbb{R}$, and semantic context $c \subset C$, where C is a context space. For example, a positive stock market experience reports a numerical return range, say $s = (2\%, 3\%)$, a marker for positive emotional valence, $v > 0$, and marker c for the semantic "stock market" context, the year, location, etc.

Similarity and cues. The mnemonic space is endowed with a function $d : E \times E \to \mathbb{R}$ that measures the discrepancy between any two experiences $e, e' \in E$, where $d(e, e')$ increases in the difference among their features. Similarity between experiences e and e' decreases in $d(e, e')$. A personal illness e' differs from a stock market experience e along the objective, subjective, and context features, entailing a high $d(e, e')$ and thus low similarity. We later measure perceived similarity. A cue is a subset $q \subseteq E$ that shapes recall, defined as the probability $r_e(q, \mathbf{f})$ of retrieving experience e given cue q and database $\mathbf{f}$:

$$r_e(q, \mathbf{f}) = \frac{S(e, q) \cdot f_e}{\sum_{e' \in E} S(e', q) \cdot f_{e'}} \tag{1}$$

The numerator captures similarity and frequency: experiences more similar to the cue and more frequent are more likely to be recalled. The denominator captures interference: even if e is similar to q and frequent, its recall can be blocked by an experience $e' \neq e$ that is more similar to q or more frequent. We now describe how RP proxies for the key interference force.

3.2 RP in The Recall Technology

To link RP to the model, we map Equation 1 to word list tasks, in two steps.

Study phase. Studying a word list affects the database. A word in position $j = 1, \ldots, n$ has an objective signal or phonetic value s_j and a context c_j capturing semantic and temporal

features (we abstract from valence). Let $L \subset E$ be the set of list traces $e_j = (s_j, c_j)$. Encoding and rehearsal during study attaches positive frequency f_j to each e_j.

Recall phase. In Equation 1, the initial cue q_0 is the word list $(s_j)_j$ and the initial context is c_0. This cue shapes the first recalled word. This recollection in turn influences the cue, which affects the second recalled word, and so on. In contextual drift models (Howard and Kahana, 2002), this is formalized using the AR(1) process $q_{t+1} = \rho q_t + (1-\rho)e_j$, $(1-\rho) \in (0, 1)$.

Word list tasks exhibit four regularities (Kahana et al., 2024): primacy (words studied earlier are more likely to be recalled), recency (words later on the list are more likely to be recalled than words in the middle), contiguity (recalling a word in position 5 fosters recall of position 4 or 6), and semantic proximity (recalling "car" promotes recall of "brake").

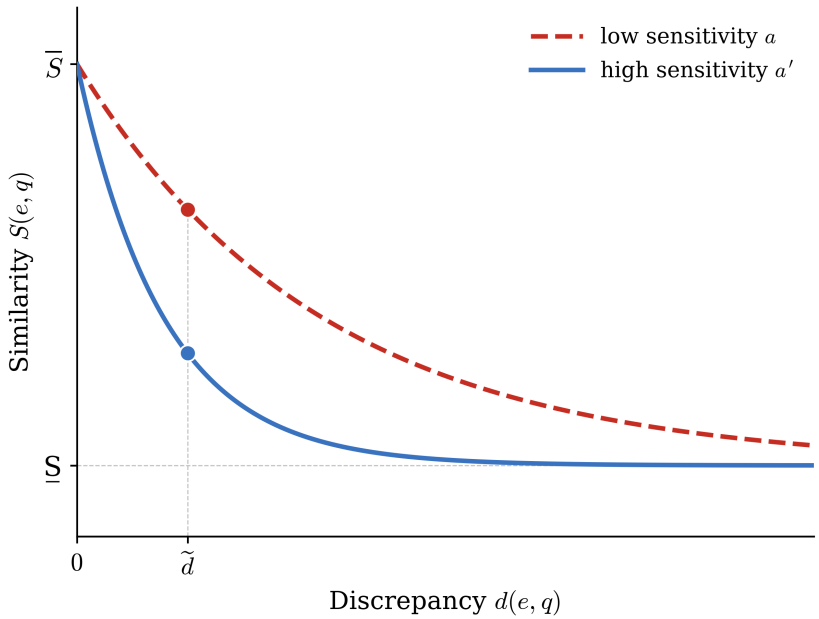
Primacy reflects frequency: the first word studied s_1 tends to be rehearsed more often than others, increasing its encoded frequency f_1 and thus its recall in Equation 1. The other three regularities reflect similarity. Consider recency: the last word studied s_n has a temporal context c_n similar to the subsequent retrieval context c_0, increasing its recall in Equation 1. Contiguity and semantic effects occur recall of s_j pollutes the cue with its context c_j, favoring recall of words studied at similar times or having related meanings.

Recall Performance. In typical experiments, the time delay between study and recall is short, so forgetting words (low RP) is not due to depreciation of stored traces but to interference: the inability of the subject to focus on words on the list compared to distracting items. By Equation 1, recall of a frequently rehearsed, recent, contiguous, or semantically similar word s_j on the list inhibits retrieval of other words s_k that do not have these properties. Crucially, randomly intruding thoughts from frequent outside-the-list experiences $e \notin L$ can also block recall from the list (Zaromb et al., 2006), particularly if perceived as similar enough to the list.

In the model, therefore, high RP reflects a sharper similarity function, that promotes selective recall of items – the studied words – that match the "word list" cue. Formally, consider the similarity function $S(e, q) = \mathcal{S}[a \cdot d(e, q)]$ where $\mathcal{S}$ is decreasing. In this parameterization, $a \in (0, 1)$ measures the sensitivity of perceived similarity to objective discrepancies in features. Figure 4 plots in blue (red) the similarity function of a person with high (low) sensitivity a.

For experiences identical to the cue, $d(e, q) = 0$, similarity is maximal regardless of a.

Figure 4: Similarity as a Function of Discrepancy, at High Versus Low Sensitivity.



Notes: We plot a similarity function $S(e,q) = \mathcal{S}[a \cdot d(e,q)]$ with $\mathcal{S}$ decreasing and bounded between $\underline{S}$ and $\overline{S}$. The blue curve reflects higher similarity sensitivity ($a' > a$) than the red curve. The functional form is $\mathcal{S}(x) = \underline{S} + (\overline{S} - \underline{S}) \exp(-x)$.

In contrast, an experience $\widetilde{e}$ with discrepancy $\widetilde{d} > 0$ is perceived to be quite similar to q if similarity has low sensitivity (red), but dissimilar if sensitivity is high (blue).[8]

The mapping to RP is immediate. Consider the probability of recalling any given word from list L in period t (with cue q_t):

$$\Pr\left(e_j \in L | q_t\right) = \frac{\sum_{e_j \in L} \mathcal{S}\left[a \cdot d\left(e_j, q_t\right)\right] \cdot f_{e_j}}{\sum_{e' \in E} \mathcal{S}\left[a \cdot d\left(e', q_t\right)\right] \cdot f_{e'}} \tag{2}$$

By differentiating Equation 2 around $a_0 = 0$ we obtain:

$$\frac{\partial \Pr\left(e_j \in L | q_t\right)}{\partial a} > 0 \quad \Longleftrightarrow \quad \frac{\sum_{e_j \in L} d\left(e_j, q_t\right) \cdot f_{e_j}}{\sum_{e_j \in L} f_{e_j}} < \frac{\sum_{e \notin L} d\left(e, q_t\right) \cdot f_e}{\sum_{e \notin L} f_e}. \tag{3}$$

Higher sensitivity a increases the probability of recalling any target $e_j \in L$ if words on the list have on average more features in common with the cue q_t than do other experiences. This condition is satisfied because items in the list share the time and place when they are studied,

[8] Bounded similarity implies in Figure 4 that a matters little at large $d(e,q)$. This property can be relaxed.

the task context, etc, compared to most experiences in the memory database. Higher a then increases the total number of retrieved words, increasing RP.[9]

Where do differences in a come from? In psychology, the sensitivity of similarity is linked to attention (Nosofsky, 1986): if two objects differ in a feature, greater attention to that feature reduces their similarity. People more attentive to the features of the list are less likely to recall distracting items, exhibiting high RP. Attention in word tasks has been studied since Murdock (1965), where it is viewed as a main driver for the primacy effect.[10] Persistent variation in attention across people is also well documented (Hirmas et al., 2024; Rosenberg et al., 2016; Unsworth et al., 2024), offering a source of stable heterogeneity in a. We will refer to parameter a interchangeably as similarity sensitivity or attention, and proxy for an individual's a_i with their measured RP_i.

3.3 Selective Recall and Beliefs

To see how variation in memory interference – captured by the sensitivity of similarity – affects beliefs, we study a model of mnemonic expectations. The DM forms beliefs about event $T_X \subset \mathbb{R}$, in domain $X = SP, HP, LE$, where SP and HP are stock and own home prices, and LE is life expectancy. Following Bordalo et al. (2025), the DM: i) recalls experiences $e = (s, v, c) \in E$, and then ii) uses recollections to imagine T_X (Hassabis et al., 2007; Schacter et al., 2007). The cue for person i at time t specifies: the current signal $s_{X,it}$ (recent stock or housing return or changes in own health), the person's emotion v_{it} (higher values stand for better mood), and the semantic context $c = X$ (we abstract from time). Formally $q_{X,it} = (s_{X,it}, v_{it}, X)$, summarizing the features that shape recall in Equation 1.

The DM uses a recalled $e \in E$ to imagine an outcome in the target space $\mathbb{R}$ based on a simulation function $\sigma_X : E \to \mathbb{R}$ that satisfies the following properties.

[9] By Equation 1, variation in RP may also arise due to differences in the encoded frequency f_j of words on the list L. This may occur if some people encode recent events more weakly (their f_j is lower) or have more numerous extralist experiences (their $\sum_{e \notin L} f_e$ is higher). It may help explain why RP declines with age: older people encode recent events less and have more irrelevant experiences than young ones (Craik and Salthouse, eds, 2008; Healey et al., 2008). This mechanism is less relevant in our context where – to form beliefs – HRS respondents must also rely on changing cues and experiences from the remote past.

[10] Also, instructing participants to classify words along a feature, such as pleasantness or size, increases subsequent retrieval of words associated with that feature (Polyn et al., 2009). Higher attention at encoding, as measured by pupil dilation, is associated with stronger recall (Miller and Unsworth, 2021).

Assumption 1 *For $e = (s, v, c)$, simulation $\sigma_X(e)$ satisfies: i) $\sigma_X(e) = s$ for DS experiences ($c = X$), ii) $\sigma_X(e)$ increases in v and does not depend on s for NDS experiences ($c \neq X$), and iii) $\sigma_X(e) = \delta_X$, a default value, for "neutral" events ($v = 0$).*

When the DM retrieves a DS experience e in $c = X$, she extrapolates the recalled realization s into the future. When she retrieves an NDS, $c \neq X$, the DM simulates based on valence, imagining a rosier future if the experience has a higher v. Recalling pleasurable financial experiences (i.e. earning a bonus) helps imagine high stock returns or growth in the price of one's own home, and conversely if bad experiences are recalled (Bordalo et al., 2026a). This occurs if higher realizations are perceived as good by the DM, and hence similar to positive NDS, an assumption we test later. If instead the recollection is neutral, $v = 0$, simulation is inhibited: the DM picks an agnostic default δ_X such as "I do not know" or an arbitrary element such as 50% in the response scale. Thinking about routine events such as walking to the office does not aid simulation. To ease notation, we set $\delta_X = 0$.[11]

The average future state $\mathbb{E}_{it}(s_X)$ simulated in X by a respondent i with database $\mathbf{f}_i$ and cue $q_{X,it}$ (where the average is taken over many memory samples) is then given by:

$$\mathbb{E}_{it}(s_X) = \sum_{e \in E} r_e(q_{X,it}, \mathbf{f}_i) \cdot \sigma_X(e) \tag{4}$$

The expectation is higher when the DM selectively recalls experiences leading to higher simulation. Beliefs are shaped by frequency and similarity, the drivers of recall in $r_e(q_{X,it}, \mathbf{f}_i)$.

Frequency causes beliefs to anchor to the database: a person is more optimistic about X (e.g. stock returns) if she had more positive DS experiences (high stock returns) or more NDS experiences that prompt stock market optimism (e.g. positive personal financial experiences). However, because the number of lived experiences is vast and many of them are too remote from the target event to aid its simulation, the same mechanism also promotes the use of the agnostic anchor δ_X, producing insensitivity to lived DS and NDS experiences. This is why similarity matters: it causes selective recall of DS experiences and of NDS ones which, being similar to the target event, promote its simulation. A sharper similarity gradient entails higher RP and

[11] We can generalize Assumption 1 and allow for simulation to flatten with context distance, as in Bordalo et al. (2026a). Our predictions are strengthened if we allow for this property.

reduces interference from a myriad of events incongruent with the cue's valence, context, or realization, leading to a higher sensitivity of beliefs to lived DS and NDS experiences.

We use Equation 4 to derive our predictions. The estimated probability of events T_X elicited in the HRS (an increase in stock or home prices, or survival for 10 years) increases in the average expected outcome in domain X, so comparative statics for Equation 4 apply also to T_X.[12] To obtain sharp predictions testable using linear regressions, we linearize $S(e,q)= \mathcal{S}\left[a \cdot d(e,q)\right]$ at the constant similarity benchmark $a = 0$, and proxy similarity sensitivity a_i with measured RP_i. We further assume the following tractable discrepancy function:

$$d\left(e, q_{X,it}\right) = \begin{cases} \left(s_{it} - s\right)^2 & if \quad c = X \\ K & if \quad c \neq X,\ v \subseteq v_{it} \\ K + V & otherwise \end{cases} \tag{5}$$

where $V > 0$, and $K > 0$ is high enough that $(s_{it} - s)^2 < K$ for realistic s. In the first row, the discrepancy of a DS experience e from the cue $q_{X,it}$ increases in the quadratic distance between their realizations s and s_{it}. If the current stock return is 10%, a 3% experience is more discrepant than a 9% one. In the second row, discrepancy jumps to K for an NDS event with congruent valence. A positive NDS such as an income windfall is less similar to the a 10% stock return than a DS event, but still somewhat similar to it due to their positive valence. Discrepancy jumps further by V if the NDS experience is also valence-incongruent with the cue. A past financial trouble is most discrepant with a 10% stock return due to low valence.

This ranking means that a DS experience is ceteris paribus most likely to be recalled, the valence incongruent NDS least likely to be recalled, and the valence congruent NDS is in the middle. Model predictions concerning DS experiences follow from the following result.

Proposition 1 (DS, RP, *and beliefs*) *Let $\overline{s}_{it}$ be the average DS signal in the database. With a linear approximation of $\sigma_X(e)$, there are positive constants $\eta_0, \eta_1, \eta_2, \eta_3$ such that beliefs satisfy:*

$$\mathbb{E}\left(s_X | q_{X,it}, \mathbf{f}_{it}\right) \approx \eta_0 + \eta_1 \cdot \overline{s}_{it} + \eta_2 \cdot \text{RP}_i \cdot \overline{s}_{it} - \eta_3 \cdot \text{RP}_i \cdot cov\left[s, \left(s - s_{it}\right)^2\right]. \tag{6}$$

[12] Results would change little if the assessed probability of the target event is modelled as the probability of recalling experiences that allow to simulate it, $\sum_{e:\sigma_X(e)\in T_X} r_e\left(q_{X,it}, \mathbf{f}_i\right)$.

The DM imagines a higher future state if she lived through a higher average DS state $\overline{s}_{it}$, as in experience based learning Malmendier and Nagel (2011, 2016). However, the corresponding coefficient η_1 – while positive – is attenuated by interference from the innumerable experiences that do not aid simulation. Positive DS experiences can be forgotten. This is why similarity, and hence RP, play a key role, giving rise to the predictions below.

Prediction 1. The marginal impact of the average DS experience $\overline{s}_{it}$ is stronger for a person with higher RP, $\eta_2 > 0$. When cued to think about the stock market, this person recalls many more stock market experiences, so beliefs are more sensitive to their average value.

Consider next the term $cov[s, (s - s_{it})^2]$ in Equation 6. It proxies for the discrepancy between the current cue s_{it} and the respondent's above average DS experiences. If this covariance is lower, a good state is more similar to the respondent's above average DS experiences, fostering their over-sampling and hence optimism. This effect is entirely channeled by similarity. It should thus be zero for a person whose similarity is flat, $\text{RP}_i = 0$, and should instead be strong for a person with high RP_i. The following prediction then obtains.

Prediction 2. The marginal impact of the cue s_{it} is stronger for a person with higher RP, $\eta_3 > 0$. With a sharper similarity gradient, seeing a 10% stock return promotes the recall of past returns close to 10%. This causes an additional belief boost if 10% is above the person's average experienced return (and conversely if 10% is below her average experienced return).

Gennaioli et al. (2025) show that the inflation expectations of US households rapidly increase if current inflation is similar to the high inflation experienced by these people in the past. In their analysis as well, this effect is captured by a negative covariance coefficient. We make a key new prediction: this effect should be stronger for people with higher RP.

Prediction 3. Higher RP people do not necessarily have more accurate beliefs. They may extrapolate too much from their average DS experiences $\overline{s}_{it}$ as well as from the time varying cue s_{it}. For instance, even if stock returns have only a mild serial correlation, a high RP person will extrapolate a current high return more sharply into the future, exhibiting higher excess optimism and predictable forecast errors.

This prediction distinguishes the hypothesis that high RP people have a sharper similarity gradient from the hypothesis that they are just more informed or statistically savvy, and hence more accurate. It is consistent with Hartzmark et al. (2021), who find that greater attention to recent stock returns, in their case driven by ownership, leads to greater belief extrapolation. A related prediction concerns NDS experiences, to which we turn next.

Prediction 4 *Suppose that neutral or v -incongruent NDS are frequent enough in the database. There is an $\eta_4 > 0$ such that having lived a v-congruent NDS affects beliefs as follows:*

$$\left.\frac{\partial \mathbb{E}\left(s_X | q_{X,it}, \mathbf{f}_{it}\right)}{\partial f_{e,it}}\right|_{v \subseteq v_{it}} \approx \frac{1}{|\mathbf{f}_i|} \cdot \sigma_X(v) + \eta_4 \cdot \mathrm{RP}_i \cdot \sigma_X(v). \tag{7}$$

The frequency condition on valence neutral or incongruent NDS experiences is easily met, given that each specific forecasting task relies on a tiny slice of our life experiences. In Equation 7, consistent with Assumption 1, NDS-based simulation only depends on v and boosts optimism if and only if the NDS experience is positive: $\sigma_X(v) > 0$ iff $v > 0$. This role of NDS experiences has already been documented (Bordalo et al., 2025; Cenzon, 2026; Taubinsky et al., 2026). The new prediction here is that this effect increases with RP. Indeed, by reducing interference, higher RP increases the effect of valence-congruent NDS experiences. At zero RP, the NDS effect may be small or zero: an individual experience would face strong interference in a large database. At high RP, the NDS is instead selectively recalled and used in simulation.

Because many NDS are uninformative about the target (e.g. a past illness is uninformative about the aggregate stock market), their use causes spurious variability in beliefs and hence inaccuracy. Bordalo et al. (2026a) show that cuing recall of NDS experiences such as personal financial or health troubles causally affects macroeconomic beliefs. Here we rely on interpersonal variation in the ability to recall these NDS experiences: the beliefs of high RP people should be more anchored to them compared to those of low RP people. Again, high RP people are not better forecasters or more sensitive to information. They are more sensitive to similar experiences, both DS and NDS ones, which may further reduce accuracy.

Finally, because a high RP person retrieves more material that helps imagining the future, her opinion is more grounded and hence her beliefs are less agnostic.

Prediction 5 (*Agnosticism*) *The probability of using the agnostic default* δ_X *falls in* RP_i.

High RP people can vividly simulate the target event, so they use the agnostic default less and are more confident than low RP people, even if their beliefs are not more accurate.

The above predictions distinguish memory from standard models in which beliefs rely only on information (in a rational or distorted way). They also distinguish memory from models where beliefs depend on tastes: motivated beliefs can account neither for a role of RP, which is measured in a task with no motives, nor for why RP symmetrically amplifies reliance on positive and negative experiences. We next bring these predictions to the data.

4 Testing the mechanism

We first describe how we proxy for the three memory drivers: the database, cues, and similarity.

The Database. In each period, a respondent's database reflects her accumulated experiences up to that period. We construct DS experiences for each domain as follows.

Stock and housing markets. We create a respondent's database of real stock returns over her lifetime, using the Shiller S&P500 dataset (https://shillerdata.com/); each entry is the trailing twelve-month log real return, computed every month from the respondent's birth through the survey month. We measure housing real returns from the Federal Housing and Finance Agency's (FHFA) home price index (https://www.fhfa.gov/data/hpi/datasets?tab=quarterly-data), based on sale and appraisal data.[13]

Health. We build indexes of childhood and current health. Childhood health is the average of the standardized values of: missed time in school for health reasons (1,0), disabled as a child (1,0), number of childhood health conditions, number of severe childhood health conditions, and number of psychological childhood conditions. Current health is the first principal component of: days spent in bed last month, number of diagnosed health conditions, number of major health conditions (out of eight), self-rated health (1-5), change in self-rated health since

[13]The census partitions the country into nine regions, each encompassing several states. The FHFA regional index is only available from 1975 onward; for earlier years, we used Robert Shiller's national real home price index (Case–Shiller) (https://shillerdata.com/) to capture the full respondent history, using variation across birth cohorts. Appendix Table B.3, Panels A and B, describes the stock-market and home price databases.

the previous survey, depressed (binary), recent heart attack (binary), new/worsening severe health condition (binary), sleep quality, number of daily tasks the respondents has difficulty doing on their own, whether the respondent has a functional helper (binary), and number of tasks a functional helper helps with on a daily basis. Both indexes are oriented so that higher values mean better health: childhood components for which a higher value means worse health (all five) are multiplied by -1 before standardizing and averaging, and, since every current-health component is coded so that a higher value means worse health (more days in bed, more conditions, worse self-rated health), we multiply the first principal component by -1. Health DS experiences are the changes in the current-health index between consecutive surveys: we initialize a respondent's health database with her childhood health and add these changes.[14]

NDS experiences. Our NDS experiences for stock and housing markets are childhood health status, described above, and childhood socioeconomic status, defined as the mean of the following standardized variables: father's years of education, mother's years of education, and whether when the respondent was a child the father was unemployed (1,0), the mother worked (0,1), the family moved due to financial difficulty (1,0), received financial aid (1,0), lived in a rural area (1,0), and spoke English at home (1,0).[15] For own life expectancy, we use as NDS experiences recent stock and housing market outcomes.

Cues. The cue in each domain X for respondent i at time t is the most recent DS experience in X (the most recent stock return, home price index return, and change in current health).

Similarity. In each domain X we compute the dissimilarity between respondent's i cue at time t and her above average DS experiences by computing $\text{cov}[s, (s - s_{it})^2]$ in the respondent's database up to period t. We measure the similarity between NDS experiences and the target by fielding a separate survey described below.

[14] See Appendix Table B.2, Panels 3 and 4, for construction of the health indices, and Appendix Table B.3, Panel C, for construction of the health database.

[15] See Appendix Table B.2, Panel 7, for construction of the childhood socioeconomic status index, which we use as a non-domain-specific experience.

4.1 Domain Specific Experiences

We estimate Equation 6 for stock prices, own home prices, and own life expectancy. DS experiences vary across these outcomes but the memory structure in Equation 6 is the same. The estimated probability $p_{X,it}$ in domain $X = SP, HP, LE$ by respondent i at time t satisfies:

$$p_{X,it} \approx \gamma_0 + \gamma_1 \cdot \overline{s}_{it} + \gamma_2 \cdot \text{RP}_i \cdot \overline{s}_{it} - \gamma_3 \cdot \text{RP}_i \cdot cov\left[s, (s - s_{it})^2\right], \tag{8}$$

where all coefficients should be positive. We compute proxies for $\overline{s}_{it}$ using the database: for the stock market it is the average yearly stock return, for home prices it is the average home price growth, and for life expectancy it is the average change in health, all lived by the respondent.[16]

Table 1 reports, for each domain, the estimates of Equation 8 in columns 1, 3, and 5. In columns 2, 4, and 6 we control for RP to rule out that it may directly affect the level of beliefs. The results support Predictions 1 and 2: in each domain, beliefs increase in "positive past DS", and more so for high RP people. Cues also matter, as indicated by the negative coefficient on the covariance term: news similar to a person's above-average experiences causes stronger optimism for high RP people. Similarity shapes beliefs through RP, consistent with less interference in the memory technology. The results are robust: Table D.2 adds survey-year fixed effects, ruling out common aggregate shocks as the driver.[17] Figure D.4 reports separate estimations in different RP quartiles, so results hold non-parametrically.

The results also show that RP is distinct from IQ. Higher IQ prompts optimism about stocks and housing, perhaps due to better information, but unlike RP it prompts pessimism about life expectancy. This suggests that higher RP need not entail normatively better beliefs: it may lead to excess extrapolation, including from improvements in personal health. We return to this issue when testing Prediction 3. As a robustness exercise, we also study IQ as a mediator of objective news by interacting it with the current state s_{it}. Our results are robust: the RP covariance term $-\gamma_3$ remains negative and significant (see Table D.1), confirming that RP is an independent driver of a person's reaction to changing conditions.

[16] The results are if anything stronger if we use the level (rather than change) in health to measure experiences.

[17] Results also carry through when adding also respondent fixed effects, but in some cases the average DS experience $\overline{s}_{it}$ loses significance because it is absorbed by respondent fixed effect. The role of cues in driving beliefs is confirmed in all specifications.

Table 1: Database Regressions for $\overline{s_{it}}$ and $cov(s_k, (s_k - s_{it})^2)$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live10+**	
Cognition						
RP	.	0.002	.	-0.013	.	0.017***
	.	[0.005]	.	[0.010]	.	[0.001]
IQ	0.024***	0.024***	0.015***	0.016***	-0.013***	-0.019***
	[0.001]	[0.001]	[0.003]	[0.003]	[0.002]	[0.002]
DS experiences						
$\overline{s_{it}}$ (γ_1)	0.000	0.001	0.029***	0.022***	0.027***	0.028***
	[0.001]	[0.003]	[0.003]	[0.006]	[0.002]	[0.002]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.005***	0.008***	0.007***	0.006***
	[0.000]	[0.001]	[0.001]	[0.003]	[0.001]	[0.001]
$RP * cov(s, (s - s_{it})^2)$ $(-\gamma_3)$	-0.004***	-0.004***	-0.031***	-0.031***	-0.001***	-0.001***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.000]	[0.000]
N	139507	139507	49940	49940	137602	137602
AR2	0.04	0.04	0.05	0.05	0.10	0.11
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

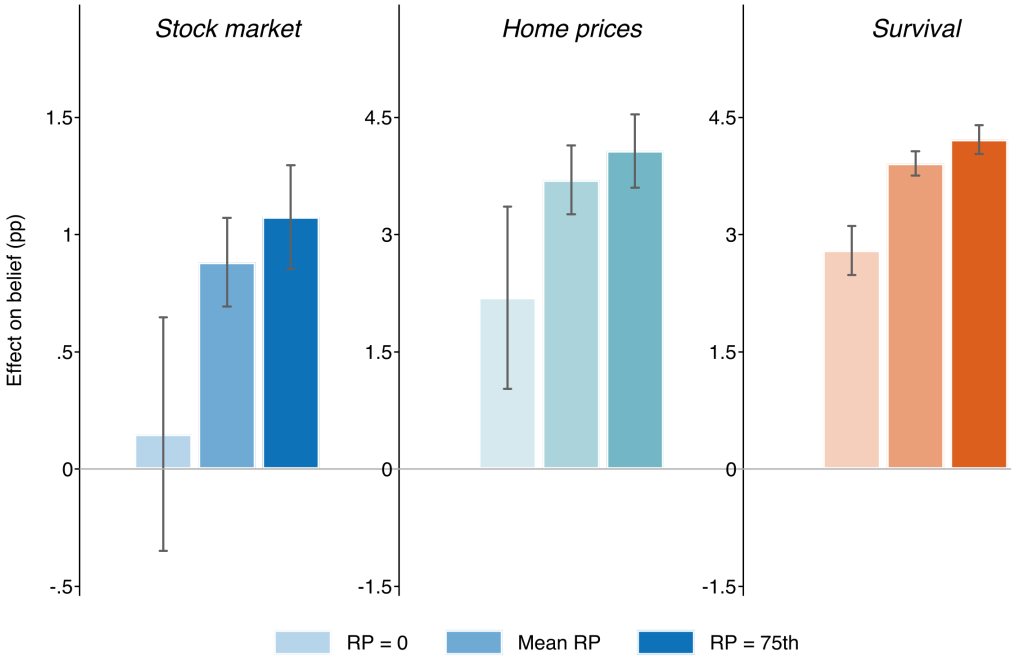
Notes: Each column reports a panel regression of subjective beliefs on Recall Performance (RP), IQ, domain-specific (DS) experience variables ($\overline{s_{it}}$, $\text{cov}(s_k, (s_k - s_{it})^2)$), and their interactions with RP as shown in Equation 8. Demographic controls include age, gender, marital status, retirement status, income, and education. Columns 1–2 use stock market beliefs; columns 3–4 home-price beliefs; columns 5–6 health beliefs. Columns 2, 4, 6 additionally include the RP base effect. RP, IQ, and the experience variables are divided by their sample IQR (the interquartile range in each column's estimation sample) before estimation. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

To quantify magnitudes, Figure 5 reports, for each domain, how strongly DS experiences shape beliefs at three levels of recall performance: zero, the average, and the 75th percentile. The figure isolates the effect of the database, i.e. the average DS experience $\overline{s}_{it}$, not the effect of cues captured by the covariance term. Bars show the effect on beliefs in percentage points.[18]

In every domain, the effect of DS experiences rises with RP, as the model predicts. For

[18]These are marginal effects of $\overline{s}_{it}$ from columns 2, 4, and 6 of Table 1. All continuous regressors ($\overline{s}_{it}$, RP, IQ) are divided by their interquartile range prior to estimation, yielding the response of beliefs to an interquartile increase in past experiences. The experience effect at recall performance level RP is $\hat{\gamma}_1 + \hat{\gamma}_2 \cdot \text{RP}$, where $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are the coefficients on $\overline{s}_{it}$ and RP $\times$ $\overline{s}_{it}$; we evaluate it at RP=0, at its mean, and at its 75th percentile.

Figure 5: Economic Magnitude of Experiences Across Belief Domains, by Recall Performance Level.



Notes: Each bar is the marginal effect on beliefs of moving past experiences from their 25th to their 75th percentile, expressed in percentage points, evaluated at three levels of recall performance: zero (no recall performance), the sample mean, and the 75th percentile. Each cluster is read off its own vertical axis: stock market on the far left, home prices in the middle, and survival on the right. The three axes share a common zero but are scaled separately. All continuous regressors are standardized by their sample interquartile range before estimation. Error bars are 95% confidence intervals. Specifications and samples match columns 2, 4, and 6 of Table 1.

the stock market, experiences barely move the subjective probability that stocks go up at zero RP (0.1 pp, 0.3% of its mean), but shift it by 0.9 pp (1.9%) at average RP and by 1.1 pp (2.3%) at the 75th RP percentile. The same gradient obtains for home prices, from 2.2 pp at zero RP (4.2% of the mean) to 3.7 pp (7.0%) at the average RP and 4.1 pp (7.7%) at the 75th percentile; as well as for life expectancy, from 2.8 pp (5.9%) to 3.9 (8.2%) to 4.2 (8.9%). For home prices and life expectancy, the direct experience effect is larger, arguably because respondents have more information about, and attend more to, own home and health conditions. These effects are comparable to those of demographics. For the stock market, an interquartile shift in RP raises the subjective probability that stocks will go up by approximately 2.1 pp, equivalent to the effect of an interquartile shift in education, five times the effect of an interquartile shift in household income, and about 40% of the effect of gender. The same ordering carries

through to home prices and life expectancy (see Figure D.1–Figure D.3). We obtain comparable magnitudes when estimating our regressions in different RP quartiles.

In sum, RP amplifies the sensitivity of beliefs to experiences across all domains, most dramatically for stocks (at the 75th percentile of RP it is roughly seven times its value at zero RP), but also in home prices and life expectancy (with 85% and 51% increases, respectively).

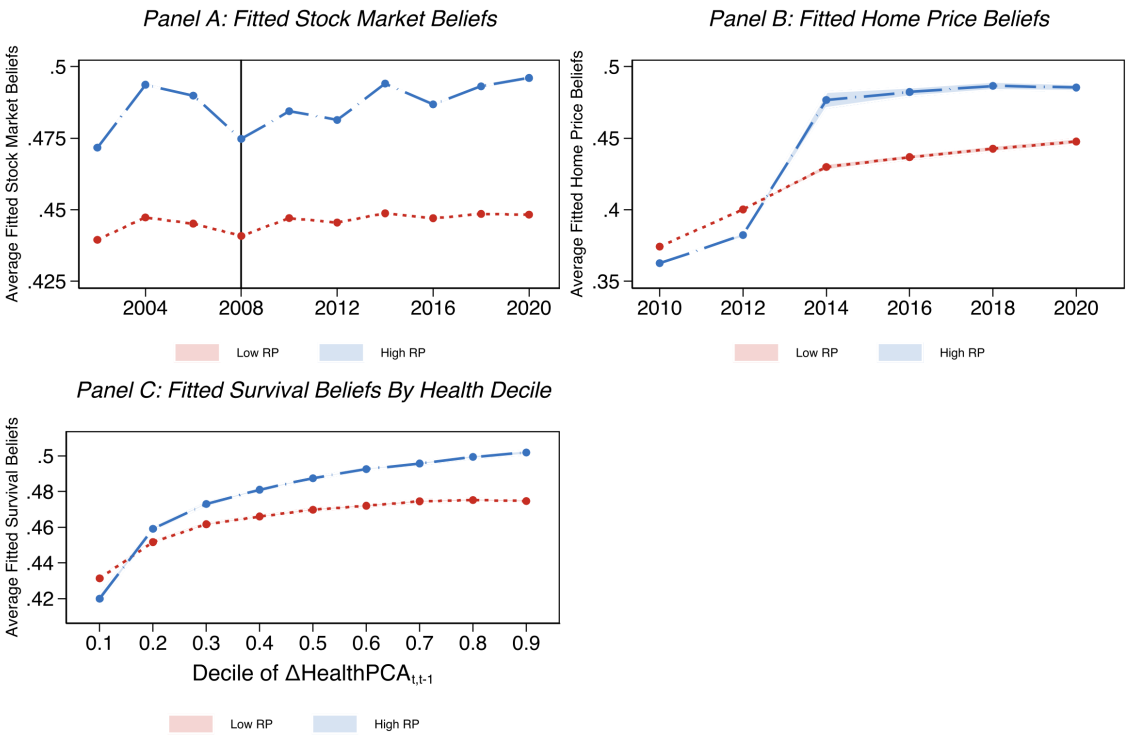
We next go back to our motivating facts. Figure 6 reports fitted beliefs from Table 1 for the top and bottom RP quartile. The figure isolates the contribution of recall performance, experiences, and cues, by holding demographics and IQ at their sample means. The model captures the sharper boom-bust in beliefs about stock market returns of higher RP people, their initial relative pessimism about home price growth followed by a steeper increase after the 2008 crisis, and the steeper gradient of survival expectations based on changes in current health. Overall, the fitted beliefs of higher RP people are more optimistic, due to the good DS experiences of the average HRS respondent. In sum, by increasing the sensitivity to DS experiences, higher RP accounts for our motivating facts.

Does RP also affect the accuracy of beliefs? According to Prediction 3, the beliefs of high RP people could be more or less accurate or volatile compared to those of low RP people, depending on their databases and cues. We saw in Table 1 that higher RP is associated with greater optimism about own survival, while higher IQ does the opposite, pointing to lower accuracy if higher IQ indeed entails better information processing. We now directly assess whether RP causes distortions via overreaction to DS experiences and cues. To do so, we construct respondent i's forecast error at $t+1$ in domain X as the probability estimate at t minus the indicator of the target event actually happening at $t+1$:

$$FE_{i,t+1} = p(\mathrm{T}_{X,t+1})_{it} - \mathbf{1}\{\mathrm{T}_{X,t+1}\}$$

For $X = SP$ the event is whether the S&P500 is higher after 12 months; for $X = HP$ it is whether the regional house price index increased after 12 months; for $X = LE$ it is whether the person is alive after 10 years, measured from the HRS Cross-Wave Tracker. A predictably positive (negative) forecast error indicates excess optimism (pessimism).

Figure 6: Fitted Beliefs Over Time for High vs. Low Recall Performance.



Notes: Fitted beliefs from the regressions in Table 1, with demographics and the IQ measure held at their sample means so the figure isolates the contribution of recall performance, experiences, and cues. Each panel shows fitted beliefs for the bottom (red) and top (blue) quartile of Recall Performance. Panel A shows fitted stock market beliefs over time from a regression of Equation 8 (column 1 of Table 1). Panel B shows fitted home-price beliefs over time using HPI moments (column 3 of Table 1). Panel C shows fitted survival beliefs by decile of $\Delta\text{HealthPCA}_{t,t-1}$ (column 5 of Table 1). Shaded bands are 95% confidence intervals. The full-regression version, which uses each respondent's actual covariates, is in Figure D.5.

To study how RP shapes the sensitivity of forecast errors to DS experiences and cues, we predict forecast errors from the regressors in Equation 8, the interactions of RP with experiences and cues, while controlling for characteristics such as income, education, or IQ, which capture other drivers of beliefs. We also include year and respondent fixed effects, capturing common shocks and baseline optimism or pessimism unrelated to experiences and RP. Regression results are reported in Table D.4. Across stocks, home prices and life expectancy, two patterns emerge. First, higher average experiences $\bar{s}_{it}$ predicts higher forecast errors (more optimism than baseline), similarly across high and low RP people. Second, high RP people react more sharply to time varying cues than low RP ones, exhibiting much more optimism

after higher current stock returns, home price growth, or changes in health. What do these effects imply for the average bias of beliefs? And what do they imply for their excess volatility?

To answer the first question, Figure 7 shows the average predicted forecast errors as a function of RP, keeping experiences and all other regressors at their mean. Panel A shows that the average respondent in our sample exhibits substantial pessimism about the stock market, but higher RP reduces this pessimism: the predicted (absolute) forecast error decreases by 1.9 pp from the bottom to the top RP quartile ($t \approx 4.8$). The mechanism goes as follows: experienced stock market returns are on average good in our sample, but interference from other experiences causes forgetting and return underestimation. High RP reduces this interference by increasing the sensitivity to stock market experiences, promoting accuracy.

Panel B shows that home price growth beliefs are also pessimistic on average. Higher RP does not significantly improve accuracy (a 0.3 pp reduction in pessimism from the bottom to the top RP quartile, $t \approx 0.3$). In line with the model, the beliefs of high RP people exhibit a sharper drop during the housing bust and a sharper boom during the recovery compared to low RP ones, creating partially offsetting effects on average forecast errors. This points to the link between RP and excess volatility, to which we return later.

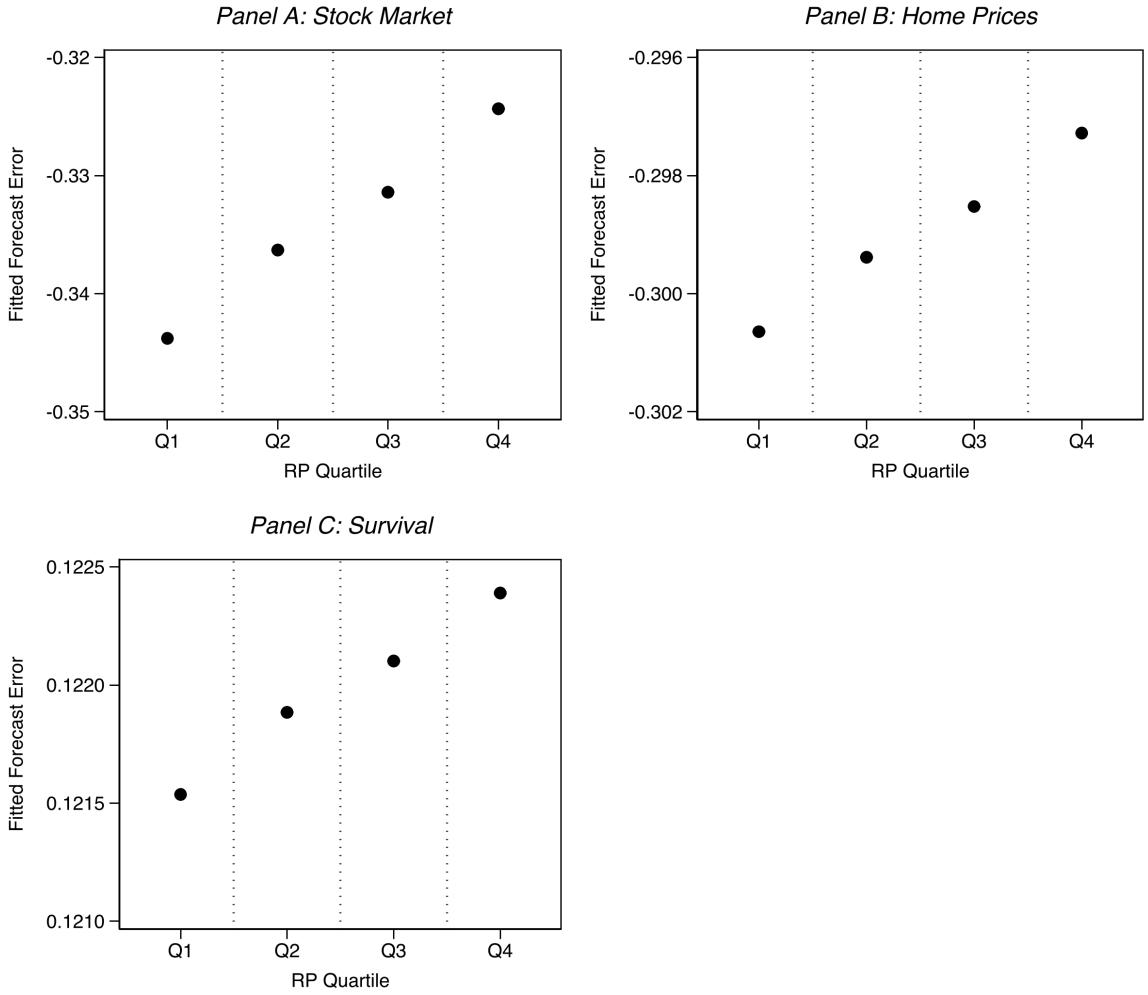
Finally, Panel C shows that survival probability is generally overestimated, and increases in RP (a 0.1 pp increase in optimism, not statistically significant, $t \approx 1.1$). In the model, a positive slope arises because higher RP causes overreaction to good health experiences.[19] IQ tempers this overreaction, moderating the estimated survival probability.

We next consider the second dimension of accuracy, namely excessive volatility based on cues. Figure 8 reports the RP gradient of fitted forecast errors for the top and bottom quartile of similarity between the cue and the respondent's above average DS experiences (that is, of the term $-cov\left[s,(s-s_{it})^2\right]$).

Panels A and C show that high RP people exhibit more volatile beliefs based on cues.

[19] Arguably, good health experiences face less interference than stock or housing market ones because they are salient in respondents' mind, thus they are less likely to be underestimated also by low RP people, causing the systematic optimism bias in Panel C. Our measure of health is concentrated in respondents' older years and likely underestimates respondents' actual database. If we allow for a direct effect of RP on beliefs, which given the other controls may further capture health conditions, then excess optimism about survival increases substantially and significantly with RP (a 2pp increase, $t \approx 2.8$, see Figure D.7), while results on stock and housing market forecast errors remain unchanged.

Figure 7: Fitted Forecast Errors by Recall Performance.

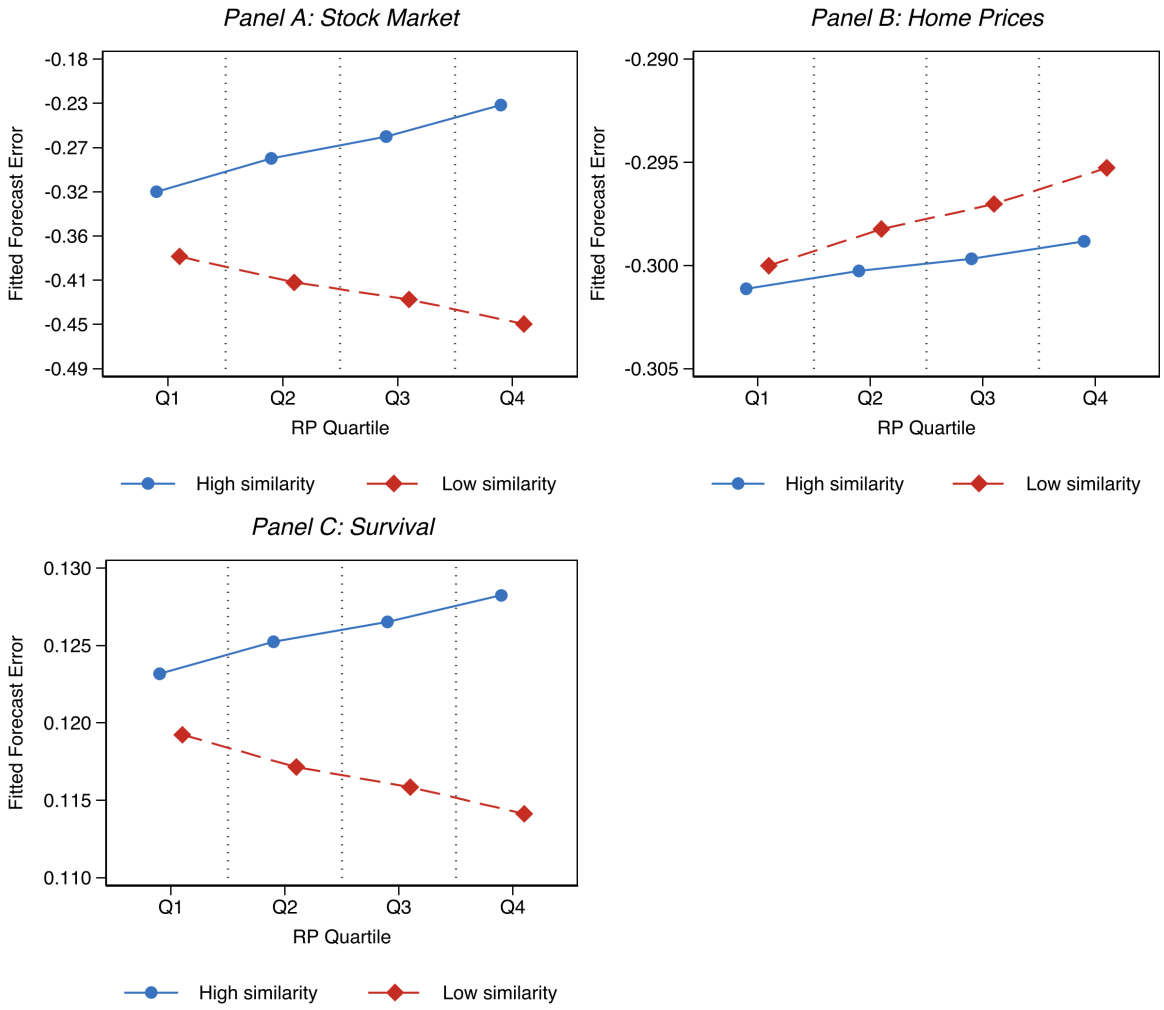


Notes: Fitted forecast errors (stated probability minus an indicator for the realized event) by quartile of Recall Performance (RP), from the regressions in Table D.4, columns 1, 3 and 5. Predictions use each respondent's own RP and sample means for all other regressors and the fixed effects; points are quartile means of the predictions. Panel A: stocks up over the next twelve months (S&P 500; N = 136,293). Panel B: home value up over the next twelve months (regional house-price index, 2010 onward; N = 47,772). Panel C: survival to the target age (HRS tracker; N = 39,422). Bars: 95% confidence intervals from the standard error of the quartile mean.

Consider beliefs about the stock market: a fall in the stock market has low similarity to above average experiences. This deepens stock market pessimism particularly among high RP people, leading to a sharp increase in their pessimism bias. Conversely, a rise in the stock market helps retrieve positive returns and leads to more optimistic beliefs, sharply reducing the pessimism bias of higher RP people.

For survival beliefs, Panel C, cue-driven volatility of high RP people leads to a sharp

Figure 8: Fitted Forecast Errors by Recall Performance for High vs. Low Similarity of the Cue to Above Average Experiences.



Notes: Fitted forecast errors from the regressions of Table D.4 (respondent and survey-year fixed effects; age and cohort fixed effects for survival), by RP quartile, for the top and bottom quartile of cue similarity. Similarity is the covariance term of Equation 8 with the sign reversed: high similarity (blue) means the current cue resembles the respondent's above-average experiences, low similarity (red) means it does not. Each respondent-wave's forecast error is predicted at its own RP with the cue term at the quartile mean of similarity and experiences, IQ, demographics and fixed effects held at their sample means; each point is the mean prediction within the RP quartile. Panel A: stock market (N = 136,293). Panel B: home prices (N = 47,772). Panel C: survival (N = 39,422). Vertical bars are 95% confidence intervals based on the standard error of the quartile mean of the predictions.

increase in their (optimism) bias when their current health improves. This significantly reduces the accuracy of high RP people. Conversely, their optimism significantly declines, and their accuracy improves, when current health declines. The magnitudes are large, and can overturn the average difference in beliefs across RP groups. This means that higher RP people can become systematically *more biased* than lower RP people: they become even more pessimistic

about stock returns when the market falls, and even more optimistic about survival following good health shocks. Beliefs about home price growth exhibit less volatility, perhaps because our measure of home price cues is more indirect.

In sum, high RP beliefs are more sensitive to DS experiences and cues, but not necessarily more accurate. We next turn to the role of NDS experiences.[20]

4.2 Non Domain Specific Experiences

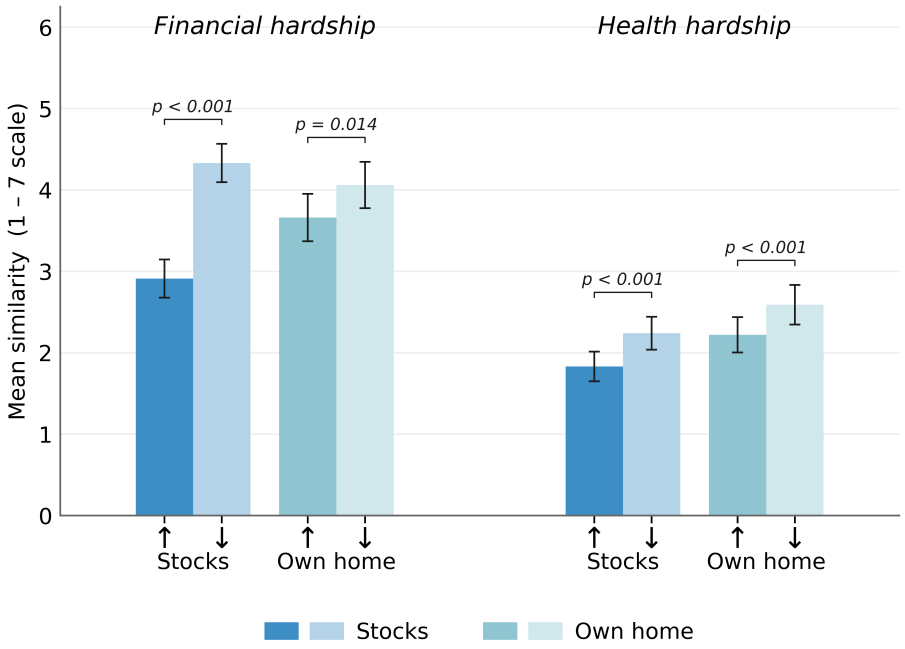
To test Prediction 4, we use as NDS experiences for stock and housing markets the participants' childhood health and socioeconomic status, denoted $ChHea$ and $ChSES$ respectively. These experiences contain no information about aggregate stock returns, and are irrelevant in statistical models, whether rational or behavioral (including experience based learning, Malmendier and Nagel (2011, 2016)). With memory, such experiences could matter by facilitating the imagination of stock return outcomes that are similar to them (Bordalo et al., 2026a; Taubinsky et al., 2026). Building on Bordalo et al. (2026a), to assess whether positive childhood financial and health experiences are more similar to stock (and own home price) increases or decreases, we elicit similarity judgments of health and financial hardships to both outcomes.[21]

Figure 9 shows that financial and health adversities are more similar to a decrease than to an increase in stock and own home prices. Prediction 4 then implies that better childhood health and SES should prompt higher RP people to exhibit more optimism for stock and own home prices. Figure 9 also shows that financial adversities are more similar than health adversities to both stock and own home price declines (both in levels and relative to increases), so $ChSES$ should have a stronger impact on beliefs, particularly for high RP people. Conversely, the figure also implies that higher stock and housing returns, which are NDS experiences from the perspective of life expectancy, should promote optimism about it.

[20] Appendix Table D.8 confirms the higher sensitivity of high RP beliefs for two aggregate outcomes for which we have limited time series dimension: the probability of a major economic depression within the next ten years (four waves, 2002-2008), and the probability that inflation will exceed five percent over the next ten years (two waves, 2006-2008). The notes to that table describe the analysis in detail.

[21] The sample comprised 400 US home owners in Prolific. Descriptive statistics are in Table D.7. The survey also assessed the perceived similarity of financial and health adversities to an *aggregate* increase in the price of homes. Contrary to the case for own home prices, home owners perceive increases of aggregate home prices as similar to adversities, particularly to financial ones. This helps reconcile our finding that positive NDS experiences drive optimism about own home prices with the finding in Bordalo et al. (2026a) that NDS adversities increases estimated growth of aggregate home prices.

Figure 9: Perceived Similarity Between Personal Hardships and Asset Outcomes.



Notes: Respondents rate, on a 1–7 scale, how similar a given personal hardship is to a state of the world in which each asset performs well or poorly. The left group rates a financial hardship, the right group a health hardship. Within each group, bars give mean similarity to stock and own home outcomes, for the asset rising (↑) and falling (↓). Darker bars refer to stock-market outcomes and lighter bars to own-home outcomes. Error bars are 95% confidence intervals.

For the stock market, childhood conditions are clearly an NDS experience, so they should play no role in a rational model. For own home prices, one cannot fully rule out a role of information: childhood experiences may be correlated with the respondent's current socioeconomic outcome and hence with the price path of her home. This concern is assuaged by our controls, which include a respondent's income and region. In addition, the key prediction of memory is that NDS experiences should matter through RP, not mechanically through the home price experiences the person faced in the past. The absence of a direct experience effect would further assuage omitted variables concerns. By the same logic, our controls should allay endogeneity concerns about NDS experiences and life expectancy beliefs.

We take the specification of Equation 8, and add to it two NDS regressors, yielding:

$$p_{X,it} \approx \gamma_0 + \gamma_1 \cdot \overline{s}_{it} + \gamma_2 \cdot \mathrm{RP}_i \cdot \overline{s}_{it} - \gamma_3 \cdot \mathrm{RP}_i \cdot cov\left[s, (s - s_{it})^2\right] + \delta_1 \cdot nds_{it} + \delta_2 \cdot \mathrm{RP}_i \cdot nds_{it} + \varepsilon_{it}, \tag{9}$$

where δ_1 captures the direct NDS effect at the bottom percentile of RP, and δ_2 captures memory's amplification. Since all regressors are IQR-normalised, each coefficient quantifies the belief change for a 25th-to-75th percentile shift in the variable.[22] According to Prediction 4, NDS experiences should matter more for high RP individuals, $\delta_2 > 0$. The usual DS effects, $\gamma_1, \gamma_2, \gamma_3 > 0$ should also continue to hold.

Table 2 reports estimates of Equation 9. NDS events matter in all domains, but almost exclusively at high RP. The base NDS coefficient $\hat{\delta}_1$ is small and statistically indistinguishable from zero (only marginally significant in column 5), while the RP-amplification term $\hat{\delta}_2$ is positive and significant throughout.

These results are consistent with the model: NDS experiences generally face strong interference, so they are used in simulation only by high RP people.[23] Compared to Table 1, the coefficients on DS experiences and their interactions with RP are very stable, consistent with our regressors capturing orthogonal memory channels rather than unobserved individual heterogeneity in information or tastes.

Effects are quantitatively large. Evaluated at mean RP, a one-IQR shift in NDS experiences shifts stock-market beliefs by about 0.5 pp (column 2) and survival beliefs by 1.0 pp (column 5).[24] The corresponding DS effect, from the same regression evaluated at mean RP, is 0.9 pp for stocks and 3.9 pp for survival, so the NDS channel corresponds to roughly a quarter

[22] For RP, after dividing by the interquartile range (which places most of the distribution between roughly -2 and $+2$), we add 2, so that RP $= 0$ corresponds to the bottom of the RP distribution—a respondent with no recalled experiences—rather than to the average respondent.

[23] These results are robust to absorbing common time trends via survey-year fixed effects, as well as to including the RP base effect (Table D.5). They are also quantitatively similar when the childhood-health and childhood-SES databases are replaced with simpler, one-dimensional indices (Table D.6).

[24] The marginal effect of NDS experiences at mean RP is $\hat{\delta}_1 + \hat{\delta}_2 \overline{\mathrm{RP}}^*$, where $\hat{\delta}_1$ and $\hat{\delta}_2$ denote the estimated coefficients on NDS_{it} and $\mathrm{RP} \times \mathrm{NDS}_{it}$ respectively, and $\overline{\mathrm{RP}}^* \approx 1.76$ is the post-standardisation sample mean of recall performance, corresponding to an average respondent with raw recall performance index of zero. Analogously, the marginal effect of DS experiences at mean RP is $\hat{\gamma}_1 + \hat{\gamma}_2 \overline{\mathrm{RP}}^*$.

Table 2: NDS Database Regressions for $\overline{s_{it}}$ and $cov(s_k, (s_k - s_{it})^2)$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live 10+**	
IQ	0.024***	0.023***	0.016**	0.015**	-0.019***	-0.022***
	[0.001]	[0.001]	[0.005]	[0.005]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	0.001	0.001	0.029***	0.029***	0.026***	0.029***
	[0.001]	[0.001]	[0.008]	[0.008]	[0.002]	[0.002]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.005***	0.004***	0.007***	0.006***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.001]	[0.001]
$RP * cov(s_k, (s_k - s_{it})^2)$ $(-\gamma_3)$	-0.004***	-0.004***	-0.031***	-0.031***	-0.001***	-0.001***
	[0.000]	[0.000]	[0.007]	[0.007]	[0.000]	[0.000]
NDS Experiences						
nds (δ_1)	-0.000	-0.003	-0.003	-0.007	-0.006*	0.001
	[0.001]	[0.002]	[0.003]	[0.007]	[0.003]	[0.001]
$RP * nds$ (δ_2)	0.001**	0.005***	0.002*	0.006*	0.009***	0.005***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.001]	[0.000]
N	134298	139504	49561	49939	137602	118162
AR2	0.04	0.04	0.05	0.05	0.11	0.11
NDS Var	ChHea	ChSES	ChHea	ChSES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Each column reports a panel regression of subjective beliefs on Recall Performance (RP), IQ, domain-specific (DS) experience variables ($\overline{s_{it}}$ and $\text{cov}(s_k, (s_k - s_{it})^2)$), non-domain-specific (NDS) experiences, and their interactions with RP. NDS variables differ by domain: childhood health and SES index for stocks and home prices; cumulative lifetime log equity and home-price returns for health. All continuous regressors (RP, IQ, $\overline{s_{it}}$, cov, NDS) are divided by their within-sample IQR prior to estimation, so each coefficient represents the change in beliefs associated with a move from the 25th to the 75th percentile of the regressor. Demographic controls include age, gender, marital status, retirement status, income, and education. Standard errors in columns 3 and 4 are clustered by census region; other columns report conventional standard errors. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

to a half of the DS channel in these domains. For home prices, the NDS channel is weaker: a one-IQR shift in childhood SES raises the subjective probability of home price appreciation by only 0.3 pp (column 4), and childhood health is negligible at 0.1 pp (column 3)—each less than one-tenth of the corresponding DS experience effect of 3.7 pp. One explanation is that, relative to the stock market, people have a larger database of home price experiences, which reduces

the effect of NDS experiences. Appendix Figure D.8 illustrates these magnitudes in absolute terms (percentage points of the dependent variable) and further decomposes the heterogeneity generated by NDS databases at different levels of RP.

Overall, high-RP people are not simply better informed or more responsive to information. They use memory more, including of events that are not informative about the domain in question, which is further evidence that high RP need not increase accuracy.

4.3 Agnosticism from Memory

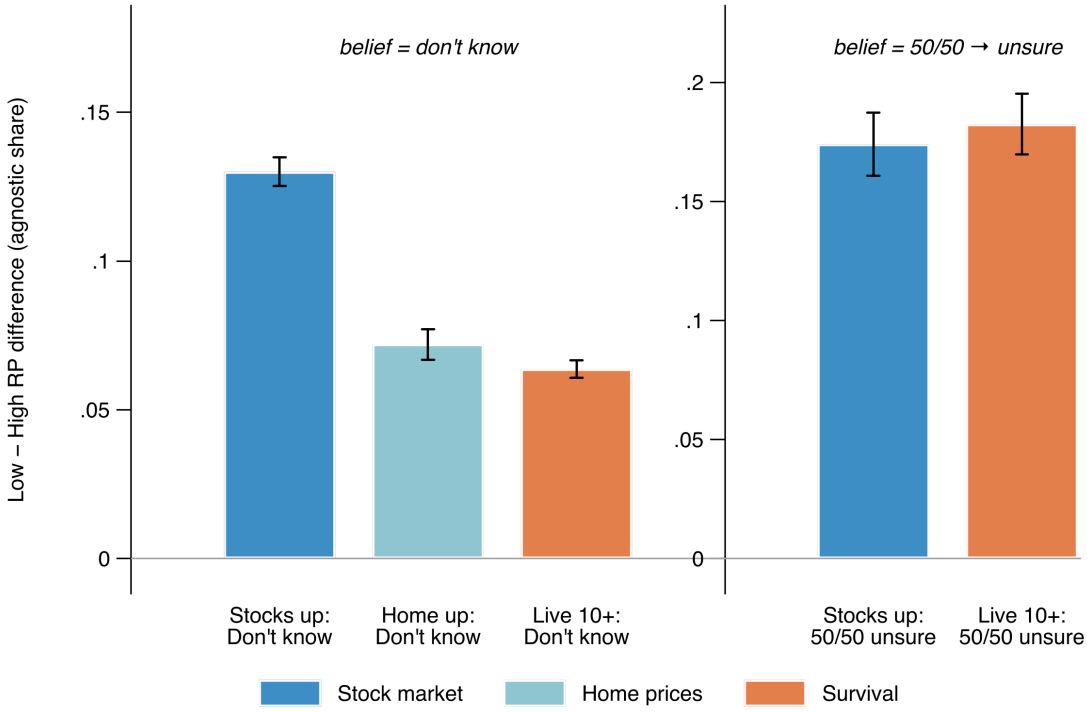
Prediction 5 says that high RP people should recall more material to simulate the future. They should then be less likely to give agnostic answers compared to low RP people, across all domains. This prediction rests on the centrality of memory for imagining the future, which is well established in cognitive science (Hassabis et al., 2007; Schacter et al., 2007).

We measure agnostic answers by the: i) share of people answering with "I do not know", and ii) share of people answering with "50-50" *and* stating that they are highly uncertain. In both cases, respondents are unsure how to think about the future.

Figure 10 shows that Prediction 5 holds strongly in the data: people in the bottom RP quartile are more likely to give agnostic answers than people in the top RP quartile. The gaps are large. Relative to low-RP respondents, high-RP respondents are about 13pp less likely to answer "I do not know" about the stock market, 7pp about their own home, and 6pp about their own survival. Among those who answer 50-50, they are about 17 and 18 pp less likely to report being unsure for stocks and survival, respectively.

The link between RP and agnosticity suggests that answers like "I do not know" may not reflect lack of information but rather cognitive bounds on retrieval. Related answers such as "50-50 and low confidence" have been traced to noise in signal perception, which causes people to stick to their prior (Enke and Graeber, 2023; Woodford, 2020). Our model suggests a different mechanism: the signal is *internally generated* through retrieval and simulation, and "noise" means that the signal fails to deliver a vivid image of the future. The DM then sticks to a belief which – unlike a Bayesian prior – does not reflect previous data. It is an agnostic

Figure 10: Agnosticism and Recall Performance: Low–High RP Difference.



Notes: Each bar is the difference between the bottom and top recall-performance quartile in the share of respondents giving an agnostic answer. For each domain-question we restrict the sample to respondents in the bottom and top recall-performance quartiles and estimate $\Pr(\text{agnostic}_i = 1) = \alpha + \beta\,\text{LowRF}_i + \varepsilon_i$, where LowRF_i equals one for the bottom quartile. The plotted bar is the low-minus-high recall-performance difference $\hat{\beta}$. Each cluster is read off its own vertical axis. The left cluster reports the difference in the share answering "Don't Know" for stock, home-price, and survival beliefs (whole sample). The right cluster reports the difference in the share who, having first answered 50/50, further state they are "unsure" rather than "equally likely", which is computed only among respondents who answer 50/50, and only for stocks and survival, since the home-price question has no 50/50 follow-up. Error bars are 95% confidence intervals.

default because nothing concrete comes to mind.[25,26] Likewise, high confidence people are not necessarily more precise and hence accurate in their beliefs. They may overweight statistically uninformative events such as NDS experiences.

[25]This mechanism could also explain why low RP answers are more dispersed and take extreme values, as shown in Appendix Figure C.9, Figure C.10, and Figure C.11. The agnostic default may in fact vary with arbitrary anchors such as the response scale etc.

[26]One potential concern is that low RP people may just be less willing to exert effort, both in laboratory recall tasks, where they recall fewer words, and in forecast elicitations, where they give agnostic low effort answers. This account does not however explain why high RP people should rely on their own DS and NDS experiences in the way shown by our previous tests, and also why this differential sensitivity should predict real world choices.

5 Recall Performance, Expectations and Choices

Does RP predict economic choices? To address this question, we leverage the HRS data on respondents' choices related to our three belief domains.

Stock Market. Two choice variables relate to beliefs about the stock market: *New Stock Market Investments 2y*, a dummy on whether the respondent made new stock market investments in the last 2 years, and *% IRA in Stocks*, the proportion of the respondent's IRA invested in stocks/mutual funds (restricted to people owning an IRA). A higher believed probability that stocks will go up should raise both measures.

Housing. The available variable is *Bought New Home 2y*, a dummy taking the value of one if the respondent bought a new home in the past two years. A higher believed probability of the respondent's own home increasing in value should promote this choice.

Life Expectancy. Here we observe three choices: *Retired*, a dummy equal to one if the respondent is retired; *% Income to Pension*, the percentage of monthly pay the respondent contributes to their pension (restricted to working people with a defined contribution plan); and *Work Past 70*, a variable from 0 to 100 indicating the respondent's expectation to work after age 70 (higher value means higher probability), restricted to working people younger than 70. Higher estimated survival probability should predict higher believed probability of working after 70, a lower probability of being retired, and a higher contribution to pensions.

Appendix Table D.9 shows that measured beliefs predict choices as expected. This is reassuring: beliefs are informative, consistent with Bailey et al. (2019); D'Acunto et al. (2023); Giglio et al. (2021); Malmendier and Nagel (2011, 2016).

Our analysis however raises a deeper question: does RP shape choices by influencing the sensitivity of beliefs to experiences and cues? Conventional economics traces choices to tastes and information. Tracing choices to memory says instead that cognition is an important source of behavior. To assess this hypothesis, we first consider a respondent's predicted beliefs $\widehat{B}_{it}$ using Equation 9, and then decompose it into three components:

$$\widehat{B}_{it} = B_{E_{it}} + B_{(\mathrm{RP}_{it} \cdot E_{it})} + B_{D_{it}}. \tag{10}$$

The first term, $B_{E_{it}}$, is a pure *experience* component, including both DS and NDS experiences, $B_{E_{it}} = \widehat{\gamma}_1 \cdot \overline{s}_{it} + \widehat{\delta}_{1,hea} \cdot ChHea + \widehat{\delta}_{1,ses} \cdot ChSES$. The second and key term captures the contribution of RP to the sensitivity of beliefs to past experiences and to their similarity-driven retrieval from current cues. It is equal to $B_{(\mathrm{RP}_{it} \cdot E_{it})} = \widehat{\gamma}_2 \cdot \mathrm{RP}_i \cdot \overline{s}_{it} + \widehat{\delta}_{2,hea} \cdot \mathrm{RP}_i \cdot ChHea + \widehat{\delta}_{2,ses} \cdot \mathrm{RP}_i \cdot ChSES - \widehat{\gamma}_3 \cdot \mathrm{RP}_i \cdot \mathrm{cov}\left[s, (s - s_{it})^2\right]$. The sum is the total contribution from memory amplification, with higher values of RP or experiences corresponding to higher predicted beliefs. The last term, $B_{D_{it}}$, is a pure *demographic*+IQ component. To link choices to RP, we regress choices on these three beliefs components. Importantly, these components are measured in the respondent's previous wave, because the choices were made before the current wave. We add our standard controls, which may affect choices independently.

Table 3 reports, for each choice, the predicted beliefs in the first column and their decomposition in the second (for parsimony, we do not report demographics). Here we also include self-reported risk aversion to benchmark the explanatory power of cognition to a standard measure of preferences.[27] Across the three domains, higher predicted beliefs affect the relevant choice probability as expected (columns 1, 3, and 5). Decomposing fitted beliefs into their three components (columns 2, 4, and 6), the memory component $B_{(\mathrm{RP}_{it} \cdot E_{it})}$ enters significantly and with the expected sign for every outcome. Crucially, the memory component is the dominant belief-driven channel in all choices except for the probability of working past 70 and retirement, where the pure-experience component $B_{E_{it}}$ is slightly larger (this component has a heterogeneous, and sometimes negative, role across domains).

Stock Market. A one-standard-deviation (SD) increase in fitted beliefs is associated with a 5.2 p.p. rise in the probability of making a new stock investment (52% of the 10% mean), and with a 6.2 p.p. rise in the share of IRA assets held in stocks (11.1% of the 56% mean). Decomposing fitted beliefs, a one-SD increase in the memory component raises new investment by 1.2 p.p. (12% of the mean) and the IRA share by 1.7 p.p. The pure-experience

[27] To measure risk aversion, we take the standardized average of: (a) risk taking for employment (choose between a job with higher expected pay but more variance vs. one with lower pay and more certainty); (b) risk taking for business inheritance (allocate an inheritance into a safe asset or invest in a risky business); and (c) self-reported risk taking, measured on a scale of 0 to 10. Respondents missing a proxy are averaged over the proxies they do have.

Table 3: Choice Regressions: Beliefs, Memory, and Economic Choices.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Stock/Home-Price Market Choices					
	New Stock Market Inv.		**% IRA in Stocks**		**Bought a New Home**	
IQ (1 sd)	0.017***	0.014***	0.038***	0.036***	0.007***	0.007***
	[0.001]	[0.001]	[0.003]	[0.003]	[0.001]	[0.001]
Lagged Beliefs *(ϕ)*						
$\widehat{B}_{i,t-2}$ (θ)	0.052***	.	0.062***	.	0.006***	.
	[0.002]	.	[0.005]	.	[0.001]	.
$B_{E_{i,t-2}}$ (γ)	.	-0.011***	.	-0.013***	.	0.001
	.	[0.001]	.	[0.003]	.	[0.001]
$B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$ (μ)	.	0.012***	.	0.017***	.	0.005***
	.	[0.001]	.	[0.002]	.	[0.001]
Avg DV	0.10	0.10	0.56	0.56	0.03	0.03
N	116454	112870	36143	35332	41785	41407
Demographic Controls	Y	Y	Y	Y	Y	Y
Risk-taking	N	Y	N	Y	N	Y
Beliefs Lagged One Wave	Y	Y	Y	Y	Y	Y
	Panel B: Health Choices					
	Work Past 70		**Retired**		**% of Income to Pension**	
IQ (1 sd)	0.014***	0.009***	-0.010***	-0.015***	0.002	0.002
	[0.002]	[0.003]	[0.001]	[0.002]	[0.001]	[0.001]
Lagged Beliefs *(ϕ)*						
$\widehat{B}_{i,t-2}$ (θ)	0.033***	.	-0.039***	.	-0.000	.
	[0.003]	.	[0.002]	.	[0.001]	.
$B_{E_{i,t-2}}$ (γ)	.	0.024***	.	-0.023***	.	-0.001
	.	[0.002]	.	[0.001]	.	[0.001]
$B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$ (μ)	.	0.007***	.	-0.013***	.	0.002*
	.	[0.002]	.	[0.001]	.	[0.001]
Avg DV	0.26	0.26	0.66	0.66	0.08	0.08
N	22470	22463	95623	92174	5025	5023
Demographic Controls	Y	Y	Y	Y	Y	Y
Risk-taking	N	Y	N	Y	N	Y
Cohort/Age FE	Y	Y	Y	Y	Y	Y
Beliefs Lagged One Wave	Y	Y	Y	Y	Y	Y

Notes: Two-stage regressions linking beliefs to economic choices, HRS 2002–2022. Stage 1 predicts beliefs from RP, IQ, domain-specific (DS) experiences and their interactions with RP, the RP × experience-covariance interaction, and the non-domain-specific (NDS) experiences and their interactions with RP. The NDS experiences are childhood health and childhood SES for the stock- and home-price choices (Panel A), and cumulative lifetime log equity and home-price returns for the health choices (Panel B). Stage 2 regresses each choice recorded at survey wave t on the fitted-belief components from that respondent's *previous* survey wave, two years earlier ($t-2$). Demographic controls, IQ, the risk-taking index, and fixed effects are all measured at wave t, contemporaneous with the choice. Odd columns (1, 3, 5) use the total lagged predicted belief $\widehat{B}_{i,t-2}$ (θ). Even columns (2, 4, 6) decompose it into an experience-driven component ($B_{E_{i,t-2}}$, γ); the total memory-amplified contribution ($B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$, μ), summing the Stage-1 fitted contributions of the RP × experience and RP × experience-covariance interactions; and a demographic component, which enters the regression but is not reported. The choice variables are defined in Section 5. Panel A is estimated by pooled OLS without fixed effects. Panel B includes age and birth-cohort fixed effects. The specification requires each respondent to be observed in two consecutive waves. Coefficients scaled to 1 SD of each regressor. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

component enters negatively for both stock outcomes, so it is the memory's amplification of the experience, not the raw experience, that shapes investment behavior.

Housing. A one-SD increase in predicted home price beliefs is associated with a 0.6 p.p. increase in the probability of buying a new home in the past two years, a 20% increase relative to the 3% baseline purchase rate. While modest, the magnitudes are economically meaningful relative to the low base rate of home purchases in the older population. The effect again works through memory, which accounts for 0.5 p.p. (16.7% of the mean); the pure-experience component is small and statistically indistinguishable from zero.

Health and Retirement. Three choice outcomes are linked to survival beliefs. For the probability of expecting to work past age 70, a one-SD increase in fitted survival beliefs raises the outcome by 3.3 p.p. (12.7% of the 26% mean), with both components contributing: the pure-experience channel accounts for 2.4 p.p. and the memory channel for 0.7 p.p. Higher survival beliefs also predict a lower probability of being retired: a one-SD increase in fitted beliefs reduces the probability of being retired by 3.9 p.p. (5.9% of the 66% baseline), driven by the pure-experience channel (-2.3 p.p.) and reinforced by a negative memory component (-1.3 p.p.). All three coefficients point in the same direction: people who are more optimistic about surviving are less likely to have retired. Finally, for the share of income contributed to a pension, the total-belief coefficient is zero, but the memory component enters positively and marginally significantly at 0.2 p.p. (2.5% of the mean, $p < 0.10$), while the pure-experience component is not individually distinguishable from zero.

To address the possibility that RP affects choices directly—through channels such as general planning capacity or self-control, rather than through its role in shaping beliefs—Table D.10 in the Appendix repeats the specification with RP included as a direct second-stage regressor. The main belief coefficients remain significant and of comparable magnitude, confirming that the belief-mediation channel documented in Table 3 is not an artifact.[28]

[28]The memory-amplified belief component remains individually significant for every choice once recall performance enters the second stage directly.

6 Conclusion

Recall Performance in a task with no economic content predicts how people think about stocks, their homes, and their own survival. We hypothesize that this connection arises because RP proxies for the recall technology people rely on to imagine the future: how well retrieved experiences match a cue. Consistent with a standard memory model, the beliefs of high RP people are more anchored to their past experiences in the forecast domain, more reactive to cues that bring back similar episodes, but also more influenced by similar yet irrelevant experiences from unrelated domains. These patterns appear in every domain we study, predicting when high RP people have more or less accurate beliefs than low RP ones. The memory-driven component of beliefs in turn predicts stock investment, home purchases, and retirement plans.

Two lessons stand out. First, memory accounts for the systematic structure of beliefs across different domains, based on a person's experiences and her ability to selectively recall them based on time-varying cues. Memory goes beyond ad-hoc domain-specific biases – the hallmark of current approaches – and offers a general testable theory of beliefs: how they are formed, when they are statistically optimal, when they are biased, and how. Memory is fundamentally distinct from Bayesian updating, which relies on information processing – including versions with frictions where some people pay attention to information or exhibit noisy beliefs. In fact, Bayesian updating and noise imply that greater availability and use of information should lead to more accurate beliefs, which is not what we see in the data. A high RP person brings more events to mind when thinking about the future, but she can be too optimistic about her own survival if she has lived through generally good health experiences, causing neglect of mortality statistics. At the same time she can become too pessimistic if her health deteriorates or based on uninformative events that cue selective recall of diseases. Good memory is not sufficient for accurate beliefs – those also require a recognition of which pieces of information are truly relevant for forecasting the future (Bordalo et al., 2026b).

The second lesson is that variation in memory faculties may help explain belief and choice heterogeneity. High RP people are more sensitive to cohort-specific or personal experiences, creating stable disagreement among them. Relative to low RP people, they are more

confident in their beliefs but also more sensitive to time-varying cues, which amplifies disagreement with low RP people during common shocks. Current analyses of heterogeneity rely on measured taste parameters such as risk aversion, present bias, and so on, to explain heterogeneity in choices such as saving, investment, and health (Barsky et al. 1997; Falk et al. 2018). Approaches based on heterogeneous preferences are also used to design optimal policies such as social insurance or commitment devices such as default (Chetty 2006, Madrian and Shea 2001). Recall performance offers a new source of heterogeneity. It is easily measurable using a simple abstract recall task, exhibits stable interpersonal differences, and its effects are theoretically disciplined by cognitively grounded memory models. Here we show that RP helps explain belief and choice heterogeneity in the HRS, and future work may assess its role in other contexts. Cognitive heterogeneity is measurable at scale, it is theoretically disciplined, so it offers new avenues for economic analysis.

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A Proofs

Proof of Proposition 1. Let $p_{it}(e) := f_{it}(e)/\mathbf{f}_{it}$ denote the frequency distribution over past experiences, with $\mathbf{f}_{it} = \sum_{z \in E} f_{it}(z)$ the total number of experiences of agent i at time t, and let

$$d(q_{X,it}) \;:=\; \sum_{e \in E} p_{it}(e)\, d(e, q_{X,it}),$$

be the p_{it}-average discrepancy to the cue. Define $\eta \;:=\; -\frac{S'(0)}{S(0)}$. By a first-order Taylor expansion of S at 0,

$$S\big(a\, d(X,e)\big) \;=\; S(0)\Big[1 - a\eta\, d(X,e)\Big] + O(a^2).$$

Substituting into the recall function $r(e \mid q_{X,it}, f_{it})$, and using $\mathbf{f}_{it}\, d(q_{X,it}) = \sum_z f_{it}(z)\, d(z, q_{X,it})$ in the denominator,

$$r(e \mid q_{X,it}, f_{it}) = \frac{1 - a\eta\, d(z, q_{X,it})}{1 - a\eta\, d(q_{X,it})}\, p_{it}(e) = p_{it}(e)\Big[1 + a\eta\big(d(X) - d(X,e)\big)\Big] + O(a^2).$$

To get beliefs, multiply by $\sigma(X,e)$ and summing over $e \in E$,

$$E_{it}[s_X] = \sum_e r(e \mid q_{X,it}, f_{it})\, \sigma(X,e) = \bar{\sigma}_{it}(X) + a\eta \sum_e p_{it}(e)\big[d(X) - d(X,e)\big]\sigma(X,e) + O(a^2). \tag{11}$$

In the limit of small a we get $S'(0)/S(0) = -1$, so $\eta = 1$.

To go from Equation 11 to Equation 6 in Prediction 1, note that $E_{it}[s_X] = \sum_{e \in E} r(e)\, \sigma(X,e)$ can be split over experiences $E = DS \cup NDS$. Writing $P(a) := \sum_{e \in DS} r(e)$ for the total recall weight on DS experiences, we obtain:

$$E_{it}[s_X] = P(a)\, E_{DS}[s] \;+\; \big(1 - P(a)\big)\, E_{NDS}[s], \tag{12}$$

where $E_{DS}[s] = \frac{\sum_{e \in DS} S(e, q_{X,it}) f_{it}(e) \sigma_X(e)}{\sum_{e \in DS} S(e, q_{X,it}) f_{it}(e)}$ is the recall normalized within DS, and similarly for $E_{NDS}[s]$. Now we can apply Proposition 1 to the DS term to derive Proposition 1, and to the NDS term to derive Prediction 4.

With discrepancy $(s - s_{it})^2$,

$$E_{DS}[s] \approx \bar{s}_X - a\eta \operatorname{Cov}_{DS}\big((s - s_{it})^2,\, s\big),$$

and similarly for $E_{NDS}[s]$. Now consider the recall-weighted share of DS experiences, which should increase in a. Denoting by ρ the share of DS experiences, a logic similar to Equation 11 yields:

$$P(a) \approx \rho + a\eta\kappa$$

where $\kappa = -\sum_e p_{it}(e)\big[d(X) - d(X,e)\big] > 0$. Keeping terms to first order in a:

$$P(a)E_{DS}[s] = \rho\bar{s}_X + a\kappa\,\bar{s}_X - a\rho \operatorname{Cov}_{DS}\big((s - s_{it})^2,\, s\big) + O(a^2),$$

as in the Proposition, with $\eta_1 = \rho > 0$, $\eta_2 = \kappa > 0$, and $\eta_3 = -\rho < 0$.[29] Predictions 1 and 2 follow immediately.

Prediction 3 also follows from Equation 6: forecasts are based on drawing on past predictions to imagine future outcomes. High RP is characterized by a stronger reliance on past DS experiences, and on the current DS cue, which need not make their forecast closer to the statistically optimal prediction, which can be qualitatively different. Examples of conditions leading to lower accuracy are given in the text and explored empirically in Section 4.1.

■

Proof of Prediction 4. Starting from Equation 12, consider an increase in the frequency of $e_0 \in NDS$. We find:

$$\frac{\partial E_{NDS}}{\partial f(e_0)} = \frac{r_{NDS}(e_0)}{f(e_0)}\big[\sigma_X(e_0) - E_{NDS}\big].$$

Since e_0 is valence-congruent, $d(X, e_0) = K$, while the NDS average discrepancy is $d_{-X} \approx K+V$ (dominated by valence-incongruent neutral experiences):

$$\frac{r_{NDS}(e_0)}{f(e_0)} \approx \frac{1}{|\mathbf{f}_{it,NDS}|}\Big[1 + a\big(d_{-X} - d(X,e_0)\big)\Big] \approx \frac{1}{|\mathbf{f}_{it,NDS}|}\big(1 + aV\big).$$

[29] A similar expression holds for NDS experiences, whose recall also changes with a.

Moreover, since neutral experiences dominate NDS frequency, $E_{NDS} \approx \bar{\sigma}_{-X} \approx \delta_X$. Substituting into the equation above, we find:

$$\frac{\partial E_{it}[s_X]}{\partial f(e_0)} \approx \frac{1}{|\mathbf{f}_{it,NDS}|}(1 + aV)\big[\sigma_X(e_0) - \delta_X\big].$$

In the limit where there are sufficiently many neutral NDS experiences, both DS and non-neutral NDS experiences become rare, so that $|\mathbf{f}_{it,NDS}| \to |\mathbf{f}_{it}|$. Setting $\delta_X = 0$ as in the text yields Prediction 3, with $\eta_4 = \frac{V}{|\mathbf{f}_{it}|}$. ■

Proof of Prediction 3. The probability of retrieving a neutral experience is

$$r(e_n|q_{X,it}, f_{it}) = \frac{S(e_n, q_{X,it}) f_{it}(e_n)}{\sum_e S(e, q_{X,it}) f_{it}(e)}$$

To first order expansion in a this is equal to

$$r(e_n|q_{X,it}, f_{it}) = \frac{f_{it}(e_n)}{|\mathbf{f}_{it}|}\left(1 - a\left(K + V\right) + ad\left(q_{X,it}\right)\right)$$

with the first term coming from the lower similarity of the numerator, and the second term from the change in similarity in the denominator (which drops by the average discrepancy $d\left(q_{X,it}\right)$). ■

B Survey Description and Variable Construction

Table B.1: Dependent Variable Survey Wording.

Variable	Exact HRS Survey Wording (by wave and HRS item code)
Chance Stock Market Up	**2002–2004 (P047):** "We are interested in how well you think the economy will do in the next year. By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?" **2006–2014 (P047):** "We are interested in how well you think the economy will do in the future. By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?" **2014 (P190, randomized branch):** "What do you think is the percent chance that the stock market will be higher in twelve months today?" **2016–2020 (P047):** "By next year at this time, what is the percent chance that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today?"
Chance Own Home Value Up	**2010 (P166):** "We are interested in how the value of your home will change in the future. On the same scale from 0 to 100, what do you think is the percent chance that by next year at this time your home will be worth more than it is today?" **2012–2020 (P166, randomized fill):** "We are interested in how the value of your home will change in the future. On the same scale from 0 to 100, what do you think is the percent chance that by next year at this time your home will be worth [more/less] than it is today?"
Chance Live 10+ Years	**2002–2004 (P029):** "(What is the percent chance) that you will live to be [X] or more?", where [X] is 80 if age 69 or younger, 85 if 70–74, 90 if 75–79, 95 if 80–84, and 100 if 85–89. **2006–2022 (P029):** "What is the percent chance that you will live to be [X] or more?", where [X] is 75 if under 65, 80 if 65–69, 85 if 70–74, 90 if 75–79, 95 if 80–84, and 100 if 85–89.
Chance Major Economic Depression	**2002–2008 (P034):** "What do you think are the chances that the U.S. economy will experience a major depression sometime during the next 10 years or so?"
Chance Cost of Living Rises >5%/yr	**2006–2008 (P116):** "Over the next 10 years what are the chances the cost of living will increase by 5 percent or more per year on average?"

Notes: This table reports the verbatim HRS Section P survey wording for each of the five dependent variables used in the paper, organized by estimation-sample wave and HRS item code. In 2014 half the sample was asked the standard P047 wording and half the alternative P190 wording (branch variable X504_4Random1_2). For *Chance Own Home Value Up*, from 2012 the question asked some respondents about "more" and others about "less" (X501 in 2012–2014, X083/P196 in 2016–2020). We recode the "less" answers so the variable is always the probability the home is worth more. For *Chance Live 10+ Years*, the target age in the question depends on the respondent's age. In 2002–2004 there were five target ages (80/85/90/95/100); from 2006 onwards there are six (75/80/85/90/95/100), since respondents under 65 are now asked about age 75. The home-value and (2014-branch) stock-market questions are asked only to the relevant subsamples (homeowners; the X504_4Random1_2 = 2 branch). Source: HRS Section P questionnaires, hrsdata.isr.umich.edu.

Table B.2: Index Variable Construction.

Variable	Description	Min	Max	Mean
Panel 1: Recall Performance Index				
Immediate Recall	Number of words recalled immediately from a 10-word list	0.00	15.00	5.40
Delayed Recall	Number of words recalled after a 5-minute delay	0.00	10.00	4.35
Retrieval Fluency	Number of unique animals named in 60 seconds, net of errors	0.00	40.00	16.25
Construction: Each component is standardized (demeaned, divided by s.d.) across the pooled sample. The index is the mean of the three standardized components. The index is then re-standardized to have mean zero and unit variance.				
Panel 2: IQ (Math) Index				
Counting Backwards	Number of correct serial-7 subtractions from 100	0.00	4.00	2.44
Number Series	Correct answers on number pattern completion (adaptive)	0.00	3.00	1.59
Numeracy	Correct answers on 3 arithmetic/probability questions	0.00	3.00	1.37
Construction: Each component is standardized (demeaned, divided by s.d.) across the pooled sample. The index is the mean of the three standardized components. The index is then re-standardized.				
Panel 3: Current Health PCA				
Self-Rated Health	Self-reported health on 5-point scale (poor to excellent)	1.00	5.00	2.91
Days in Bed	Days spent in bed due to illness in the past month *(inverted)*	-31.00	0.00	-0.77
Health Conditions (Full)	Count of all diagnosed health conditions *(inverted)*	-11.00	0.00	-2.57
Depressed	Indicator for depression diagnosis *(inverted)*	-1.00	0.00	-0.17
Health Change	Self-reported health change since last wave (3-point scale) *(inverted)*	-3.00	-1.00	-2.15
Health Conditions (Basic)	Count of 8 major diagnosed conditions *(inverted)*	-8.00	0.00	-2.16
New Heart Attack	New heart attack since last wave *(inverted)*	-1.00	0.00	-0.01
New/Worsening Severe	New or worsening severe health condition since last wave *(inverted)*	-1.00	0.00	-0.33
Sleep Quality	1st principal component of 4 sleep difficulty items *(inverted)*	-3.55	1.81	0.00
Functional Limitations	Count of functional limitations (0–15 ADL/IADL scale) *(inverted)*	-15.00	0.00	-3.28
Has Helper	Indicator for receiving help with daily activities *(inverted)*	-1.00	0.00	-0.16
Tasks Helped With	Number of daily tasks receiving help (0–6) *(inverted)*	-6.00	0.00	-0.26
Construction: The first principal component is taken of the 12 components listed above. Components where a higher raw value indicates worse health are multiplied by -1 (marked *(inverted)*) so that higher values of every component, and of the resulting index, indicate better health. Missing values are imputed with the sample median.				
Panel 4: Childhood Health Index				
Missed School	Missed 1+ month of school due to health before age 16 *(inverted)*	-1.00	0.00	-0.11
Disabled as Child	Disabled for 6+ months before age 16 *(inverted)*	-1.00	0.00	-0.04
Childhood Conditions	Count of 16 childhood diseases/conditions *(inverted)*	-12.00	0.00	-0.64
Severe Conditions	Count of 5 severe childhood conditions *(inverted)*	-4.00	0.00	-0.21
Psychological Conditions	Count of 3 psychological conditions before 16 *(inverted)*	-3.00	0.00	-0.07

Continued on next page

Table B.2 – continued from previous page

Variable	Description	Min	Max	Mean
Construction: The index is the mean of the standardized values of the 5 subcomponents listed above. Childhood measures are asked once per respondent and carried forward across waves using the within-person mode. Components where a higher raw value indicates worse health are multiplied by −1 (marked *(inverted)*) so that higher values indicate better childhood health. Missing values are imputed with the sample median.				
Panel 5: Risk-Taking Index				
Subjective Risk Taking	Willingness to take risks, 0–10 self-report scale	0.00	10.00	5.37
Risk Taking (Employment)	Prefers risky high-pay job over safe lower-pay job	0.00	1.00	0.15
Risk Taking (Business)	Prefers risky business gamble over safe cash-out	0.00	1.00	0.28
Construction: Each component is standardized (demeaned, divided by s.d.); the statistics above are for the components as asked, before standardization. The subjective risk-taking item is asked from 2014; pre-2014 values are imputed with the respondent's modal response. The index is the mean of the three standardized components, each entering once; respondents missing a component are averaged over the components they do have, so the index is defined for more respondents than any single component.				
Panel 6: Current Socioeconomic Status PCA				
Education	Years of education completed	0.00	17.00	12.60
Income	Household income residualized on retirement status (millions USD)	-0.08	0.95	0.00
Wealth	Total household wealth (millions USD)	0.00	35.50	0.19
Home Ownership	Indicator for owning a home	0.00	1.00	0.60
Not Unemployed	Indicator: not currently unemployed or laid off	0.00	1.00	0.97
Construction: The first principal component is taken of the 5 subcomponents listed above. Missing values are imputed with the sample median.				
Panel 7: Childhood Socioeconomic Status Index				
Father's Education	Father's years of education	0.00	17.00	9.67
Mother's Education	Mother's years of education	0.00	17.00	9.91
Father Unemployed	Father had extended unemployment before 16 *(inverted)*	0.00	1.00	0.79
Moved (Fin. Difficulty)	Family moved due to financial difficulty before 16 *(inverted)*	0.00	1.00	0.82
Family Received Aid	Family received financial aid before 16 *(inverted)*	0.00	1.00	0.84
Lived Rural	Lived in rural area before age 16 *(inverted)*	0.00	1.00	0.57
Spoke English	English spoken at home as a child	0.00	1.00	0.91
Mother Worked	Mother worked outside home during childhood *(inverted)*	0.00	1.00	0.43
Construction: The index is the mean of the standardized values of the 8 subcomponents listed above. Childhood measures are asked once per respondent and carried forward across waves using the within-person mode. Components where a higher raw value indicates lower childhood SES are multiplied by −1 (marked *(inverted)*) so that higher values indicate higher childhood SES. Missing values are imputed with the sample median.				

Notes: This table documents the construction of all index variables used in the paper. Components marked *(inverted)* are multiplied by −1, so that higher values indicate better health or higher socioeconomic status. The reported min, max, and mean reflect this inversion. Income and wealth are reported in millions of dollars.

Table B.3: Construction of the experience and cue-similarity regressors

Variable	Construction
Panel A. Stock-market beliefs (DV: subjective chance the stock market rises)	
$\bar{s}_{it}^{\text{stk}}$ Average experienced stock return	Mean of the experienced stock-return signals over the respondent's lifetime, $\bar{s}_{it}^{\text{stk}} = \frac{1}{K}\sum_k s_k$. The signal s_k is the trailing twelve-month log return on the S&P 500 price index, $s_k = \log P_k - \log P_{k-12}$, evaluated at every month k from the respondent's birth through the survey month. A higher value means the respondent has lived through higher average stock returns.
$cov^{\text{stk}}\big(s_k, (s_k - r_t)^2\big)$ Cue-similarity covariance	Within-window sample covariance between each experienced return s_k and its squared distance from the current cue, $cov^{\text{stk}} = \frac{1}{K}\sum_k \big(s_k - \bar{s}\big)\big(s_k - r_t\big)^2$, where r_t is the trailing one-year S&P 500 return as of the survey month. It is low (negative) when the returns the respondent experienced most often lie close to today's return – i.e. when the recalled past is similar to the current cue.
Panel B. House-price beliefs (DV: subjective chance home prices rise)	
$\bar{s}_{it}^{\text{hpi}}$ Average experienced house-price growth	Mean of the experienced house-price signals over the lifetime, $\bar{s}_{it}^{\text{hpi}} = \frac{1}{K}\sum_k s_k$. The signal s_k is the trailing four-quarter (one-year) log return on the regional house-price index, evaluated each quarter k from birth through the survey quarter. The index is the FHFA house-price index for the respondent's *childhood* Census division, available from 1975. We back-fill years before 1975 using the national Case–Shiller real return.
$cov^{\text{hpi}}\big(s_k, (s_k - r_t)^2\big)$ Cue-similarity covariance	Within-window sample covariance between each experienced house-price return s_k and its squared distance from the current cue, $cov^{\text{hpi}} = \frac{1}{K}\sum_k \big(s_k - \bar{s}\big)\big(s_k - r_t\big)^2$, where r_t is the current trailing-year house-price return for the respondent's childhood Census division at the survey quarter. It is low (negative) when the house-price growth rates the respondent experienced most often lie close to today's growth rate – i.e. when the recalled past is similar to the current cue.
Panel C. Survival beliefs (DV: subjective chance of living ten or more years)	
$\bar{s}_{it}^{\text{hlth}}$ Average experienced health change	Mean of the respondent's experienced health changes, $\bar{s}_{it}^{\text{hlth}} = \frac{1}{K}\sum_k s_k$. The signal s_k is the wave-to-wave change in the respondent's current-health index, $s_k = \text{health}_k - \text{health}_{k-1}$, taken over all HRS waves k in which the respondent is observed, up to and including the survey wave. The current-health index is the first factor of a single-factor model fitted to a battery of current-health measures, as described in the factor-loadings table above. At the first observed wave the change is the current-health index minus the childhood-health index, which is the analogous single-factor index over childhood-health measures.
$cov^{\text{hlth}}\big(s_k, (s_k - r_t)^2\big)$ Cue-similarity covariance	Within-window sample covariance between each experienced health change s_k and its squared distance from the current cue, $cov^{\text{hlth}} = \frac{1}{K}\sum_k \big(s_k - \bar{s}\big)\big(s_k - r_t\big)^2$, where r_t is the respondent's current health change at the survey wave. It is low (negative) when the health changes the respondent experienced most often lie close to the current change – i.e. when the recalled past is similar to the current cue.

Notes. The table shows the two domain-specific regressors in Equation 8 for each belief domain: the average experienced signal $\bar{s}_{it}$ and the cue-similarity covariance $cov(s_k, (s_k - r_t)^2)$. The experience window spans from the respondent's year of birth (for stocks and house prices) or the first observed HRS wave (for health) up to the survey period. $\{s_k\}$ collects the K period signals, $\bar{s}$ is their mean, and r_t is the contemporaneous signal that serves as the recall cue. We divide both regressors by their interquartile range before estimation, so that a one-unit change corresponds to a 25th-to-75th-percentile move.

Table B.4: Variable Summary Statistics

	Mean	St.d	IQR	P10	P25	P50	P75	P90
Cognition								
Recall Performance	0.01	0.90	1.14	-1.16	-0.55	0.04	0.59	1.10
– Immediate Memory	5.40	1.75	3.00	3.00	4.00	5.00	7.00	8.00
– Delayed Memory	4.35	2.08	3.00	1.00	3.00	4.00	6.00	7.00
– Retrieval Fluency	16.25	6.93	10.00	8.00	11.00	16.00	21.00	25.00
IQ	-0.01	0.88	1.54	-1.21	-0.74	0.05	0.81	1.00
– Counting Backwards	2.44	1.56	3.00	0.00	1.00	3.00	4.00	4.00
– Number Series	1.59	1.06	2.00	0.00	1.00	2.00	3.00	3.00
– Numeracy	1.37	1.02	2.00	0.00	0.00	1.00	2.00	3.00
Experiences (DS)								
$\overline{s_{it}}^{STOCKS}$	0.03	0.01	0.01	0.02	0.03	0.03	0.03	0.04
$\overline{s_{it}}^{HOUSE}$	0.03	0.01	0.01	0.02	0.02	0.03	0.03	0.04
$\overline{s_{it}}^{HEALTH}$	0.02	0.35	0.27	-0.32	-0.12	0.01	0.14	0.36
$cov(s,(s-s_{it})^2)^{STOCKS}$	-0.00	0.01	0.01	-0.01	-0.01	-0.00	0.00	0.01
$cov(s,(s-s_{it})^2)^{HOUSE}$	0.00	0.00	0.00	-0.00	-0.00	-0.00	0.00	0.00
$cov(s,(s-s_{it})^2)^{HEALTH}$	0.30	2.92	0.18	-0.31	-0.03	0.01	0.16	0.91
Experiences (NDS)								
Childhood Health PCA	-0.04	0.71	0.41	-0.94	-0.10	0.31	0.31	0.36
Childhood SES PCA	0.06	0.47	0.60	-0.63	-0.22	0.16	0.37	0.62
Δ_L S&P 500	1.68	0.55	0.97	0.82	1.16	1.83	2.13	2.30
Δ_L HPI	1.72	0.35	0.46	1.37	1.46	1.64	1.92	2.21
Beliefs								
Chance Stock Increase [0-1]	0.47	0.26	0.35	0.10	0.25	0.50	0.60	0.80
Chance Home Value Increase [0-1]	0.53	0.33	0.60	0.10	0.20	0.50	0.80	1.00
Chance Live 10+ Years [0-1]	0.48	0.32	0.55	0.00	0.20	0.50	0.75	0.95
Controls								
Age	67.63	11.39	17.00	54.00	59.00	67.00	76.00	83.00
Female [0,1]	0.59	0.49	1.00	0.00	0.00	1.00	1.00	1.00
Married [0,1]	0.63	0.48	1.00	0.00	0.00	1.00	1.00	1.00
Retired [0,1]	0.54	0.50	1.00	0.00	0.00	1.00	1.00	1.00
Income (1000s)	64.50	129.64	50.80	7.33	14.80	30.95	65.60	124.00
Years Education	12.60	3.28	3.00	8.00	12.00	12.00	15.00	17.00

Notes: HRS biennial waves 2002–2022. *Recall Performance*: standardized mean of immediate recall (10-item list), delayed recall, and retrieval fluency (unique animals in 60s). *IQ*: standardized mean of Counting Backwards (serial subtraction of 7), Number Series, and Numeracy. DS: experience variables $\overline{s}_{it}$: average lifetime realized return (monthly S&P 500 for stocks, quarterly FHFA regional HPI for home prices, wave-to-wave changes in a health PCA for health). The covariance term $\text{cov}(s,(s-r_t)^2) \equiv \frac{1}{T_i}\sum_k (s_k-\bar{s}_i)(s_k-r_t)^2$ governs the direction of belief distortion under selective retrieval. NDS: *Childhood Health PCA* and *Childhood SES PCA* are as described in Table B.2. Δ_L S&P 500 and Δ_L HPI are the lifetime log return on each index, i.e. the change in log price from childhood to the survey date. Beliefs are elicited on a 0–100 scale and rescaled to $[0,1]$: chance stocks rise next year, home values rise locally, and respondent survives 10+ more years. Controls measured at interview; income in 1000s USD; female indicator $=1$ if female, 0 if male. IQR denotes the interquartile range (P75 − P25).

Supplementary Material

Online Appendix

Recall Performance, Beliefs and Behavior

October 16, 2026

C Recall Performance and Beliefs: descriptive results

Table C.1: Correlation of Recall Performance with the Components of IQ.

	IQ Index (overall)	Counting Backwards	Number Series	Numeracy
Recall Performance	0.41***	0.33***	0.38***	0.37***
Observations	191,274	191,273	75,735	109,744

Notes: Each cell reports the pairwise correlation between Recall Performance and a measure of IQ, pooled across all respondent-waves for which both items are non-missing. The first column uses the overall IQ (math) index; the remaining columns use its three individual components: counting backwards (serial-7 subtraction from 100), number series (number-pattern completion), and numeracy (arithmetic and probability questions); see Table B.2 for definitions. Observation counts differ across columns because the number-series and numeracy items are administered to restricted subsamples. Stars denote significance of the correlation: *** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Variance Decomposition of Cognitive Indices and Sub-Components.

	Memory Index	Math Index	Immed. Mem.	Delayed Mem.	Named Animals	Count. Back.	Number Ser.	Numeracy
Variance								
$\text{Var}(y_{it})$	0.802	0.770	1.000	1.000	1.000	1.000	1.000	1.000
$\text{Var}_i(\overline{y}_i)$ (cross-section)	0.495	0.520	0.550	0.568	0.749	0.609	0.693	0.680
$\text{E}_i[\text{Var}_t(y_{it})]$ (over time)	0.308	0.250	0.450	0.432	0.251	0.391	0.307	0.320
Proportion								
$\text{Var}(y_{it})$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
$\text{Var}_i(\overline{y}_i)$ (cross-section)	0.617	0.676	0.550	0.568	0.749	0.609	0.693	0.680
$\text{E}_i[\text{Var}_t(y_{it})]$ (over time)	0.383	0.324	0.450	0.432	0.251	0.391	0.307	0.320

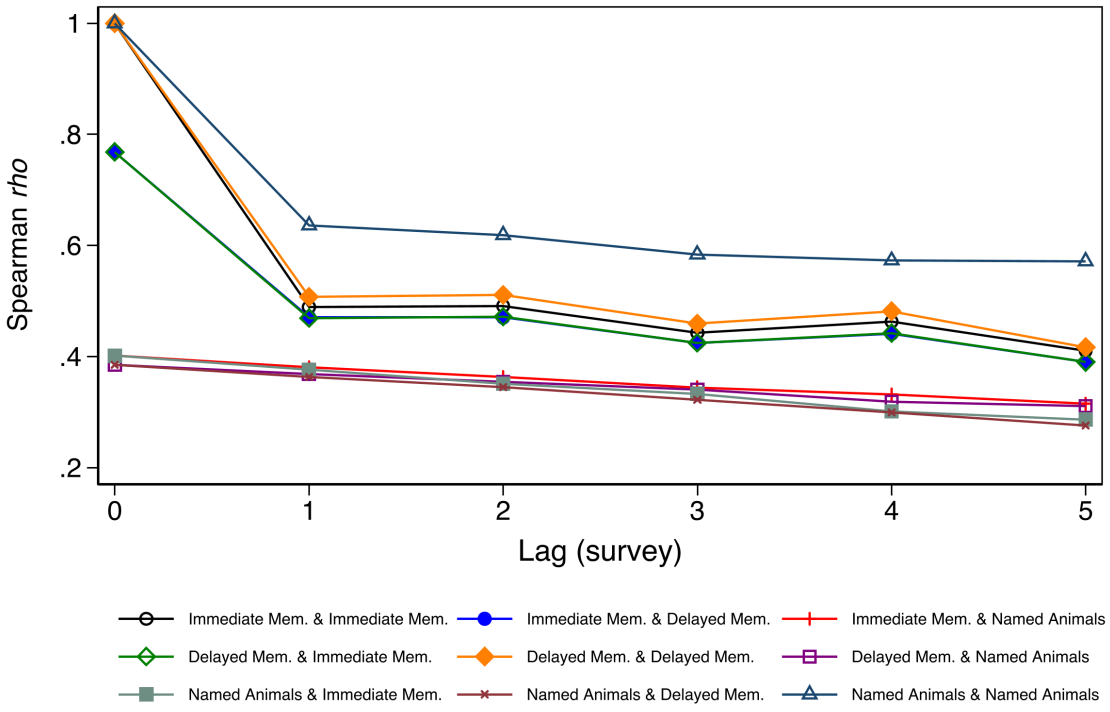
Notes: Variance decomposition of the Memory Index (Recall Performance), the Math Index (IQ), and all individual cognitive sub-components across HRS waves 2002–2022. $\text{Var}(y_{it})$ is total variance. $\text{Var}_i(\overline{y}_i)$ is the between-person component: the variance of respondent means, weighting each respondent by her number of non-missing observations. $\text{E}_i[\text{Var}_t(y_{it})]$ is the within-person variance over time, averaged across respondents. The proportion rows give each component as a share of total variance. All sub-components are standardized (mean zero, unit variance) before entering the indices.

Table C.3: Within- vs Between-Person Variance Decomposition of Experience Signals and Covariance Terms.

	Stocks		Home Prices		Health	
	$\overline{s_{it}}$	$\text{cov}(s, (s - s_{it})^2)$	$\overline{s_{it}}$	$\text{cov}(s, (s - s_{it})^2)$	$\overline{s_{it}}$	$\text{cov}(s, (s - s_{it})^2)$
Total variance	1.000	1.000	1.000	1.000	1.000	1.000
$\text{Var}_i(\overline{y}_i)$ (cross-section)	0.766	0.139	0.908	0.127	0.808	0.792
$\text{E}_i[\text{Var}_t(y_{it})]$ (over time)	0.234	0.861	0.092	0.873	0.192	0.208
$\text{Corr}(\overline{s_{it}}, \text{cov}(s, (s - s_{it})^2))$	0.253		-0.007		0.425	

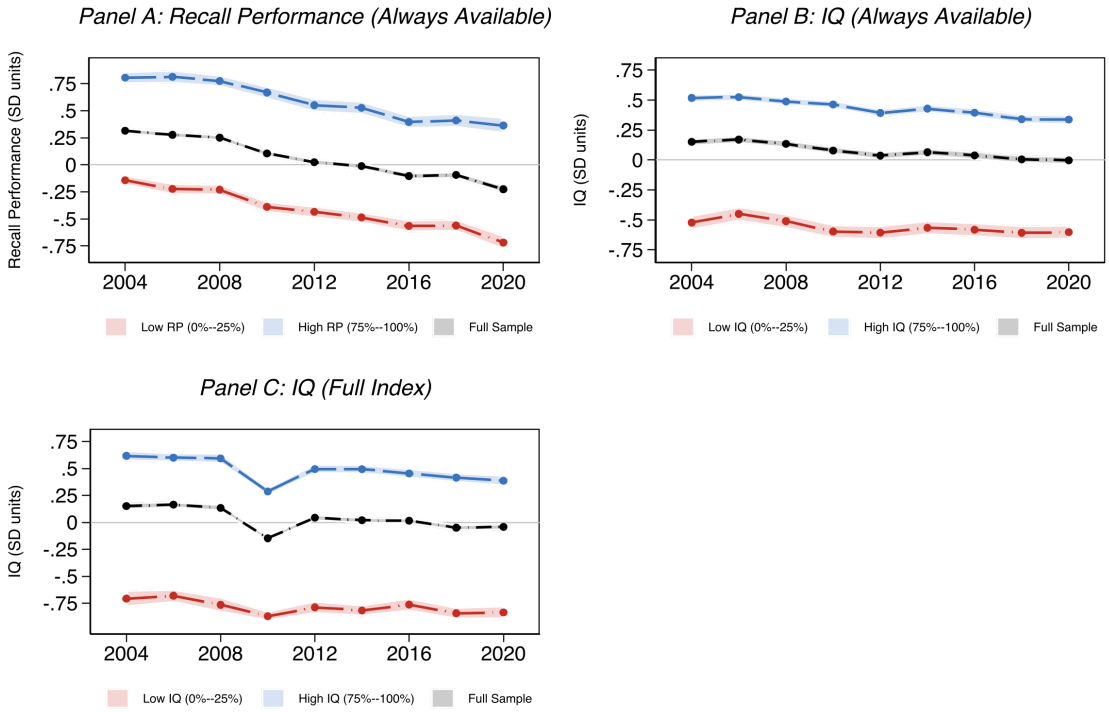
Notes: Proportional variance decomposition of the domain-specific experience signal $\overline{s_{it}}$ and the covariance term $\text{cov}(s, (s - s_{it})^2)$ for stocks, home-price, and health. The table parallels Table C.2 for the Memory and Math indices. Rows report shares of total variance, defined as in Table C.2. The bottom row reports the pooled person-wave correlation between $\overline{s_{it}}$ and the covariance term for each category. Stocks series are monthly returns, home-price series are quarterly HPI returns, and health series are built from HRS-wave health indices.

Figure C.1: Memory Trait Stability: Spearman Correlation by Lag.



Notes: Each series plots the Spearman rank correlation ρ (y-axis) between a pair of memory sub-components at increasing survey lags (x-axis; lag 0 = same survey wave, lags 1–5 = up to ten years apart). There are nine series, one for each pair of the three sub-components. At lag 0 the correlation of a component with itself is one by construction. The correlations remain high at all lags, indicating that rankings of respondents by memory is stable over time.

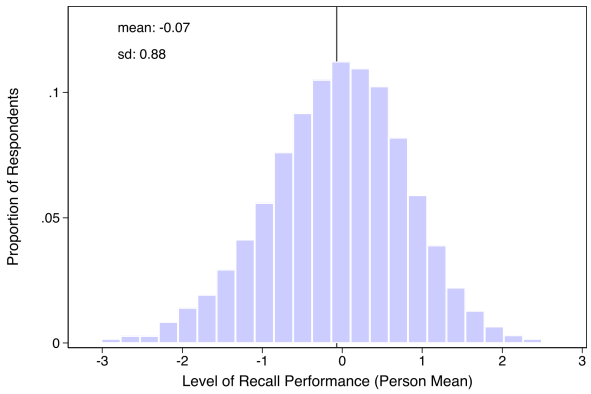
Figure C.2: Recall Performance and IQ Over Time, by Quartile: Always-Available Indices



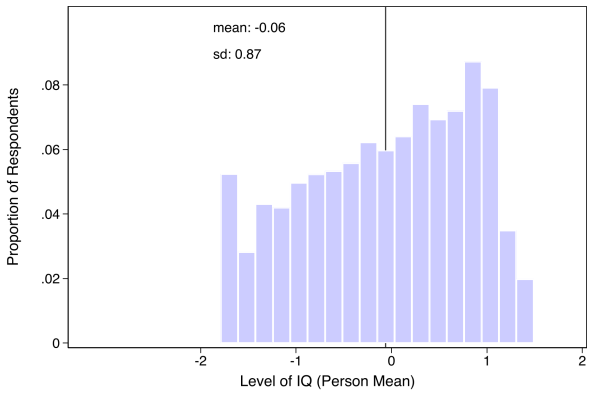
Notes: Each panel plots the average level of an index in each survey wave from 2004 to 2020, by quartile of that index in the first wave. Panel A plots Recall Performance using only the immediate- and delayed-recall components, which are administered in every wave. Panel B plots IQ using only the counting-backwards task, the single IQ component administered in every wave. Panel C plots IQ using the full index, which additionally includes the number series and numeracy components (numeracy is not re-administered at re-interview). Blue lines show respondents in the top quartile of the respective index in the first wave; red lines show the bottom quartile; black lines show the full sample. Shaded bands are 95% confidence intervals. The sample is restricted to the 3,169 respondents present in all ten survey waves (2002–2020). The first wave is omitted as it is the base year for group assignment. Memory and IQ are expressed in standard deviation units.

Figure C.3: Distribution of Recall Performance and IQ (Raw, Un-Residualized).

Panel A: Recall Performance



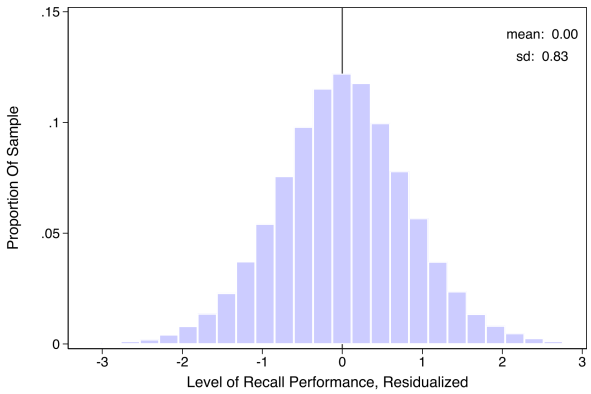
Panel B: IQ



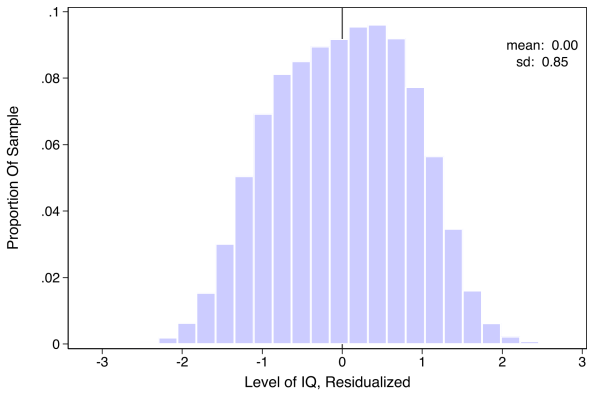
Notes: The figure shows the raw (un-residualized) counterpart of Figure 1. Each panel plots the distribution across respondents of a person-level average, where for each respondent the relevant index is averaged over all survey waves in which they appear (one observation per respondent). Panel A shows person-level average Recall Performance (RP); Panel B shows person-level average IQ.

Figure C.4: Distribution of Recall Performance and IQ (Pooled Person-Wave, Residualized).

Panel A: Recall Performance (Residualized)



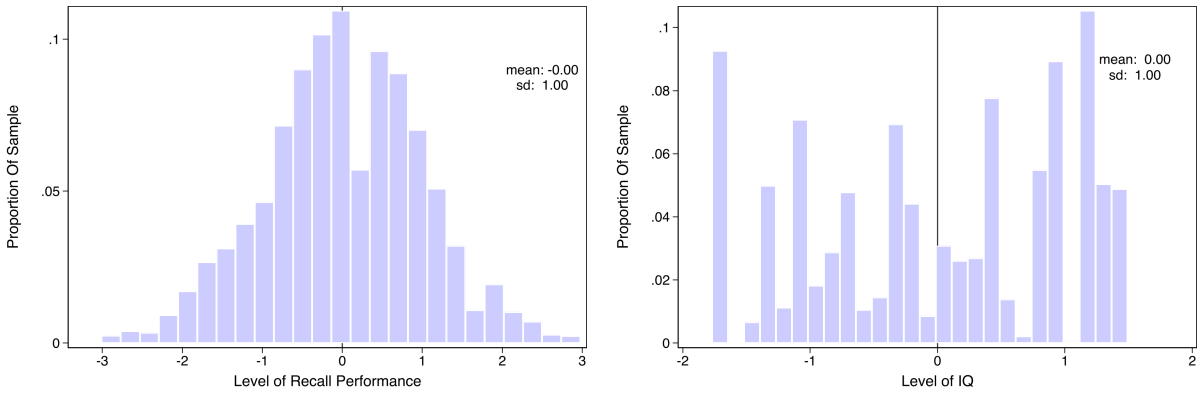
Panel B: IQ (Residualized)



Notes: The figure shows the pooled (person-wave) residualized counterpart of Figure 1; here each index is taken at the person-wave level rather than averaged within respondent. Panel A shows the distribution of Recall Performance (RP) after residualizing on IQ, age, income, education, gender, marital status, and retirement status. Panel B shows the distribution of IQ after residualizing on RP and the same demographics. The cross-sectional standard deviations reported in the main text (Section 2) refer to this pooled sample. The raw (un-residualized) pooled distributions are reported in Figure C.5.

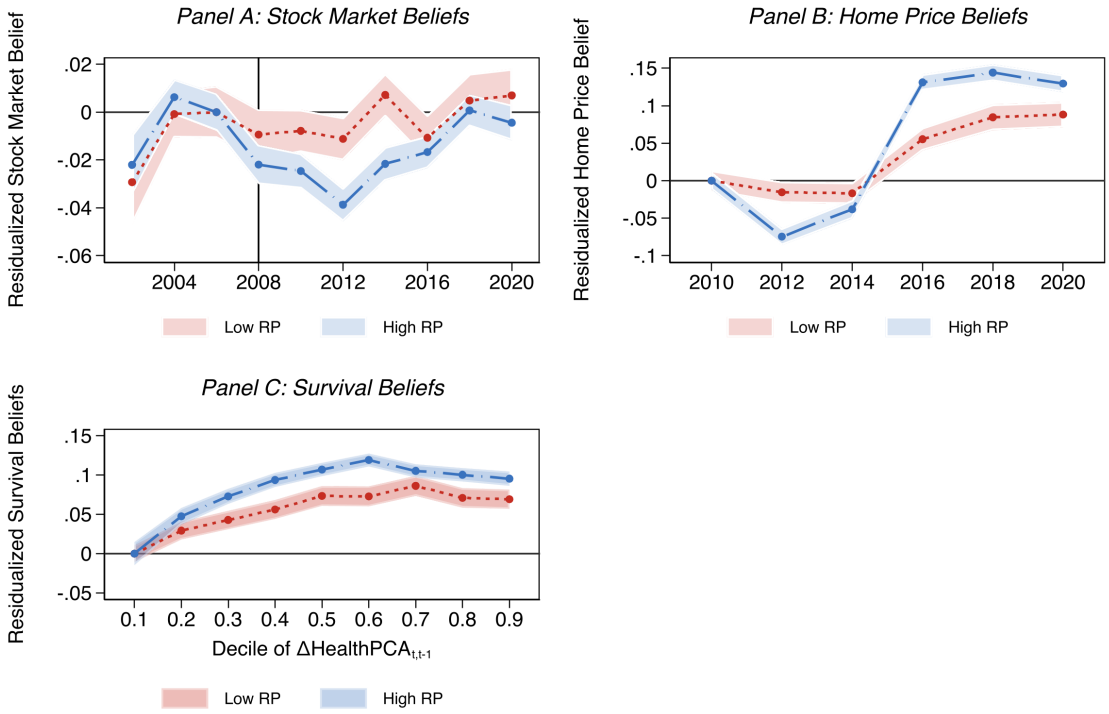
Figure C.5: Distribution of Recall Performance and IQ (Pooled Person-Wave, Raw).

Panel A: Recall Performance *Panel B: IQ*



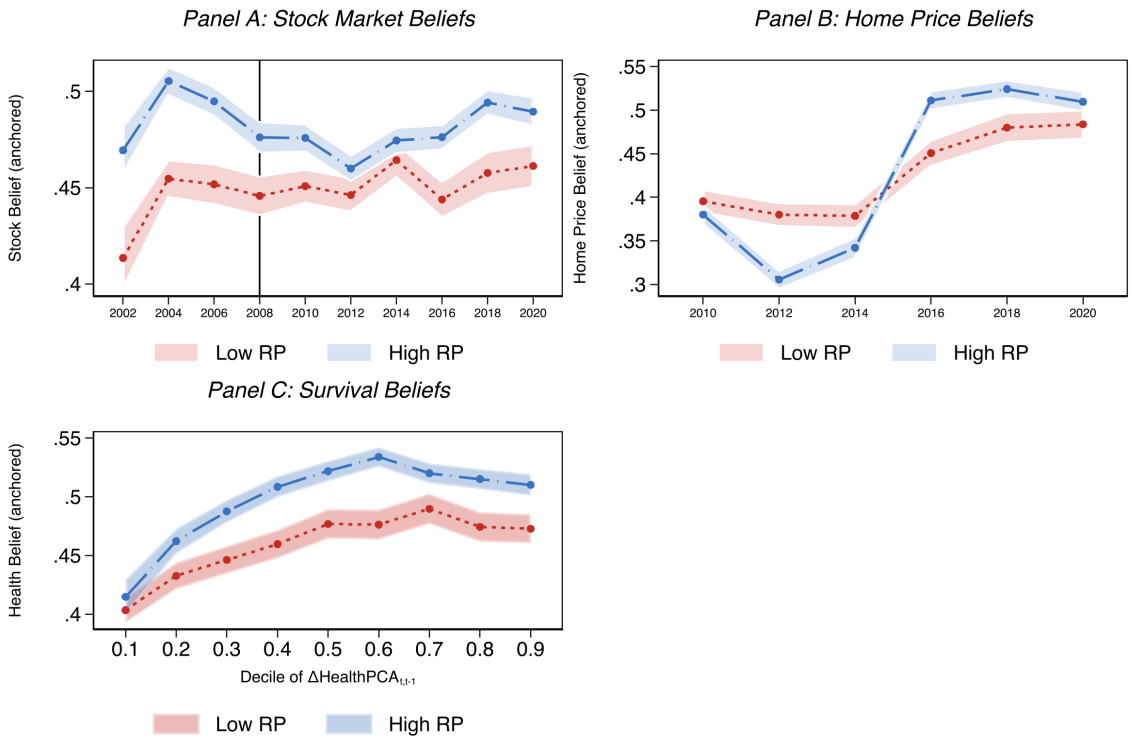
Notes: The figure shows the raw (un-residualized) pooled (person-wave) counterpart of Figure 1. Panel A shows the raw distribution of Recall Performance (RP) across all respondents and survey waves; Panel B shows the raw distribution of IQ. The residualized pooled distributions are reported in Figure C.4.

Figure C.6: Residualized Beliefs Over Time, by Recall Performance (Scaled).



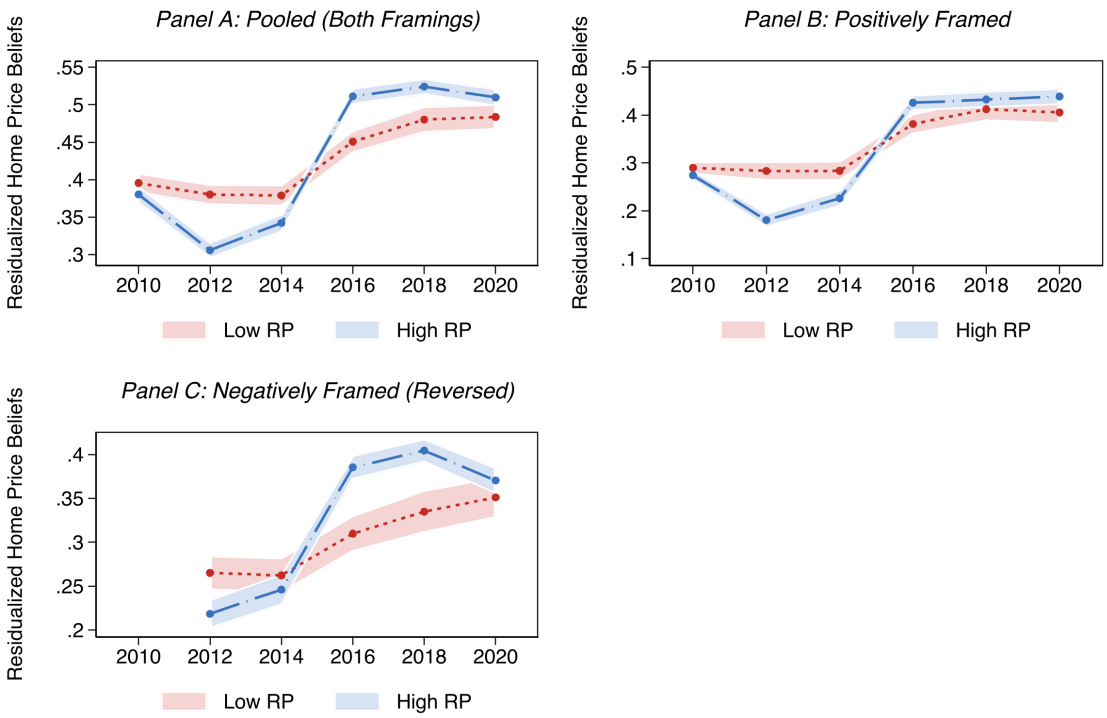
Notes: Each panel shows average residualized beliefs by Recall Performance (RP) group. Panel A plots stock market beliefs ("chance stocks will be worth more in one year") by survey year (2002–2020). Panel B plots home-price beliefs ("chance home prices will go up") by survey year (2010–2020). Panel C plots survival beliefs ("chance of surviving 10+ years") by decile of health change between survey waves. In all panels, beliefs are residualized on: age, gender, marital status, retirement status, income, education, and the Math Index (IQ). The stock market series additionally controls for whether the respondent actively follows the stock market. Each RP group is normalized to zero at its baseline: 2006 for stocks (just prior to the financial crisis), 2010 for home prices (first year the question was asked), and the lowest health-change decile for survival. Low RP (High RP) is respondents in the bottom (top) quartile of the RP distribution. Shaded areas are 95% confidence intervals.

Figure C.7: Average Residualized Beliefs Over Time by Recall Performance.



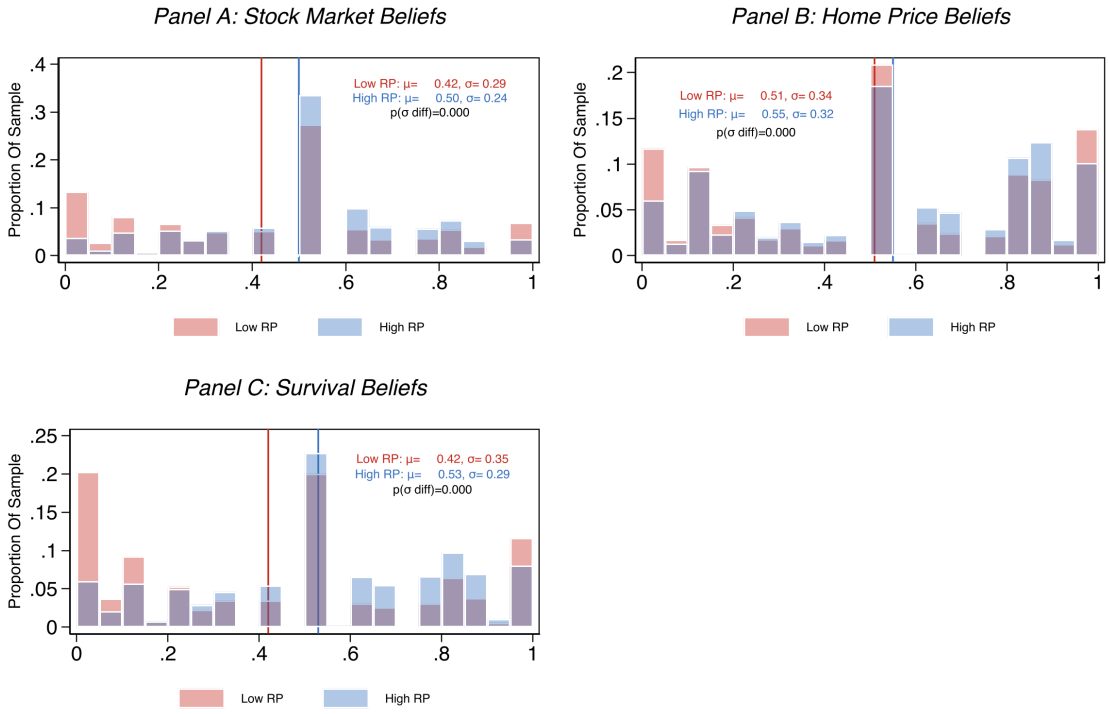
Notes: Each panel plots beliefs residualized on IQ, age, income, education, gender, marital status, and retirement status, then re-anchored to the overall raw sample mean so that the vertical axis is in interpretable belief units, for the top (blue) and bottom (red) quartile of Recall Performance (RP). Panel A shows the average stock market belief ("chance stocks will be worth more in one year") by survey wave. Panel B shows the average home-price belief ("chance home will be worth more in one year") by survey wave from 2010 onward. Panel C shows the average survival belief ("chance live 10+ years") by decile of health change ($\Delta\text{HealthPCA}_{t,t-1}$). Shaded bands are 95% confidence intervals. The dispersion of these residualized beliefs across RP groups is shown in Appendix Figure C.10; the corresponding raw (un-residualized) beliefs over time are shown in Figure 3.

Figure C.8: Home Price Beliefs by Question Framing, by Recall Performance.



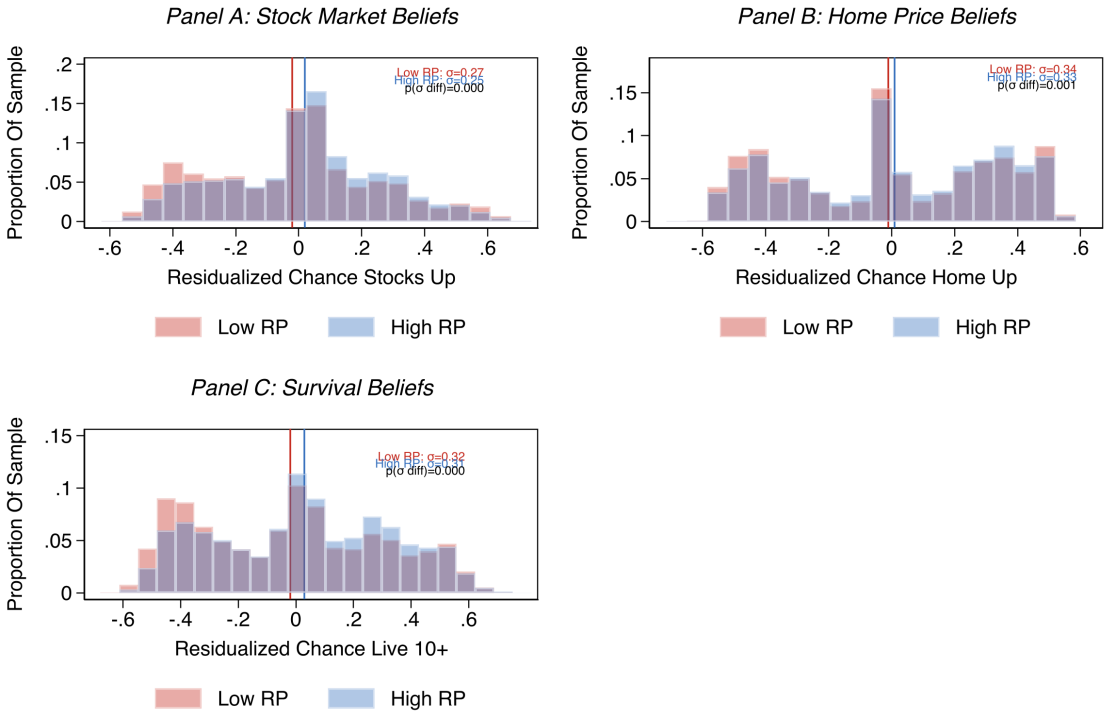
Notes: Each panel reproduces the home-price belief series of Panel B of Figure C.7 for a different subsample defined by how the home-price question was framed. Home-price beliefs are residualized on IQ, age, income, education, gender, marital status, and retirement status, then anchored to a subsample mean, by survey wave from 2010 onward, for the top (blue) and bottom (red) quartile of Recall Performance. Respondents asked about the probability that the home loses value are reverse-coded onto the probability-of-increase scale, and we add a constant so that their mean matches that of respondents asked about a gain. This constant enters Panels A and C and does not change variation within either framing. Panel A pools both framings, Panel B uses only respondents asked about a gain, and Panel C only respondents asked about a loss. Shaded bands are 95% confidence intervals.

Figure C.9: Raw Belief Distributions, by Recall Performance.



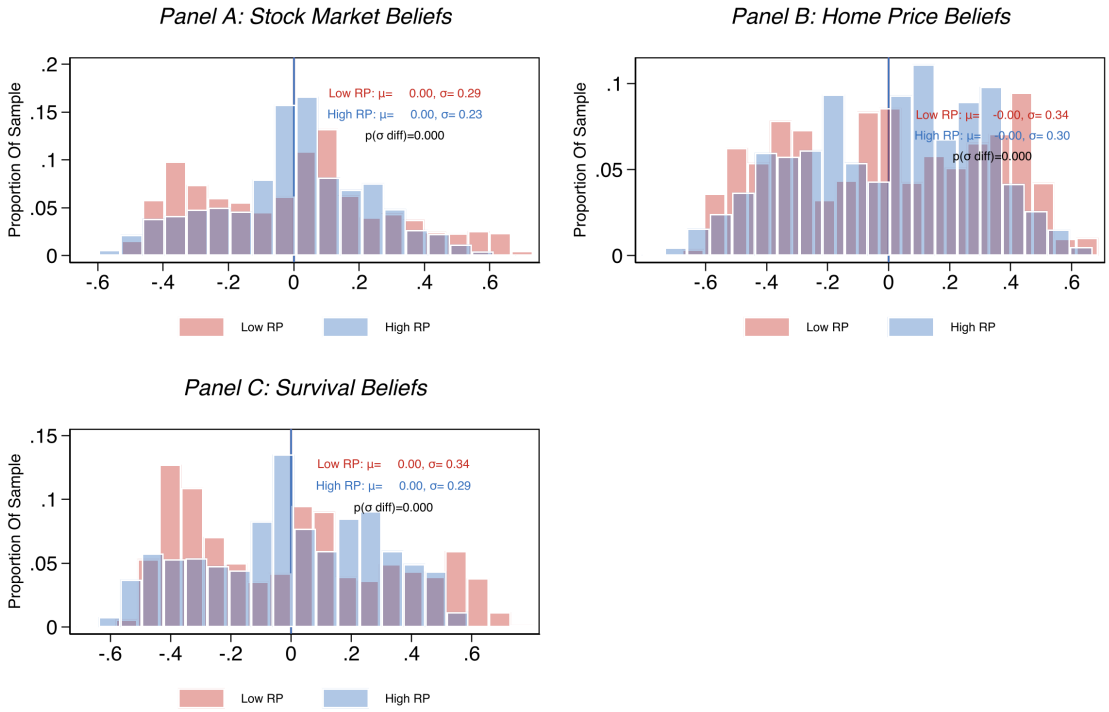
Notes: Each panel shows the distribution of raw (un-residualized) beliefs by Recall Performance (RP) group. Panel A: stock market beliefs ("chance stocks will be worth more in one year"). Panel B: home-price beliefs ("chance home prices will go up"; negative-framed responses reversed). Panel C: survival beliefs ("chance of surviving 10+ years"). Low RP (High RP) is respondents in the bottom (top) quartile of the raw RP distribution. Vertical lines show group means. Text on each panel reports group means (μ), standard deviations (σ), and the p-value for the Brown-Forsythe test of equality of variances.

Figure C.10: Dispersion of Residualized Beliefs, by Recall Performance.



Notes: Each panel shows the distribution of beliefs residualized on IQ, age, income, education, gender, marital status, and retirement status, for the high (blue) and low (red) quartile of Recall Performance (RP), plotted on a common bin grid; the inset reports each group's standard deviation and the p-value of an equality-of-variance test. Panel A: stock market beliefs ("chance stocks will be worth more in one year"). Panel B: home-price beliefs ("chance home will be worth more in one year"). Panel C: survival beliefs ("chance live 10+ years"). These demographic-residualized distributions sit between the raw distributions of Appendix Figure C.9 and the demographic- and time-residualized distributions of Appendix Figure C.11.

Figure C.11: Demographic- and Time-Residualized Belief Distributions, by Recall Performance.



Notes: Each panel shows the distribution of beliefs residualized on both demographics and time. Panel A: stock market beliefs ("chance stocks will be worth more in one year"). Panel B: home-price beliefs ("chance home prices will go up"; negative-framed responses reversed). Panel C: survival beliefs ("chance of surviving 10+ years"). Beliefs are first residualized on: age, gender, marital status, retirement status, income, education, and the Math Index (IQ); the stock market series additionally controls for stock market participation. The memory index is also residualized on the same demographics and used to form RP quartile groups. Beliefs are then further residualized on survey-year dummies separately within each RP group, removing common time variation. Vertical lines show group means. Text on each panel reports group means (μ), standard deviations (σ), and the p-value for the Brown-Forsythe test of equality of variances.

D Recall Performance and Beliefs: regression analysis

Table D.1: DS Regressions With IQ × Experience and IQ × Signal Interactions.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live10+**	
Cognition						
RP	.	0.003	.	-0.021*	.	0.017***
	.	[0.006]	.	[0.012]	.	[0.001]
IQ	0.020**	0.018**	0.008	0.019	-0.014***	-0.020***
	[0.008]	[0.009]	[0.015]	[0.017]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	-0.000	0.000	0.016**	0.009	0.022***	0.023***
	[0.003]	[0.003]	[0.007]	[0.007]	[0.002]	[0.002]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	0.005***	0.004**	0.003***	0.008***	0.006***	0.004***
	[0.000]	[0.002]	[0.001]	[0.003]	[0.001]	[0.001]
$RP * cov(s, (s - s_{it})^2)$ $(-\gamma_3)$	-0.003***	-0.003***	-0.007***	-0.007***	-0.001***	-0.001***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.000]
IQ Interactions						
$IQ * \overline{s_{it}}$ (κ_1)	0.001	0.001	-0.002	-0.006	0.007***	0.008***
	[0.002]	[0.002]	[0.004]	[0.005]	[0.001]	[0.001]
$IQ * s_{it}$ (κ_2)	-0.003***	-0.003***	0.062***	0.062***	-0.002***	-0.002***
	[0.001]	[0.001]	[0.002]	[0.002]	[0.001]	[0.001]
N	118535	118535	44840	44840	135681	135681
AR2	0.04	0.04	0.07	0.07	0.10	0.11
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

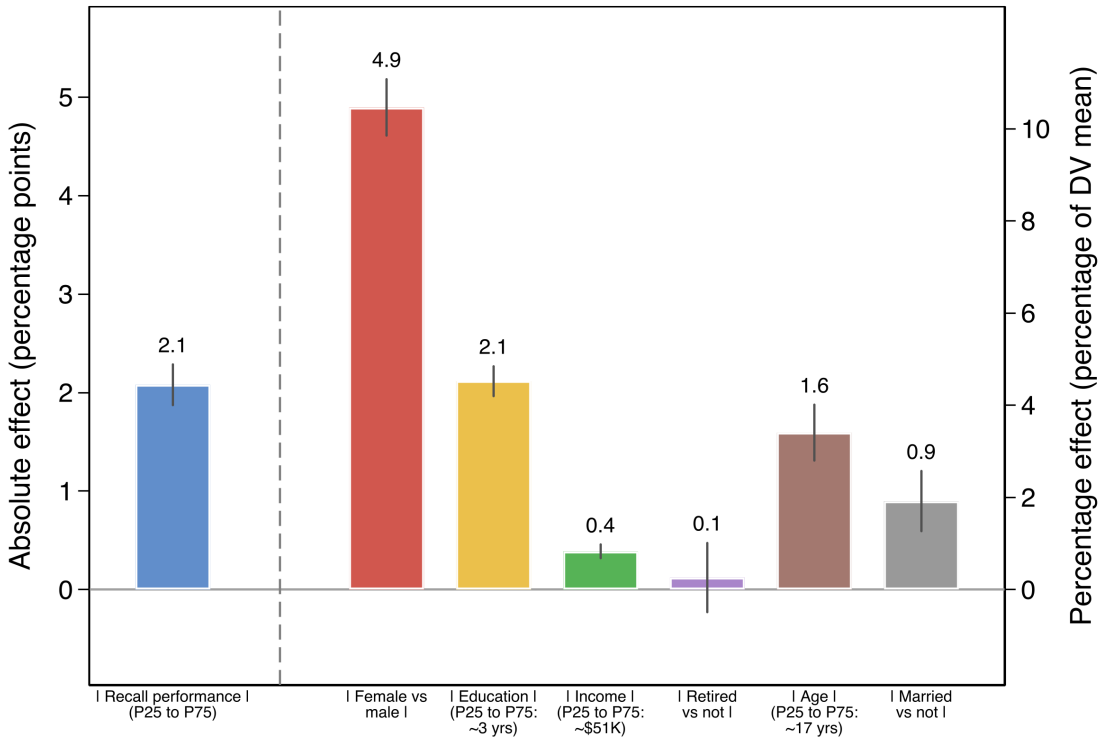
Notes: Each column replicates the corresponding column of the main Table 1 with the addition of IQ × mean-experience and IQ × current-signal interactions (IQ $\times\overline{s_{it}}$ and IQ $\times s_{it}$). The signal is the most recent realized return that enters the covariance term. Its main effect is omitted by design; only its interaction with IQ enters. Dependent variables: cols 1–2 chance stocks rise; cols 3–4 chance home prices rise; cols 5–6 chance of living 10+ years. Columns 2, 4, 6 additionally include RP as a main effect. All regressors standardized by sample IQR prior to estimation. Demographic controls include age, gender, marital status, retirement status, income, and education; home-price regressions also include home ownership. Survey-year fixed effects are absorbed in every column; columns 5–6 additionally absorb age and birth-cohort fixed effects. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table D.2: DS Regressions With Survey-Year Fixed Effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live10+**	
Cognition						
RP	.	0.003	.	-0.043***	.	0.016***
	.	[0.005]	.	[0.010]	.	[0.001]
IQ	0.023***	0.023***	0.001	0.002	-0.015***	-0.021***
	[0.001]	[0.001]	[0.003]	[0.003]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	-0.004***	-0.003	0.006**	-0.017***	0.027***	0.028***
	[0.001]	[0.003]	[0.003]	[0.006]	[0.002]	[0.002]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.003***	0.014***	0.007***	0.006***
	[0.000]	[0.001]	[0.001]	[0.003]	[0.001]	[0.001]
$RP * cov(s, (s - s_{it})^2)$ $(-\gamma_3)$	-0.003***	-0.003***	-0.005***	-0.005***	-0.001***	-0.001***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.000]	[0.000]
N	139507	139507	49940	49940	137602	137602
AR2	0.04	0.04	0.10	0.10	0.11	0.11
Demographic Controls	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

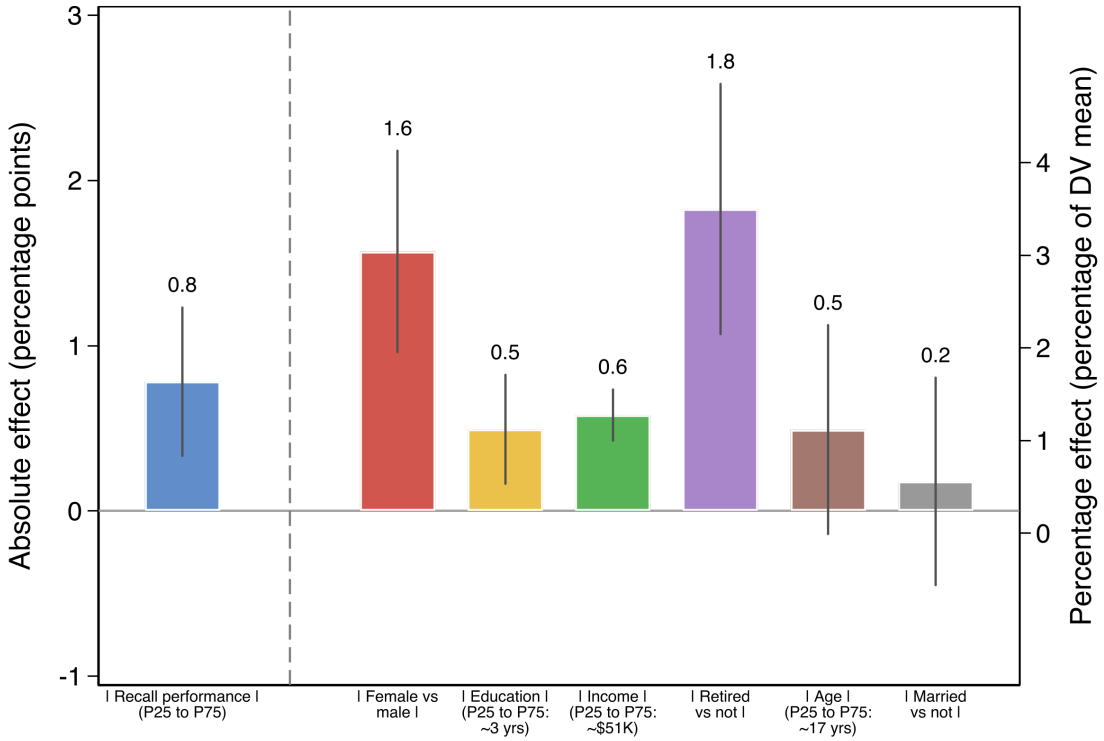
Notes: Each column replicates the corresponding column of the main Table 1 with the addition of survey-year fixed effects throughout. Year fixed effects absorb shocks common to all respondents in a survey wave, such as macroeconomic news or market-wide returns. Persistence of the RP × experience interactions (γ_2, γ_3) after absorbing year effects shows that results reflect individual memory reactions to personal past experiences rather than common aggregate information. All other specification choices are identical to Table 1: dependent variables, regression specification, controls, and (non-year) fixed effects structure. Dependent variables: cols 1–2 chance stocks rise; cols 3–4 chance home prices rise; cols 5–6 chance of living 10+ years. Columns 2, 4, 6 additionally include RP as a main effect. Demographic controls include age, gender, marital status, retirement status, income, and education; home-price regressions also include home ownership. RP, IQ, and the experience variables are divided by their sample IQR (the interquartile range in each column's estimation sample) before estimation. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Figure D.1: Economic Magnitude of Recall Performance Versus Main Demographic Controls: Chance Stocks Will Go Up.



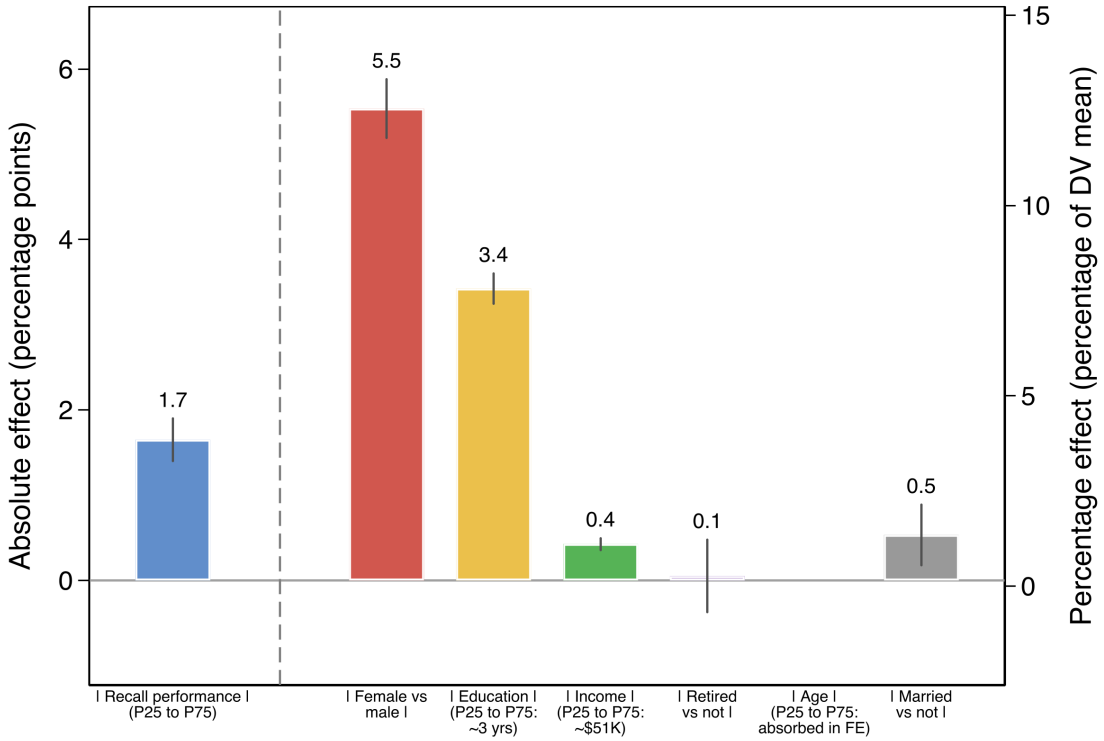
Notes: Bars show the change in the predicted belief (in percentage points) implied from moving each regressor by its interquartile range, holding the other regressors constant. The estimates are computed from column 1 of Table 1 (outcome: Chance Stocks Will Go Up; $N = 139{,}507$). The interquartile ranges are computed on the full HRS panel rather than on each column's estimation sample, so the three figures plot the same shift. Binary regressors move from 0 to 1 instead of an interquartile shift. The recall-performance bar is $\hat{\gamma}_2 \cdot \overline{s}_{it}$ at the sample mean of $\overline{s}_{it}$, following Equation 8. Error bars are 95% confidence intervals.

Figure D.2: Economic Magnitude of Recall Performance Versus Main Demographic Controls: Chance Home Prices Will Go Up.



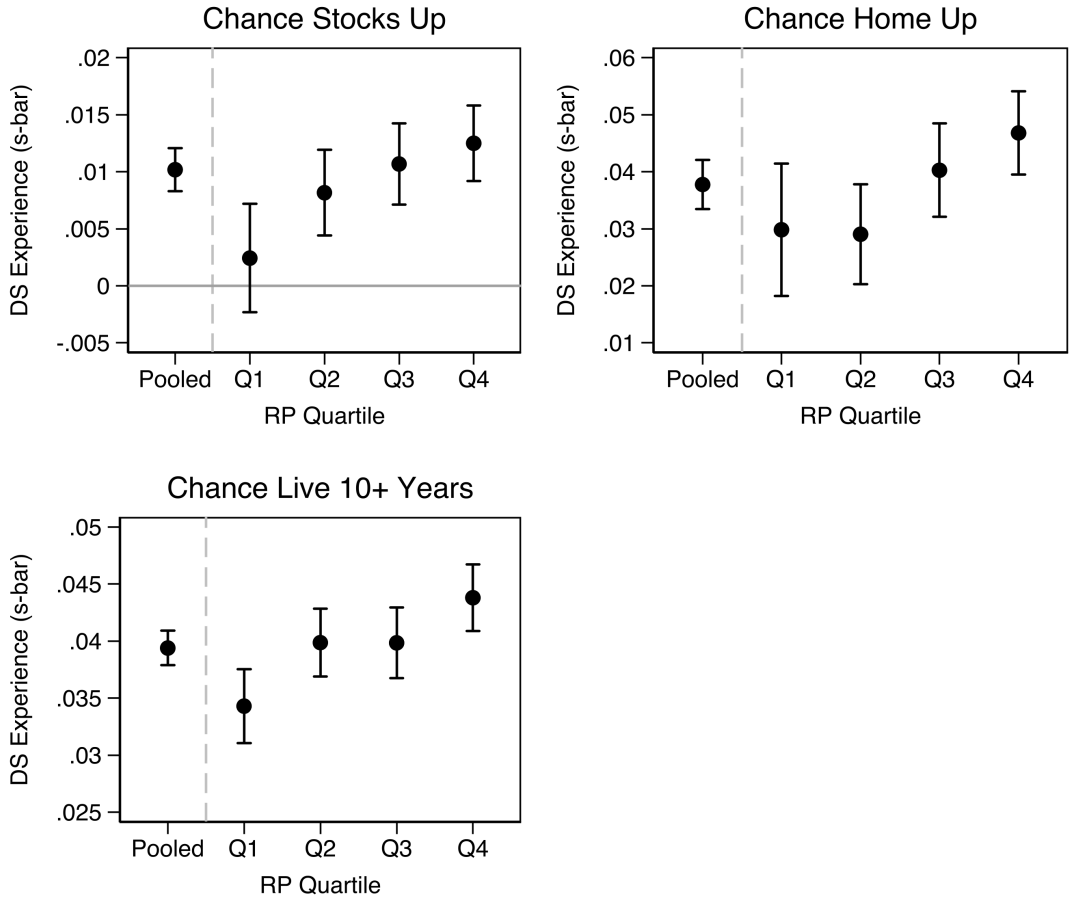
Notes: Bars show the change in the predicted belief (in percentage points) implied from moving each regressor by its interquartile range, holding the other regressors constant. The estimates are computed from column 3 of Table 1 (outcome: Chance Home Prices Will Go Up; $N = 49{,}940$). The interquartile ranges are computed on the full HRS panel rather than on each column's estimation sample, so the three figures plot the same shift. Binary regressors move from 0 to 1 instead of an interquartile shift. The recall-performance bar is $\hat{\gamma}_2 \cdot \overline{s}_{it}$ at the sample mean of $\overline{s}_{it}$, following Equation 8. Error bars are 95% confidence intervals.

Figure D.3: Economic Magnitude of Recall Performance Versus Main Demographic Controls: Chance of Living 10+ Years.



Notes: Bars show the change in the predicted belief (in percentage points) implied from moving each regressor by its interquartile range, holding the other regressors constant. The estimates are computed from column 6 of Table 1 (outcome: Chance of Living 10+ Years; $N = 137{,}602$). The interquartile ranges are computed on the full HRS panel rather than on each column's estimation sample, so the three figures plot the same shift. Binary regressors move from 0 to 1 instead of an interquartile shift. The recall-performance bar is the RP base effect plus $\hat{\gamma}_2 \cdot \overline{s}_{it}$ at the sample mean of $\overline{s}_{it}$, following Equation 8. Age is absorbed in the column fixed effects. Error bars are 95% confidence intervals.

Figure D.4: Non-Parametric RP-Quartile Experience Coefficients, by Domain.



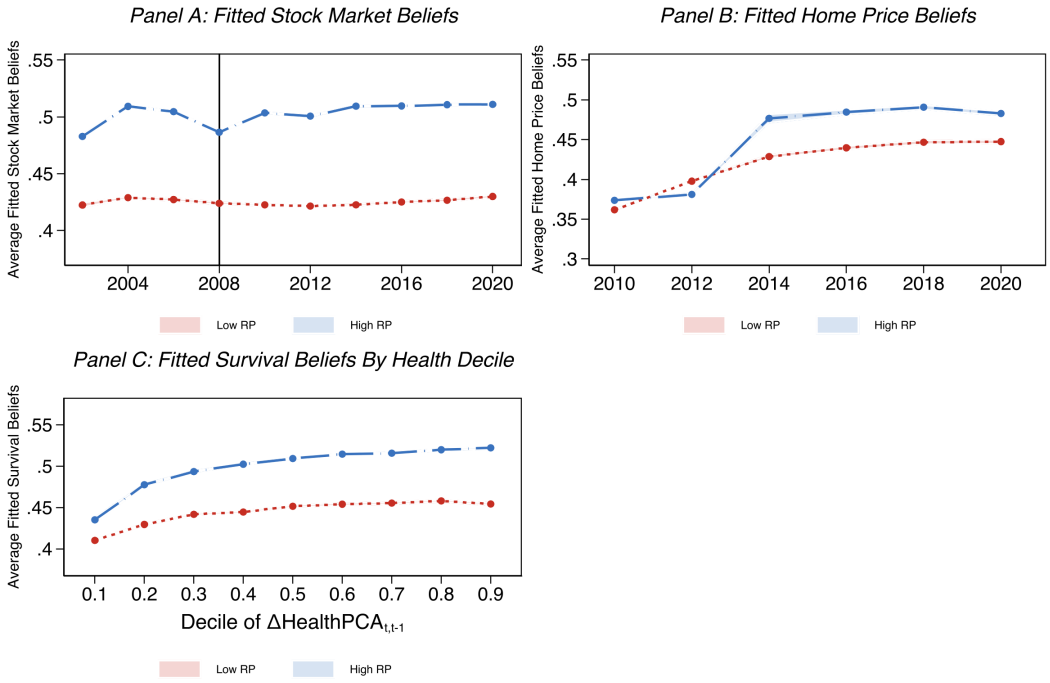
Notes: Each panel reports, for one belief domain (top-left: stock market; top-right: home prices; bottom: survival), the DS experience coefficient ($\hat{\gamma}_1$ on $\bar{s}_{it}$) and its 95% confidence interval from the specification in columns 1, 3, and 5 of Table 1, estimated separately within each recall performance (RP) quartile. The RP $\times$ Experience interaction is dropped, so each estimate is the experience effect within that quartile. Pooled is the full-sample estimate. Continuous regressors are divided by their interquartile range in the full sample. Demographic controls are included throughout (IQ, age, gender, marital status, retirement status, income, and education; the home-price panel also includes home ownership, and the survival panel absorbs age through age and cohort fixed effects).

Table D.3: Stock-Market Belief Regressions, by Whether Respondent Follows the Stock Market.

	(1)	(2)	(3)	(4)
	Follow Stocks		**Don't Follow Stocks**	
Cognition				
RP	.	0.005	.	-0.004
	.	[0.008]	.	[0.008]
IQ	0.018***	0.018***	0.017***	0.017***
	[0.002]	[0.002]	[0.002]	[0.002]
DS Experiences				
$\overline{s_{it}}$ (γ_1)	0.006***	0.009**	0.000	-0.001
	[0.002]	[0.004]	[0.002]	[0.003]
RP Interactions				
$RP * \overline{s_{it}}$ (γ_2)	0.004***	0.003	0.004***	0.005***
	[0.000]	[0.002]	[0.000]	[0.002]
$RP * cov(s, (s - s_{it})^2)$ $(-\gamma_3)$	-0.004***	-0.004***	-0.003***	-0.003***
	[0.001]	[0.001]	[0.001]	[0.001]
N	71154	71154	67818	67818
AR2	0.03	0.03	0.02	0.02
Demographic Controls	Y	Y	Y	Y

Notes: Each column reports a panel regression of the subjective probability that the stock market will rise on Recall Performance (RP), IQ, stock-return experience variables ($\overline{s_{it}}$, $cov(s_k, (s_k - s_{it})^2)$), and their interactions with RP. The specification mirrors columns 1–2 of Table 1. Columns 1 and 3 omit the RP main effect, and columns 2 and 4 add it back. Columns 1–2 restrict the sample to respondents who report following the stock market; columns 3–4 restrict to respondents who do not. Demographic controls include age, gender, marital status, retirement status, income, and education. All regressors IQR-standardized within the estimation sample. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Figure D.5: Fitted Beliefs Over Time for High vs. Low Recall Performance: Full Regression with Respondent Own Covariates.



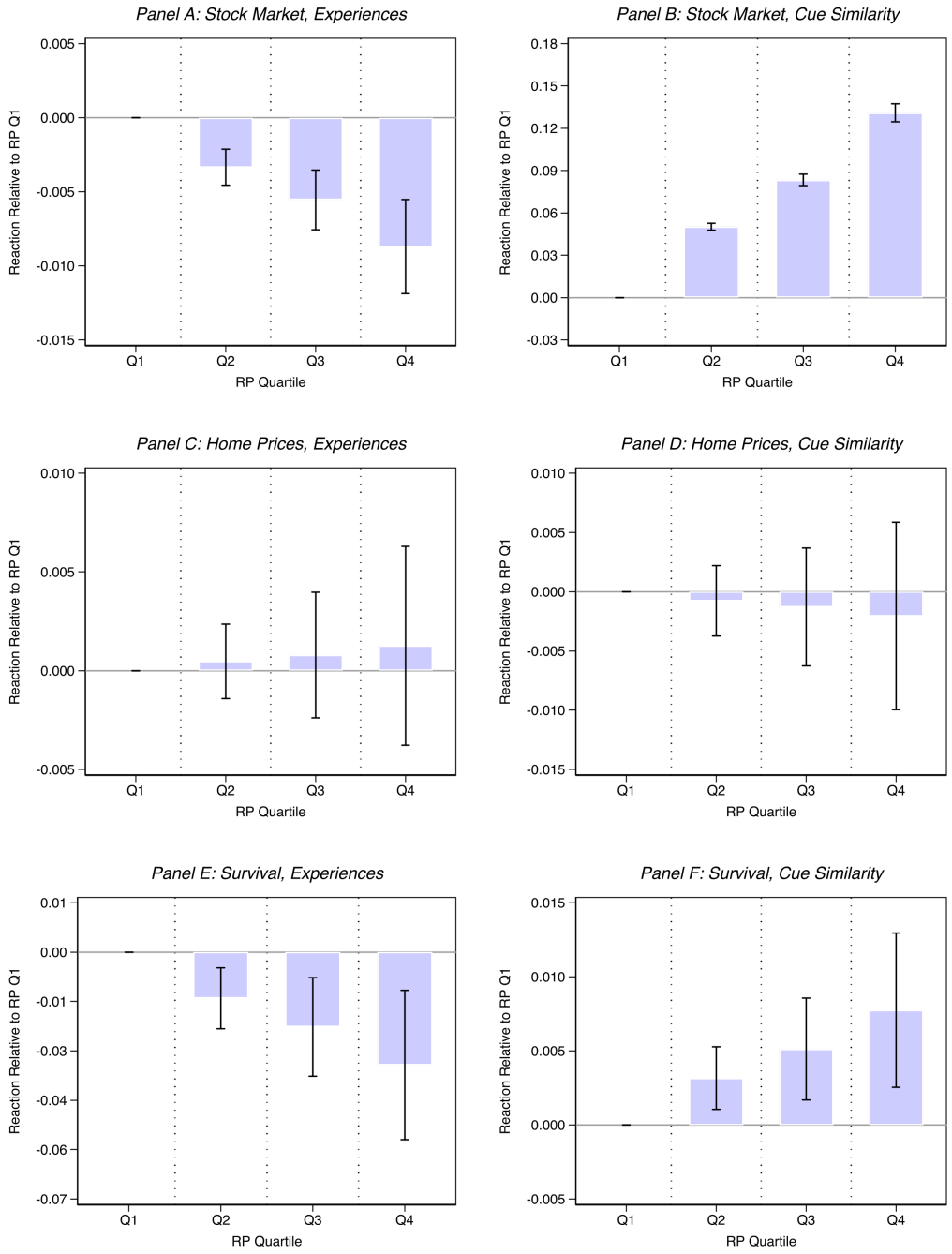
Notes: Fitted beliefs from the full database regressions of Table 1 (columns 1, 3 and 5 for Panels A, B and C), computed with each respondent's actual covariates, for the bottom (red) and top (blue) quartile of Recall Performance. Panels A and B plot fitted stock market and home-price beliefs over time; Panel C plots fitted survival beliefs by decile of $\Delta\text{HealthPCA}_{t,t-1}$. Shaded bands are 95% confidence intervals. Compare Figure 6, which holds demographics and IQ at their sample means.

Table D.4: Forecast Errors, DS Experiences and Cues, by Recall Performance.

	Dependent variable: forecast error (belief − realization)					
	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live10+**	
Cognition						
RP	.	0.044***	.	-0.046***	.	0.010***
	.	[0.011]	.	[0.017]	.	[0.004]
IQ	-0.001	-0.001	0.008	0.008	0.008	0.007
	[0.003]	[0.003]	[0.005]	[0.005]	[0.005]	[0.005]
DS experiences						
$\overline{s_{it}}$ (γ_1)	0.897***	0.924***	0.045***	0.021	0.042***	0.041***
	[0.011]	[0.013]	[0.013]	[0.015]	[0.006]	[0.006]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	-0.003***	-0.013***	0.0005	0.013***	-0.008***	-0.008***
	[0.001]	[0.003]	[0.0009]	[0.005]	[0.003]	[0.003]
$RP * cov(s, (s - s_{it})^2)$ $(-\gamma_3)$	-0.041***	-0.041***	0.001	0.0005	-0.0005***	-0.001***
	[0.001]	[0.001]	[0.001]	[0.0011]	[0.0002]	[0.000]
N	136,293	136,293	47,772	47,772	39,422	39,422
AR2	0.43	0.43	0.51	0.51	0.64	0.64
Demographic Controls	Y	Y	Y	Y	Y	Y
Survey-year FE	Y	Y	Y	Y	Y	Y
Respondent FE	Y	Y	Y	Y	Y	Y
Age FE	N	N	N	N	Y	Y
Cohort FE	N	N	N	N	Y	Y

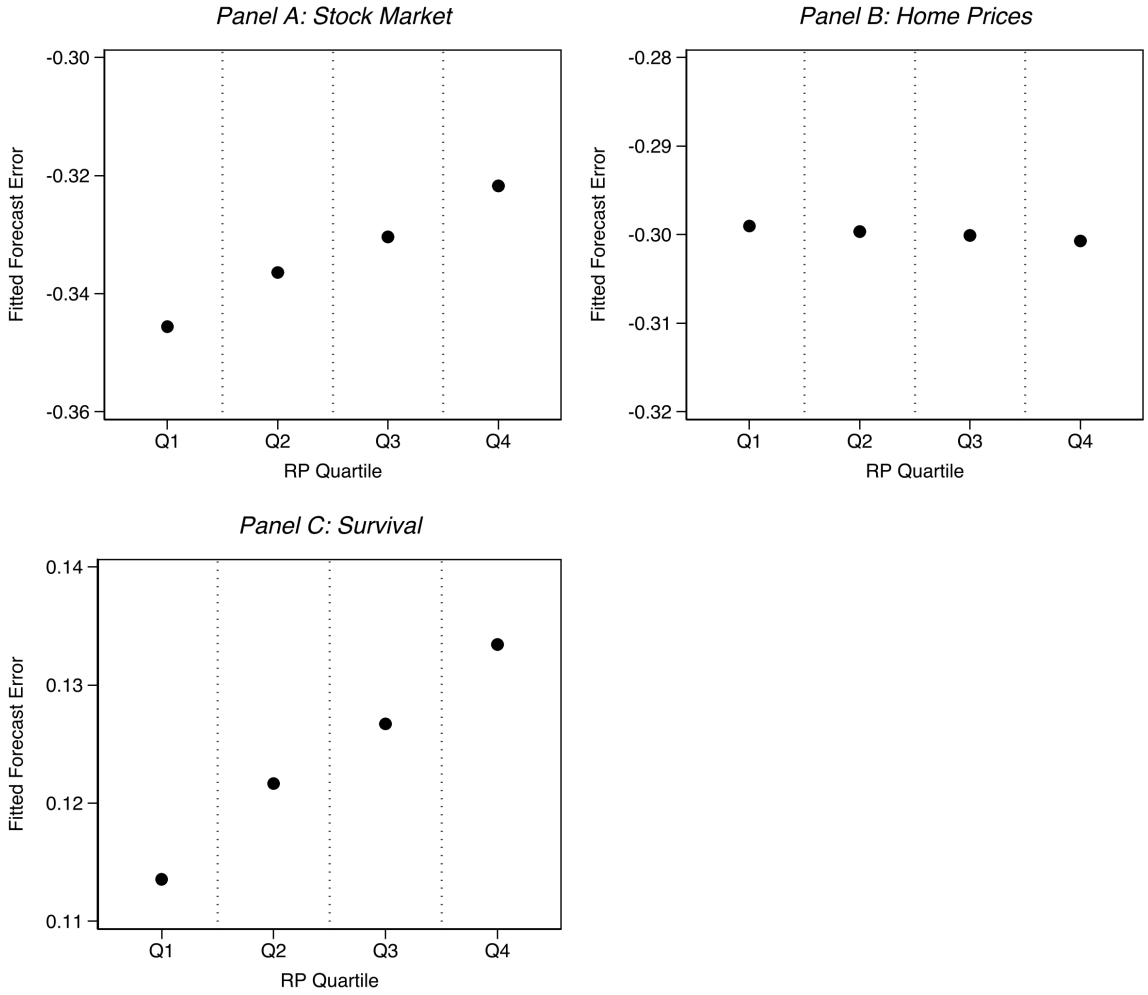
Notes: Each column reports a panel regression of the forecast error on Recall Performance (RP), IQ, domain-specific (DS) experience variables ($\overline{s_{it}}$, $\text{cov}(s_k, (s_k - s_{it})^2)$), and their interactions with RP as shown in Equation 8. The forecast error is the stated probability minus an indicator for the realized outcome: the S&P 500 rising over the twelve months after the interview (stocks), the regional house-price index rising over the twelve months after the interview (home prices), and survival to the target age in the survey question according to the HRS mortality tracker (health; horizons ending by 2022). Demographic controls include age, gender, marital status, retirement status, income, and education (plus homeownership for home prices); age is absorbed by the fixed effects. Columns 1–2 use stock market beliefs; columns 3–4 home-price beliefs (excluding 2008); columns 5–6 health beliefs. All columns include respondent and survey-year fixed effects; columns 5–6 also include age and cohort fixed effects. Columns 2, 4, 6 additionally include the RP base effect; columns 1, 3, 5 are the regressions behind Figure 7 and Figure D.6. RP, IQ, and the experience variables are divided by their sample IQR (the interquartile range in each column's estimation sample) before estimation. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Figure D.6: Reaction of Fitted Forecast Errors to Experiences and Cue Similarity, by Recall Performance.



Notes: Each bar is a difference in differences from the regressions of Figure 7: the fitted forecast error at the top tercile of the variable minus that at the bottom tercile, in each RP quartile, minus the same difference in the bottom RP quartile. The bottom quartile is therefore zero by construction. The left panels use lifetime experience, $\overline{s}_{it}$; the right panels use the similarity between the current cue and the respondent's above-average past experiences, the covariance term of Equation 8 with its sign flipped, so that a positive bar means that higher RP respondents react more strongly to a cue that resembles their above-average experiences. Panels A and B: stock market beliefs. Panels C and D: home-price beliefs. Panels E and F: survival beliefs. Vertical bars are 95% confidence intervals.

Figure D.7: Fitted Forecast Errors by Recall Performance: Including the RP Base Effect.



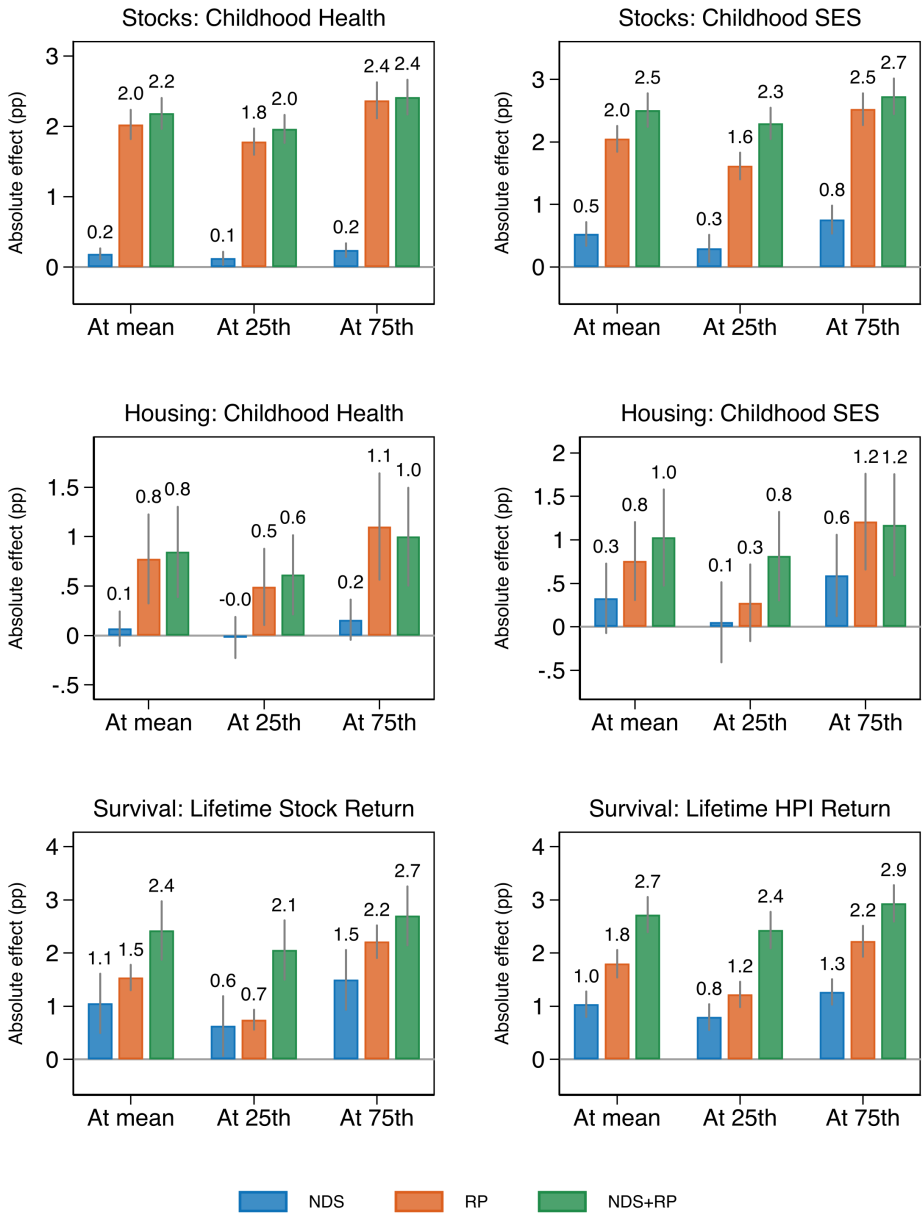
Notes: Fitted forecast errors (stated probability minus an indicator for the realized event) by quartile of Recall Performance (RP), from the regressions in Table D.4, columns 2, 4 and 6. These regressions add the RP base effect to the specification of Figure 7, so the RP gradient combines the base effect with the RP interactions. Predictions use each respondent's own RP and sample means for all other regressors and the fixed effects; points are quartile means of the predictions. Panel A: stocks up over the next twelve months (S&P 500; N = 136,293). Panel B: home value up over the next twelve months (regional house-price index, 2010 onward; N = 47,772). Panel C: survival to the target age (HRS tracker; N = 39,422). Bars: 95% confidence intervals from the standard error of the quartile mean.

Table D.5: NDS Database Regressions With Survey-Year Fixed Effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live 10+**	
Cognition *(η)*						
RP	0.005	0.003	-0.047***	-0.043**	-0.003	-0.013**
	[0.006]	[0.005]	[0.014]	[0.013]	[0.004]	[0.006]
IQ	0.023***	0.023***	0.002	0.001	-0.019***	-0.022***
	[0.001]	[0.001]	[0.004]	[0.004]	[0.002]	[0.002]
DS Experiences *(θ)*						
$\overline{s_{it}}$ (γ_1)	-0.002	-0.002	-0.019*	-0.016*	0.029***	0.034***
	[0.003]	[0.003]	[0.009]	[0.008]	[0.002]	[0.002]
RP Interactions *(β)*						
$RP \times \overline{s_{it}}$ (γ_2)	0.003**	0.004***	0.015***	0.014***	0.005***	0.002**
	[0.001]	[0.001]	[0.004]	[0.004]	[0.001]	[0.001]
$RP \times \text{cov}(s, (s - s_{it})^2)$ ($-\gamma_3$)	-0.003***	-0.003***	-0.005**	-0.005**	-0.001***	-0.001***
	[0.001]	[0.001]	[0.002]	[0.002]	[0.000]	[0.000]
NDS Experiences *(δ, τ)*						
nds (δ_1)	-0.000	-0.004	-0.002	-0.004	0.034***	-0.009***
	[0.001]	[0.002]	[0.002]	[0.006]	[0.005]	[0.003]
$RP \times nds$ (τ)	0.001**	0.005***	0.002*	0.005*	0.015***	0.010***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.002]	[0.002]
N	134298	139504	49561	49939	137609	118169
AR2	0.04	0.04	0.10	0.10	0.07	0.07
NDS Var	Chld.Health	Chld.SES	Chld.Health	Chld.SES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y

Notes: Replicates Table 2 adding survey-year fixed effects throughout. Year fixed effects absorb common time trends, so the RP–experience interactions are identified from within-year, cross-person variation. Columns 5–6 therefore replace the age fixed effects used in the health columns of Table 2 with survey-year fixed effects, since age is determined by survey year and birth cohort. Age remains a linear demographic control in every column. NDS variables: childhood health and SES for stocks and home prices; lifetime equity and home-price returns for health. All continuous regressors (RP, IQ, $\overline{s_{it}}$, $\text{cov}(s_k, (s_k - s_{it})^2)$, NDS) divided by within-sample IQR; coefficients represent a move from p25 to p75. Demographic controls: age, gender, marital status, retirement status, income, education. Standard errors in columns 3 and 4 are clustered by census region; other columns report conventional standard errors. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Figure D.8: NDS Impact Coefficient Sizes: Stock, Home-Price, and Survival Beliefs.



Notes: Each panel reports the absolute marginal effect, in percentage points of the dependent variable, at three evaluation points—*At mean*, *At 25th*, *At 75th*—denoting where all four covariates (RP, NDS, IQ, Experiences) are held when computing marginal effects. Within each group, three bars: NDS = marginal effect of the non-domain-specific experience holding RP constant; RP = marginal effect of recall performance holding NDS constant; NDS+RP = combined effect of moving both NDS and RP from the 25th to the 75th percentile. Rows correspond to the three dependent variables (chance stocks rise; chance home prices rise; chance of living 10+ years). The two columns within each row report the two non-domain-specific experience measures used for that outcome: childhood health and childhood SES indices for the stock and home-price outcomes, and lifetime S&P 500 and HPI returns for survival. Error bars: 95% CIs. Variables standardized by sample IQR.

Table D.6: NDS Database Regressions: Robustness of Simple NDS Indexes.

NDS index components used in this version:

Childhood Health (4 vars): severe childhood conditions, disabled as a child, total childhood conditions, missed school for health.

Childhood SES (2 vars): father's education, mother's education.

	(1)	(2)	(3)	(4)	(5)	(6)
	Chance Stocks ↑		**Chance Home ↑**		**Chance Live 10+**	
Cognition						
IQ	0.024***	0.024***	0.016**	0.013*	-0.019***	-0.022***
	[0.001]	[0.002]	[0.005]	[0.006]	[0.002]	[0.002]
DS Experiences						
$\overline{s_{it}}$ (γ_1)	0.001	0.002*	0.029***	0.029***	0.026***	0.029***
	[0.001]	[0.001]	[0.008]	[0.007]	[0.002]	[0.002]
RP Interactions						
$RP * \overline{s_{it}}$ (γ_2)	0.005***	0.004***	0.005***	0.004***	0.007***	0.006***
	[0.000]	[0.000]	[0.001]	[0.001]	[0.001]	[0.001]
$RP * cov(s_k, (s_k - s_{it})^2)$ $(-\gamma_3)$	-0.004***	-0.004***	-0.031***	-0.031***	-0.001***	-0.001***
	[0.000]	[0.000]	[0.007]	[0.007]	[0.000]	[0.000]
NDS Experiences						
nds (δ_1)	-0.001	0.000	-0.004	-0.007	-0.006*	0.001
	[0.001]	[0.002]	[0.002]	[0.007]	[0.003]	[0.001]
$RP * nds$ (δ_2)	0.002***	0.005***	0.002**	0.007**	0.009***	0.005***
	[0.001]	[0.001]	[0.001]	[0.003]	[0.001]	[0.000]
N	134298	124690	49561	46083	137602	118162
AR2	0.04	0.04	0.05	0.05	0.11	0.11
NDS Var	Chld.Health	Chld.SES	Chld.Health	Chld.SES	$\Delta_L sp500$	$\Delta_L hpi$
Demographic Controls	Y	Y	Y	Y	Y	Y
Cohort FE	N	N	N	N	Y	Y
Age FE	N	N	N	N	Y	Y

Notes: Robustness version of the main NDS table with simpler childhood indexes. The childhood-health index in columns 1 and 3 uses four objective components and drops psychological childhood conditions. In columns 2 and 4 the childhood SES index uses parental education only. Columns 5 and 6 use the same lifetime stock-return and lifetime home-price-return NDS variables as the main table. Everything else (sample, demographic controls, fixed effects, IQR rescaling of continuous regressors, standard errors) is as in the main NDS Table 2. All continuous regressors (RP, IQ, the DS experience mean and covariance, and the NDS index) are divided by their within-sample IQR prior to estimation, so each coefficient is the change in beliefs associated with a P25-to-P75 move in the regressor. Demographic controls include age, gender, marital status, retirement status, income, and education. Standard errors in columns 3 and 4 are clustered by census region; other columns report conventional standard errors. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table D.7: Descriptive statistics of similarity ratings, by between-subjects group.

Outcome	n	vs. Illness / accident	vs. Financial difficulties
Group 1: personal economic outcomes			
Own home price: future increase of 7% or more	199	2.22 (1.57)	3.66 (2.09)
Own home price: future decrease of 7% or more	199	2.59 (1.75)	4.06 (2.05)
U.S. economy: major depression in the next 10 years	199	3.60 (1.83)	5.75 (1.43)
Cost of living: increase of 5%+ per year over next 5 years	199	2.95 (1.73)	5.28 (1.60)
Group 2: macro-market outcomes			
General U.S. home prices: future increase of 7% or more	203	2.35 (1.55)	4.96 (1.59)
General U.S. home prices: future decrease of 7% or more	203	2.12 (1.54)	3.36 (1.96)
Blue-chip stocks (DJIA): increase of 7% or more	203	1.83 (1.33)	2.91 (1.71)
Blue-chip stocks (DJIA): decrease of 7% or more	203	2.24 (1.47)	4.33 (1.72)

Notes: $N = 402$ U.S. homeowners (screened for owner-occupied housing). Each respondent was randomly assigned to one of two between-subjects groups ($n = 199$ and $n = 203$) and rated the four outcomes in their group. Similarity is measured on a 1–7 scale (1 = not similar at all, 7 = extremely similar). The two reference experiences are (a) a serious or chronic illness or accident, and (b) struggling with financial difficulties. Cell entries are mean perceived similarity, with the standard deviation in parentheses.

Table D.8: Database Regressions for Macroeconomic Beliefs: Baseline and Non-Domain-Specific Experiences.

	Chance Depression				**Chance Inflation > 5%**			
	Baseline		+ NDS		Baseline		+ NDS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cognition								
RP		-0.101***	-0.037	-0.097***		-0.036	-0.045	-0.036
		[0.022]	[0.025]	[0.022]		[0.099]	[0.100]	[0.099]
IQ	-0.028***	-0.028***	-0.028***	-0.028***	-0.000	-0.000	-0.002	-0.000
	[0.003]	[0.003]	[0.003]	[0.003]	[0.004]	[0.004]	[0.004]	[0.004]
DS experiences								
$\overline{s_{it}}$ (γ_1)	0.004**	-0.005**	0.002	-0.005*	-0.001**	-0.002	-0.002	-0.002
	[0.002]	[0.002]	[0.003]	[0.002]	[0.000]	[0.001]	[0.001]	[0.001]
RP Interactions								
$RP * \overline{s_{it}}$ (γ_2)	0.001***	0.006***	0.003*	0.006***	0.000***	0.000	0.000	0.000
	[0.000]	[0.001]	[0.001]	[0.001]	[0.000]	[0.001]	[0.001]	[0.001]
$RP * cov(s,(s-s_{it})^2)$ ($-\gamma_3$)	-0.058***	-0.058***	-0.058***	-0.058***	-0.013***	-0.013***	-0.010***	-0.013***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
NDS experience								
nds (δ_1)			-0.004***	0.001			-0.002	0.007
			[0.001]	[0.004]			[0.001]	[0.005]
$RP * nds$ (δ_2)			-0.000	-0.007***			-0.000	-0.006**
			[0.000]	[0.002]			[0.001]	[0.003]
NDS Var	—	—	Chld.Health	Chld.SES	—	—	Chld.Health	Chld.SES
N	46576	46576	42984	46576	29682	29682	28452	29682
AR2	0.05	0.05	0.06	0.05	0.01	0.01	0.01	0.01
Min year	2002	2002	2002	2002	2006	2006	2006	2006
Max year	2008	2008	2008	2008	2008	2008	2008	2008
Demographic Controls	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Each column reports a panel regression of subjective macroeconomic beliefs on Recall Performance (RP), IQ, domain-specific (DS) experience variables ($\overline{s_{it}}$ and $cov(s,(s-s_{it})^2)$), and their RP interactions, mirroring Table 1. For each outcome the first two (*Baseline*) columns add cognition and RP interactions without and then with the RP base effect; the next two (*+ NDS*) columns add a single non-domain-specific experience entered directly (δ_1) and interacted with RP (δ_2), using childhood health and then childhood socioeconomic-status (SES), as reported in the *NDS Var* row. Childhood health and childhood SES are never included in the same regression. **Chance Depression** is the subjective probability that the U.S. economy will experience a major depression within the next 10 years, paired with lifetime NBER recession experience. **Chance Inflation > 5%** is the subjective probability that the cost of living will rise by more than 5% on average over the next 10 years, paired with lifetime inflation experience. All continuous regressors are divided by their within-sample IQR prior to estimation. Min/Max year report the earliest and latest HRS wave year in each column's estimation sample. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table D.9: Choices on Previous-Wave Measured Beliefs.

	(1)	(2)	(3)	(4)	(5)	(6)
	New Stock Market Inv.	% IRA in Stocks	Bought a New Home	Work Past 70	Retired	% Income to Pension
Cognition						
RP (1 sd) (κ^M)	0.017***	0.022***	0.004***	0.016***	-0.010***	0.001
	[0.001]	[0.003]	[0.001]	[0.002]	[0.001]	[0.001]
IQ (1 sd)	0.022***	0.041***	0.007***	0.010***	-0.007***	0.003**
	[0.001]	[0.003]	[0.001]	[0.002]	[0.001]	[0.001]
Beliefs *(ϕ)*						
$B_{i,t-2}$ (θ)	0.021***	0.053***	0.002*	0.040***	-0.005***	0.002*
	[0.001]	[0.002]	[0.001]	[0.002]	[0.001]	[0.001]
Avg DV	0.10	0.56	0.03	0.26	0.60	0.08
N	118874	36555	42163	35033	137464	7375
Demographic Controls	Y	Y	Y	Y	Y	Y
Fixed Effects	N	N	N	Age/Cohort	Age/Cohort	Age/Cohort

Notes: Each column regresses a single choice outcome on the survey-measured belief relevant to that choice, taken from the respondent's *previous* interview ($B_{i,t-1}$, θ), recall performance (RP, κ^M), IQ, and demographic controls (age, gender, marital status, income, education; the household-retirement control is omitted for the health outcomes). The belief is the respondent's stated probability that stock prices rise over the next year (for the two stock outcomes), that the value of their own home rises (for home purchase), and of surviving at least ten more years (for the three health outcomes). The sample is smaller than in the contemporaneous specification because it requires the respondent to be observed in two consecutive waves. All regressors are scaled to one standard deviation. Columns 1–3 (*New Stock Market Inv.*, *% IRA in Stocks*, *Bought a New Home*) are estimated by pooled OLS without fixed effects; columns 4–6 (*Work Past 70*, *Retired*, *% Income to Pension*) include age and birth-cohort fixed effects. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.

Table D.10: Robustness: Choice Regressions Including RP Directly in the Second Stage.

	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Stock/Home-Price Market Choices					
	New Stock Market Inv.		**% IRA in Stocks**		**Bought a New Home**	
Cognition						
RP (1 sd) (κ^M)	0.011***	0.012***	0.017***	0.017***	0.003**	0.002**
	[0.001]	[0.001]	[0.003]	[0.003]	[0.001]	[0.001]
IQ (1 sd)	0.015***	0.012***	0.036***	0.033***	0.006***	0.006***
	[0.001]	[0.001]	[0.003]	[0.003]	[0.001]	[0.001]
Lagged Beliefs (ϕ)						
$\widehat{B}_{i,t-2}$ (θ)	0.046***	.	0.052***	.	0.006***	.
	[0.002]	.	[0.005]	.	[0.001]	.
$B_{E_{i,t-2}}$ (γ)	.	-0.012***	.	-0.014***	.	0.001
	.	[0.001]	.	[0.003]	.	[0.001]
$B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$ (μ)	.	0.009***	.	0.012***	.	0.005***
	.	[0.001]	.	[0.003]	.	[0.001]
Avg DV	0.10	0.10	0.56	0.56	0.03	0.03
N	116203	112638	36085	35277	41728	41351
Demographic Controls	Y	Y	Y	Y	Y	Y
Risk-taking	N	Y	N	Y	N	Y
	Panel B: Health Choices					
	Work Past 70		**Retired**		**% of Income to Pension**	
Cognition						
RP (1 sd) (κ^M)	0.012***	0.012***	-0.004***	-0.006***	0.001	0.000
	[0.002]	[0.002]	[0.002]	[0.002]	[0.001]	[0.001]
IQ (1 sd)	0.011***	0.007**	-0.009***	-0.014***	0.002	0.002
	[0.003]	[0.003]	[0.001]	[0.002]	[0.001]	[0.001]
Lagged Beliefs (ϕ)						
$\widehat{B}_{i,t-2}$ (θ)	0.031***	.	-0.038***	.	-0.000	.
	[0.003]	.	[0.002]	.	[0.001]	.
$B_{E_{i,t-2}}$ (γ)	.	0.024***	.	-0.022***	.	-0.002
	.	[0.002]	.	[0.001]	.	[0.001]
$B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$ (μ)	.	0.005***	.	-0.011***	.	0.002*
	.	[0.002]	.	[0.001]	.	[0.001]
Avg DV	0.26	0.26	0.66	0.66	0.08	0.08
N	22408	22401	95380	91955	5013	5011
Demographic Controls	Y	Y	Y	Y	Y	Y
Risk-taking	N	Y	N	Y	N	Y
Cohort/Age FE	Y	Y	Y	Y	Y	Y

Notes: Robustness version of Table 3. The two-stage structure is identical: Stage 1 predicts beliefs from RP, IQ, experience interactions, and the RP × experience-covariance interaction (same specification as Table 2). Stage 2 decomposes fitted beliefs into three additive components: experience ($B_{E_{i,t-2}}$, γ); total memory-amplified contribution to fitted belief ($B_{(\mathrm{RP}_{i,t-2}\cdot E_{i,t-2})}$, μ), summing the Stage-1 fitted contributions of the RP × experience and RP × experience-covariance interactions; and a controls component ($B(Dems)$). As in Table 3, Stage 2 uses the fitted-belief components from the respondent's previous survey wave, two years earlier ($t-2$). The only difference from Table 3 is that RP (Recall Performance) is also included as a direct second-stage regressor (κ^M), allowing it to affect choices through channels beyond its belief-shaping effect. Odd columns (1, 3, 5): total lagged predicted belief $\widehat{B}_{i,t-2}$ (θ). Even columns (2, 4, 6): full decomposition with risk taking. The choice variables are defined in Section 5. Panel A is estimated by pooled OLS without fixed effects. Panel B includes age and cohort fixed effects. Demographic controls in all columns. Coefficients scaled to 1 SD of each regressor. Standard errors in brackets; * p<0.10, ** p<0.05, *** p<0.01.